

## Highlights

**A Cartesian multipole hierarchy for the voxel: closed-form single-site  $T$ -matrices, and the error of elastic volume-integral scattering isolated on a space-filling layer of cubes**

T. M. Nestor

- A Cartesian multipole hierarchy gives the cube's single-site  $T$ -matrix in closed form
- A voxel needs two bases: one for the medium and one for the wavefield
- The order is  $\min(2p+2, 2r+2)$  for field degree  $p$  and medium degree  $r$
- The single-site  $T$ -matrix's error is the medium's relative projection error
- The error is known in advance, from the medium and the grid alone

A Cartesian multipole hierarchy for the voxel:  
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space-filling layer of cubes

T. M. Nestor

*Independent researcher, Australia*

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**Abstract**

To model seismic waves efficiently in a realistically complicated medium, volume-integral methods divide it into cubic voxels coupled through the background Green's tensor. Each voxel must represent the medium within it and the wavefield within it. We ask which basis each needs and what error a given choice leaves. We answer on a layer of voxels: because cubes tile space,  $n$  planes of voxels are the layer itself, and its exact solution isolates the discretisation error. With the wavefield expanded to Legendre degree  $p$  and the medium's contrast to degree  $r$  in each voxel, the scheme converges at order  $\min(2p + 2, 2r + 2)$ , with leading error  $c_p(kd)^{2p+2}$  in closed form; the orders hold for SV and SH waves and at oblique incidence. The single-site  $T$ -matrix's second-order error equals the relative projection error of the medium at degree  $\min(p, r)$ : the two bases must be matched, and the error is known from the medium before anything is solved. In a plane-stratified background the orders persist: the background changes the field the voxels must represent, not how well they represent it. The single site is the first member of a

Cartesian multipole hierarchy, an elastodynamic extension of the equivalent inclusion method, whose coefficients are moments of the Green's tensor over the cube, in closed form on four constants. Its blocks gain accuracy in pairs: with third gradients it is 300 times more accurate than the uniform-strain closure against the exact sphere, and as a voxel scheme it is fourth order on a graded sphere.

*Keywords:* Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

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## 1. Introduction

*The problem this paper serves.* Seismic waves in the crust are scattered by heterogeneity on every scale, and the aim of the work this paper begins is an efficient model of the wavefield in a medium of realistic complication. The models that describe such a medium are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation divides the medium into cells and treats every cell as a scatterer: a voxel carrying a single-site  $T$ -matrix, the operator that turns the field incident on the voxel into the sources it radiates, and coupled to every other voxel through the Green's tensor of the background, with the multiple-scattering series summed.

*The question: two bases.* A voxel is a small region in which two functions must be represented by a finite number of coefficients. One is the *medium*: the contrast of the elastic moduli and the density from the background,

16 which the model prescribes. The other is the *wavefield*: the displacement and  
 17 strain inside the voxel, which the calculation must find. Each is expanded  
 18 in a basis on the cell, and the two bases are separate choices. Here both  
 19 are Legendre polynomials, the wavefield to degree  $p$  and the contrast to  
 20 degree  $r$ ; the usual scheme, a uniform field in a cell of constant properties,  
 21 is  $p = r = 0$ . Neither choice is new in itself. In higher-order volume integral  
 22 methods for electromagnetics the material within an element is represented  
 23 by polynomials whose orders are independent of those of the field [1], as it  
 24 is in finite elements, and voxels that carry a linear field are in use [2]. What  
 25 has been missing is the law that relates the two bases: the order a given pair  
 26 attains, and the error it leaves. The answer decides the cost of everything  
 27 built on it, because a voxel that represents its contents more accurately can  
 28 be larger, and the number of voxels, not the size of each one's state, sets the  
 29 cost of the multiple-scattering solve.

30 *Why the question is hard to answer.* For the media such a scheme is built for,  
 31 there is no exact solution. Its error cannot be measured against the answer.  
 32 Comparison with another numerical method mixes the errors of both. And a  
 33 refinement study can mislead, because a voxel scheme can converge smoothly  
 34 to the wrong limit. §4 shows how that happens with the most natural choice  
 35 of coupling, the Green's tensor evaluated between voxel centres: it carries  
 36 a local term whose size per unit volume does not shrink as the voxels do,  
 37 and unless the single voxel's response is matched to cancel it, refinement  
 38 converges to the wrong answer without revealing it.

39 *The method: a layer of voxels.* This paper therefore answers the question  
 40 in the simplest configuration in which a voxel scheme runs in full (every

41 voxel, every coupling, the whole multiple-scattering series, at normal and  
 42 at oblique incidence) and the exact answer is nevertheless known: a layer  
 43 filled with voxels. Because cubes tile space, a layer of identical voxels does  
 44 not approximate a homogeneous layer; it *is* one. Any difference between  
 45 its reflection and transmission and those of the continuous layer is therefore  
 46 error of the discretisation alone, with no modelling error and no shape error  
 47 behind which to hide. A sphere, the usual test object for volume-integral and  
 48 discrete-dipole methods [3], cannot give this: its error mixes the discretisation  
 49 with the staircase approximation of a curved surface by cubes. The layer  
 50 plays the part of the patch test in finite elements: reproducing a uniform  
 51 medium correctly is necessary, though not sufficient, for a scheme to be  
 52 trusted in a heterogeneous one. The experiment is then made harder in  
 53 three steps, each with an exact solution of its own: a randomly stratified  
 54 layer, in which every cell is homogeneous but the cells differ; a layer whose  
 55 properties vary smoothly within every cell, which is where the basis for the  
 56 medium is tested; and a sphere whose contrast falls smoothly to zero at its  
 57 surface, which carries the layer's prediction into three dimensions without a  
 58 staircase. The coupling and the single site that the layer validates are the  
 59 same in any medium; a laterally heterogeneous medium remains to be tested.

60 *The answer.* (1) *The order is set by the weaker basis.* The scheme converges  
 61 at order  $\min(2p + 2, 2r + 2)$  in the cell size (§9). A contrast held constant in  
 62 each cell therefore caps every voxel at second order in a medium that varies  
 63 within the cells, however rich the basis of its field; and a contrast of higher  
 64 degree than the field is wasted. (2) *The leading error is in closed form, for*  
 65 *every grid.* For a voxel of side  $d$  and a wave of wavenumber  $k$  it is  $c_p(kd)^{2p+2}$

66 with  $c_0 = 1/12$ ,  $c_1 = 1/720$  and  $c_2 = 1/100800$  (§8); the orders 2, 4 and 6 of  
 67  $p = 0, 1$  and 2 hold for P, SV and SH waves at normal and oblique incidence.  
 68 For the collocation scheme, the uniform-field voxel in its simplest form, the  
 69 error is  $+(kd)^2/8$  in reflection and  $-(kd)^2/24$  in transmission (§7), and the  
 70 opposite signs explain why no correction of the single site can improve both  
 71 (§12). (3) *The error that remains is the single site's, and it is the projection*  
 72 *error of the medium.* The single-site  $T$ -matrix sums the rescattering within a  
 73 voxel to all orders and the coupling between voxels is exact, but the  $T$ -matrix  
 74 is that of the voxel as represented, its contrast projected to degree  $r$  and its  
 75 field held to degree  $p$ . The error of the term of second order in the contrast  
 76 equals the *relative projection error* of the medium, the fraction of the squared  
 77 norm of its profile that polynomials of degree  $\min(p, r)$  on the cells cannot  
 78 represent (Eq. (43)): exactly in the long-wave limit, and to within 1.3% for  
 79 all nine pairs with  $p, r \leq 2$  (§9). The wavefield basis must therefore carry  
 80 what the medium imprints on the field, not only the wave, and the two bases  
 81 must be matched. (4) *The error is known in advance.* The projection error  
 82 is a property of the medium and the grid alone, so the cell size and the  
 83 degree that a required accuracy needs are chosen before anything is solved.  
 84 (5) *Where the medium is piecewise constant on the cells the projection error*  
 85 *vanishes*, a constant suffices for the medium, and the degree of the field  
 86 alone sets the order: in a randomly stratified model one first-moment voxel  
 87 per model layer reaches a reflection accuracy of  $10^{-6}$ , where the uniform-field  
 88 voxel needs about forty (§9), and in the uniform layer it allows voxels 58 times  
 89 larger in each direction (Appendix Appendix I). (6) *In three dimensions* the  
 90 smoothly graded sphere converges at second order with the mean-only voxel,

91 as the layer predicts (§10), and at fourth order with the first-moment voxel  
 92 (§11).

93 *What makes the experiment exact.* The answer above can be trusted only if  
 94 nothing but the two bases is approximated. Three results, set against the  
 95 coupled-dipole method described below, secure that. (a) *An exact coupling.*  
 96 The integrated Green tensor and the filtered coupled dipoles approximate the  
 97 field of a cell; for cubes the source-cell average is exact, because the cube's  
 98 form factor vanishes on the reciprocal lattice (the tiling identity, §5), and  
 99 at normal incidence the lattice kernel is in closed form with no lattice sum.  
 100 (b) *No polarisability to choose.* The polarisability, a modelling choice in the  
 101 coupled-dipole method, is here a derived object (§6): the moments of the  
 102 Green's tensor over the cube, taken as distributions, yield Eshelby's  $\delta$  term  
 103 without adding it by hand, the cube's departure from the sphere as a pure  
 104 cubic anisotropy, and the closure's error against exact scattering by a sphere  
 105 as a real  $(ka)^2$  term in closed form. The self term the single site carries is the  
 106 one the averaged kernel omits, so the discrete system is a collocation of the  
 107 continuum equation. (c) *The elastic state.* A voxel carries a force (through  
 108 its density contrast) and a moment (through its stiffness contrast), so its  
 109 state has nine components, displacement and strain, the coupling is a  $9 \times 9$   
 110 tensor, and the single site's self-interaction is the elastic depolarisation of a  
 111 cube, the discrete counterpart of the inclusion tensor of Eshelby [4]. Cubes  
 112 serve all three: they tile space, so a uniform region cut into cubes is still that  
 113 region, and their regular lattice makes the coupling a convolution, solved by  
 114 fast transforms.

115 *The coupled-dipole method.* Replacing a heterogeneous body by a lattice of  
 116 small scatterers that interact through the background Green's function is the  
 117 multiple-scattering description of Foldy [5] and Lax [6]. In electromagnetics  
 118 it became a numerical method as the coupled-dipole method, or discrete  
 119 dipole approximation (DDA): each cell of a cubic grid is a polarisable point  
 120 dipole driven by the incident field and by the fields of every other dipole  
 121 [7, 8]. The method is a collocation of the volume integral equation for the  
 122 electric field, with the cell's self-interaction absorbed into its polarisability  
 123 [9], and Yurkin and Hoekstra [10] review it in that form. Its accuracy turns  
 124 on two ingredients, and this paper takes up both for elasticity. The first is the  
 125 polarisability. The Clausius–Mossotti value is the static response of a sphere;  
 126 the lattice dispersion relation of Draine and Goodman [11] adjusts it so that  
 127 an infinite lattice propagates waves as the continuum does to a given order  
 128 in  $kd$ ; Rahmani et al. [12] instead derive it from the exact static solution for  
 129 the whole body, so that the method is exact in the long-wavelength limit.  
 130 The second is the coupling. Integrating the Green tensor over the source  
 131 cell rather than sampling it at the centre improves the accuracy, most of  
 132 all for strong contrasts [13], and the filtered coupled dipoles of Piller and  
 133 Martin [14], revived by Yurkin et al. [15] for large refractive indices, replace  
 134 the point dipole by a source of finite extent whose spectrum is band-limited  
 135 to the grid. The coupling summed over a lattice extends the method to  
 136 periodic structures [16]. Its error converges at second order in the dipole  
 137 spacing when shape errors are absent [3], and for a cube, the one shape  
 138 a cubic grid represents without a staircase, Yurkin and Kahnert [17] reach  
 139 relative uncertainties of  $10^{-7}$  to  $10^{-3}$  by extrapolation in the grid. On the

140 mathematical side, Costabel et al. [18] show that the method is a finite section  
 141 of an infinite block Toeplitz matrix, stable for a large range of the contrast  
 142 but not for all of it, and Costabel [19] that in a one-dimensional model it  
 143 converges at second order in the interior, with a boundary layer in which  
 144 the consistency error does not vanish. The layer studied here is the elastic  
 145 counterpart of that cube: a body the grid represents exactly, so that the  
 146 discretisation error can be isolated. What this paper adds is the closed form:  
 147 for cells that tile space, the coupling that makes the lattice reproduce the  
 148 continuum exactly, and the order and the constant of the error that remains,  
 149 as functions of the two bases. In the terms used here the coupled-dipole  
 150 method is the pair  $p = r = 0$ , and its refinements improve the polarisability  
 151 and the coupling while leaving both bases constant in the cell.

152 *Its elastic counterparts.* The elastic problem has the same volume integral  
 153 equation [20], and every cell-based solver of it is a coupled-dipole method  
 154 whether or not it carries the name. Nestor [21] divided a two-and-a-half-  
 155 dimensional medium into cells that interact through the background Green's  
 156 tensor, each carrying a contrast renormalised by its own near field, with  
 157 the multiple-scattering series summed. Kanaun and Levin [22] discretise the  
 158 inclusion problem with Gaussian approximating functions whose matrix ele-  
 159 ments are analytic. Yang et al. [23] solve a stress-velocity form of the integral  
 160 equation on a grid by conjugate gradients with fast Fourier transforms, and  
 161 Touhei [24] does the same for a half space with a generalised Fourier trans-  
 162 form. Jakobsen [25] and Jakobsen et al. [26] build the seismic forward and  
 163 inverse problems on the transition operator of the discretised Lippmann-  
 164 Schwinger equation, the first in the acoustic approximation and the second

165 in arbitrary anisotropic elastic media with a nine-component state vector.

166 *This paper.* It extends the formulation of Nestor [21] to cubic voxels, and is  
 167 ordered as the experiment is built. §§2–3 set out the discrete medium and the  
 168 continuous layer it must reproduce; §4 shows why the point coupling fails,  
 169 and §5 why the source-averaged coupling succeeds exactly. §6 derives the  
 170 single-site  $T$ -matrix as the first member of a Cartesian multipole hierarchy in  
 171 closed form, and sets out what that gives (§6.7). §§7–8 take up the wavefield  
 172 basis, §9 the medium basis and its projection error, §10 the sphere, and  
 173 §11 the hierarchy as a voxel scheme. §12 discusses what the results imply  
 174 and §13 concludes. Every result is derived symbolically in Mathematica and  
 175 checked against an independent Python/Fortran implementation or an exact  
 176 solution.

## 177 2. Setting

178 *The discrete medium.* Space is tiled by cubes of side  $d$  (half-side  $h = d/2$ ).  
 179 Cube  $i$  carries a contrast  $\Delta\lambda, \Delta\mu, \Delta\rho$  from an isotropic background with  
 180 Lamé parameters  $\lambda, \mu$ , density  $\rho$ , P and S speeds  $\alpha, \beta$  and P-wave modulus  
 181  $M_P = \lambda + 2\mu$ . Time dependence is  $e^{-i\omega t}$ ;  $k_P = \omega/\alpha$ ,  $k_S = \omega/\beta$ . Coordinates  
 182 are ordered  $(z, x, y)$  with  $z$  down. The field at a voxel is the nine-component  
 183 state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

184 displacement followed by engineering strain. A voxel responds with a force  
 185 and a moment (stress dipole); the  $9 \times 9$  single-site  $T$ -matrix maps the exciting  
 186 state to them, and the  $9 \times 9$  lattice Green’s tensor  $\mathbf{K}(i, j)$  maps a source

187 voxel's force and moment to the state at voxel  $i$ . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

188 with  $\psi^0$  the incident state.

189 *Where the kernel comes from.* Write the medium as the background plus  
 190 the contrast. With time dependence  $e^{-i\omega t}$  the equation of motion is  $\partial_c \sigma_{bc} +$   
 191  $\rho \omega^2 u_b = 0$ , with  $\sigma_{bc} = c_{bcde} \varepsilon_{de}$  and summation over repeated indices. Split-  
 192 ting  $\rho = \rho_0 + \Delta \rho$  and  $c = c^0 + \delta c$ , with  $\delta c_{bcde} = \Delta \lambda \delta_{bc} \delta_{de} + \Delta \mu (\delta_{bd} \delta_{ce} + \delta_{be} \delta_{cd})$   
 193 the contrast stiffness, and moving the contrast terms to the right leaves the  
 194 background operator on the left and the contrast on the right as a body force,

$$f_b = \omega^2 \Delta \rho u_b + \partial_c m_{bc}, \quad m_{bc} = \delta c_{bcde} \varepsilon_{de}, \quad (3)$$

195 a force from the density contrast, and the divergence of the contrast stress  $m$ ,  
 196 the moment, from the stiffness contrast. With  $\mathbf{G}$  the Green's tensor of the  
 197 background, the displacement is  $u = u^0 + \int \mathbf{G} f \, d\mathbf{y}$ , and the moment term  
 198 integrates by parts,

$$\int G_{ab}(\mathbf{x}-\mathbf{y}) \frac{\partial m_{bc}}{\partial y_c} \, d\mathbf{y} = - \int \frac{\partial G_{ab}}{\partial y_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y} = + \int \frac{\partial G_{ab}}{\partial x_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y}, \quad (4)$$

199 the boundary term vanishing because  $m$  is confined to the scatterers, and the  
 200 last step holding because  $\mathbf{G}$  depends only on the separation. The moment  
 201 therefore radiates through the *receiver* derivative of  $\mathbf{G}$ , with a plus sign, when  
 202 the moment is the contrast stress itself. With the moment tensor defined  
 203 instead through its equivalent body force  $-\partial_c [M_{bc} \delta(\mathbf{x} - \mathbf{y})]$ , as in Aki and  
 204 Richards [27], the same field reads  $M_{bc} \partial G_{ab} / \partial y_c$ , the source derivative, with

205  $M = -m$ . The two are one convention written two ways; what matters is that  
 206 the derivative in the kernel and the sign in the contrast operator are chosen  
 207 together. The nine-component kernel follows (Fig. 1): its force columns are  
 208  $\mathbf{G}$  and its moment columns the receiver derivatives  $\partial_c G_{ab}$ , symmetrised over  
 209  $(b, c)$  with a factor  $\frac{1}{2}$  on the shear components; its displacement rows are  
 210 those entries and its strain rows their receiver derivatives, symmetrised in  
 211 the same way and doubled on the shear rows to give engineering strain. In  
 212 that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathcal{C}), \quad (5)$$

213 where  $\Delta \mathcal{C}$  is the  $6 \times 6$  Voigt form of  $\delta c$ , the contrast stiffness with  $\Delta \lambda + 2\Delta \mu$   
 214 and  $\Delta \lambda$  in the normal block and  $2\Delta \mu$  (not  $\Delta \mu$ ) on the shear diagonal.

215 *Averaging, and collocation.* Three operations in this paper are averages over  
 216 a cell, and they are distinct. The *source-cell average* integrates the kernel  
 217 over the cell that radiates:  $\langle \mathbf{K} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{K}(\mathbf{r} - \mathbf{y}) d\mathbf{y}$  is the field at  $\mathbf{r}$   
 218 of a uniform polarisation of the source cell. Every scheme in this paper uses  
 219 it, and §5 shows it to be exact. *Collocation* fixes where the discrete equation  
 220 is imposed: at the receiving cell's centre, with the internal field of each cell  
 221 uniform, so that the unknown is the field at the centre. The *receiver-cell*  
 222 *average* imposes the equation in the mean over the receiving cell instead;  
 223 with a uniform internal field that is the Galerkin scheme of degree zero,  
 224 called the mean-only voxel below. The two are both *uniform-field* voxels, a  
 225 term that refers to the internal field and not to the test, and they differ only  
 226 in the test function (a delta at the centre against the cell's indicator) and  
 227 in the constant of their  $O((kd)^2)$  error (Table 7). The *cell-mean contrast*,

228 finally, is an average of the material, not of the field: where the medium  
 229 varies within a cell, the contrast is projected onto the cell's basis, its mean  
 230 for degree  $r = 0$ , rather than sampled at the centre (§§9, 10). In the language  
 231 of the coupled-dipole method, the point dipole is collocation with the point  
 232 kernel, the integrated Green tensor adds the source-cell average, and the  
 233 filtered coupled dipoles replace the point source by one of finite extent.

234 *The laterally uniform layer.* A layer of thickness  $D$  is filled with  $n$  planes  
 235 of voxels,  $d = D/n$ , and illuminated at normal incidence. The field is then  
 236 laterally uniform, only the zero-lateral-wavenumber (specular) component of  
 237 the lattice coupling acts, and Eq. (2) becomes exactly a chain of  $n$  planes  
 238 coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (6)$$

239 where the sum runs over the lattice vectors  $\mathbf{R} = d(0, p, q)$ ,  $p, q$  integers, that  
 240 join the centres of the voxels in one plane (a square lattice of pitch  $d$ ), and  
 241  $m$  is the separation of the two planes in units of  $d$  (Fig. 2a,b). The question  
 242 is whether, and how fast, this chain converges to the continuous layer as  
 243  $n \rightarrow \infty$  at fixed  $D$ .

244 *Implementations.* Every result is derived symbolically in Mathematica and  
 245 compared with an independent numerical implementation or an exact so-  
 246 lution. “The package” below is the Python/Fortran implementation of the  
 247 voxel scheme (`cubic_scattering`): its cube  $T$ -matrix, its lattice kernels and  
 248 its stratified propagator, which serve both as the object under test and, where  
 249 validated, as a second implementation.

### 250 3. The continuous layer and its thin-layer expansion

251 At normal incidence a plane force drives a one-dimensional problem per  
 252 wave type, with modulus  $M$  ( $M_P$  for P,  $\mu$  for S), wavenumber  $k$  and plane  
 253 Green's function  $g(z) = \frac{i}{2Mk} e^{ik|z|}$ . The layer's reflection and transmission  
 254 follow from continuity of displacement and traction at its faces; they agree  
 255 with the stratified propagator used as the reference in this work to  $2.5 \times 10^{-8}$ ,  
 256 P and S (`ContinuumLimit_Reference.wl`).

257 For a layer of thickness  $D$  with modulus  $M_1$  and density  $\rho_1$  in a back-  
 258 ground  $M_0, \rho_0$ , write  $Z_j = \sqrt{M_j \rho_j}$  for the impedances,  $\zeta = Z_1/Z_0$  for their  
 259 ratio and  $k_1 = \omega \sqrt{\rho_1/M_1}$  for the layer's wavenumber. The reflection coeffi-  
 260 cient expands as

$$\begin{aligned} R = & \frac{ik_1 D}{2} (\zeta - \zeta^{-1}) + \frac{(k_1 D)^2}{4} (\zeta^{-2} - \zeta^2) - \frac{i(k_1 D)^3}{24} (\zeta - \zeta^{-1}) (3\zeta^{-2} + 2 + 3\zeta^2) \\ & + \frac{(k_1 D)^4}{48} (\zeta^2 - \zeta^{-2}) (3\zeta^2 - 2 + 3\zeta^{-2}) + O(D^5), \end{aligned} \quad (7)$$

261 exact in the contrast (`ContinuumLimit_ThinLayer.wl`, which checks each  
 262 coefficient symbolically and the truncated series against the exact layer).  
 263 The first term is

$$\frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[ \frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left( \frac{1}{M_1} - \frac{1}{M_0} \right) \right] : \quad (8)$$

264 to first order a thin layer is an interface across which the traction jumps by  
 265 the excess inertia and the displacement by the excess compliance. At oblique  
 266 incidence the same statement holds with the first-order system  $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$   
 267 for the P-SV state  $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$ : the layer acts to  $O(D)$  as the jump  
 268  $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$ , which reproduces the layer's  $O(D)$  reflection coefficients

269  $R_{PP}, R_{PS}$  to  $10^{-12}$ ; the series truncated at  $O(D^k)$  converges at exactly order  
 270  $k$ ; and a soft thin layer with shear modulus  $\mu_1 = D/\eta$  at fixed compliance  $\eta$   
 271 tends to the linear-slip interface as  $D \rightarrow 0$ . These expansions are the target  
 272 a single plane of voxels must reproduce order by order. A plane of voxels  
 273 that carry Legendre moments of their internal field to degree  $p$  reproduces  
 274 Eq. (7) term by term through  $O(D^{2p+2})$  and first differs at  $D^{2p+3}$  (§8): the  
 275 mean-only voxel reproduces the equivalent interface and the  $D^2$  term, the  
 276 first-moment voxel all four terms shown.

#### 277 **4. The specular sums of the point Green's tensor**

278 *Why transform the sum.* The specular sum (6) adds, at a receiver, the field  
 279 of every voxel of a plane  $m$  planes away. In that form it is a poor object to  
 280 study. The Green's tensor decays slowly with distance, so the sum converges  
 281 slowly, and it hides the one question that matters here: which part of it is  
 282 the continuum, and which part is the discretisation. The Poisson summation  
 283 formula [28] answers both at once, and is the standard tool for lattice sums  
 284 of wave fields [29]. For a function  $f$  on the plane and the square lattice of  
 285 pitch  $d$  it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \quad \hat{f}(\mathbf{g}) = \int f(\mathbf{x}) e^{-i\mathbf{g} \cdot \mathbf{x}} d^2x, \quad \mathbf{g} = \frac{2\pi}{d}(p, q),$$
(9)

286 with  $p, q$  integers: a sum of samples of  $f$  on the lattice equals a sum of samples  
 287 of its Fourier transform on the reciprocal lattice, whose pitch is  $2\pi/d$  (Fig. 3).  
 288 Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md),$$
(10)

289 with  $\hat{\mathbf{G}}(\mathbf{g}; z)$  the two-dimensional (lateral) transform of the Green's tensor  
 290 between planes a distance  $z$  apart. Each term has a physical meaning. Trans-  
 291 formed laterally, the field of a plane of sources is a superposition of plane  
 292 waves, one for each lateral wavenumber, and the lattice selects the wavenum-  
 293 bers  $\mathbf{g}$ . The order  $\mathbf{g} = 0$  is the laterally uniform plane wave that a continuous  
 294 layer radiates: it is the continuum's term. Every other order has  $|\mathbf{g}| \geq 2\pi/d$ ,  
 295 far beyond the wavenumber of the field when the voxels are small compared  
 296 with the wavelength, so it is evanescent and decays away from the plane as  
 297  $e^{-|\mathbf{g}|md} = e^{-2\pi m\sqrt{p^2+q^2}}$ , independently of  $d$ . The transform therefore splits  
 298 the lattice sum exactly into the continuum and a discretisation correction,  
 299 and the correction converges after a few orders.

300 *The local term that survives.* In the strain-from-moment block the evanescent  
 301 orders carry two powers of  $|\mathbf{g}| \propto 1/d$ , so their contribution per voxel volume  
 302 does *not* vanish as  $d \rightarrow 0$ : at  $m = 1$  it is eleven times the continuum  
 303 term (`ContinuumLimit_SpecularSums.wl`, matched to the package's Ewald  
 304 lattice sum to  $5 \times 10^{-13}$ ). This surviving term is static. Summed over all  
 305 planes in slab order, the static lattice sum of the point kernel is exactly  
 306 the depolarisation of a uniform plate ( $-1/M_P$  on  $\varepsilon_{zz} \leftarrow M_{zz}$  and  $-1/(2\mu)$   
 307 on the normal shears, per unit volume) plus a cubic-symmetric remainder  
 308 (`ContinuumLimit_StaticLocal.wl`, to  $2 \times 10^{-10}$ ). The point kernel therefore  
 309 reproduces the continuum only if that remainder is cancelled by the voxel's  
 310 own self term.

## 311 5. The tiling identity

312 Replace the point coupling by the *source-cell average*  $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r} -$   
 313  $\mathbf{y}) d\mathbf{y}$ , the field at a voxel centre of a uniform polarisation of the source voxel.  
 314 Each Poisson order in Eq. (10) is then multiplied by the cube’s form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (11)$$

315 which vanishes at every  $\mathbf{g} \neq 0$  of the reciprocal lattice since  $\text{sinc}(p\pi) = 0$  for  
 316 integer  $p \neq 0$ . This is the Fourier form of the fact that cubes tile the plane,  
 317  $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$  for the cube’s indicator  $\chi$ : a partition of unity (Fig. 4).  
 318 Two exact statements follow.

319 *Between planes.* For  $m \neq 0$  every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (12)$$

320 the continuum’s plane-wave term times the vertical form factor. The pack-  
 321 age’s cell-averaged kernel agrees in all 81 entries to  $2.8 \times 10^{-16}$  (`ContinuumLimit_`  
 322 `AveragedSums.wl`).

323 *Within a plane.* The sum over *all* cells of the plane, the voxel’s own included,  
 324 is the field at the mid-plane of a uniform plate of thickness  $d$  (Fig. 2c):

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (13)$$

325 where  $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$  is elementary: in the one-dimensional  
 326 transform each entry is  $c_0 + A/(Mk_z^2 - \rho\omega^2)$ , so it averages to  $c_0/d + A(e^{ikh} -$   
 327  $1)/(dMk^2)$ . Its only nonzero entries are  $u \leftarrow f$  on the diagonal and the  
 328 three normal strain-from-moment entries, whose local weights  $-1/M_P$  and

329  $-1/(2\mu)$  are the plate depolarisation of §4. Statically the identity holds for  
 330 all 21 independent components of the strain-from-moment tensor to  $4 \times 10^{-9}$ ,  
 331 with every cell integral evaluated analytically (below).

332 *Consequence: an exact kernel with no lattice sum.* At normal incidence the  
 333 same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (14)$$

334 with no Ewald splitting, near shell or asymptotic tail. The route it replaces  
 335 (Ewald summation with a near shell averaged by quadrature and an  $O(d^2)$   
 336 Taylor tail beyond it) carries a static strain-from-moment error of  $9.4 \times 10^{-4}$   
 337 at its default settings, falling to  $2.6 \times 10^{-5}$  only when both the shell radius  
 338 and the quadrature order are raised; Eq. (14) is now the package’s kernel at  
 339 normal incidence.

340 *Cell integrals without quadrature.* Write the static (Kelvin) tensor as  $G_{bc} =$   
 341  $(c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$  with  $c_B = (1/\mu - 1/M_P)/8\pi$  and  $c_A = 1/(4\pi\mu) - c_B$ .  
 342 Every cell integral of its derivatives is a cell integral of derivatives of  $1/r$  and  
 343  $r$ . One application of the divergence theorem moves it onto faces  $x_p = c_p \pm h$ ,  
 344 which never contain the origin, and there the two-dimensional antiderivative  
 345 in the face coordinates is continuous, so the face integral is a sum over the  
 346 face’s corners. The distributional term at  $r = 0$  is carried by the divergence  
 347 theorem itself. (A triple antiderivative evaluated at the cube’s corners is  
 348 *not* usable for repeated derivatives: its arctangent terms jump across the  
 349 coordinate planes through the origin, and differentiating twice there discards  
 350 exactly that term.) The self term so obtained reproduces the classical closed  
 351 form of the cube’s depolarisation [30, 31] to machine precision.

## 352 6. The single site: the cube $T$ -matrix from moment integrals

353 Every error of a voxel scheme is made at the single site (§12), so the single  
354 site is derived here rather than imported. The classical route imports it. For  
355 an ellipsoid, Eshelby’s uniformity theorem makes the internal strain under a  
356 uniform load uniform, with a depolarisation tensor in closed form [4, 32, 33];  
357 its dynamic generalisation is known for spheres and ellipsoids [34, 35, 36].  
358 For a cube or a general polyhedron the static field of a uniform eigenstrain  
359 is also in closed form [30, 37, 31], and it has been computed for a cuboid in  
360 an anisotropic matrix [38], but the strain it produces is not uniform, so there  
361 is no exact uniform-strain inclusion to import. Closed-form Eshelby tensors  
362 for polyhedral inclusions with polynomial eigenstrains have since been given  
363 [39], the dynamic inclusion problem has been formulated for shapes other  
364 than the ellipsoid, cubes among them [40], and the Green’s tensor has been  
365 integrated analytically over a cuboid for the coupled-dipole method, to an  
366 error of  $O((kd)^4)$  [41]; these are the static and electromagnetic counterparts of  
367 individual moments below. In scattering theory the volume integral equation  
368 of an elastic inclusion [20] is closed in the same spirit by a self-consistent  
369 or effective-field approximation [42, 43]: a first-order (Born) scatterer [44]  
370 renormalised by the geometric series of its own near field, the step that turns  
371 a contrast into an effective contrast. Nestor [21] introduced the single site  
372 in this form, summing the series of the cell’s self integral, with the singular  
373 near field evaluated over an infinitesimal cell. The moments below evaluate  
374 the same self integral over the finite cube, as a distribution. The elastic  $T$ -  
375 matrix of Waterman [45] represents a scatterer in spherical multipoles, the  
376 natural basis for a sphere; a voxel in a lattice needs one compatible with the

377 lattice, and the moments over the cube provide it. The construction below  
 378 makes the approximation of a voxel explicit rather than implicit: it derives  
 379 the cube's  $T$ -matrix from the moments of the elastodynamic Green's tensor  
 380 over the cube, with the uniform-strain closure as the first truncation of a  
 381 hierarchy.

382 *A Cartesian multipole hierarchy.* We call the construction the Cartesian mul-  
 383 tipole hierarchy (Fig. 5). The contrast acting on the voxel's field is a source:  
 384 the density contrast a body force  $\omega^2 \Delta \rho \mathbf{u}$ , the stiffness contrast a stress  
 385  $\delta \mathbf{c} : \boldsymbol{\varepsilon}$ . Each voxel radiates through the Cartesian multipole moments of  
 386 these sources over the cube, a force monopole and a stress dipole from the  
 387 uniform field (the cell of the coupled-dipole method) and higher multipoles  
 388 from the field's gradients, and receives the field through its gradients at its  
 389 centre. The name needs one caution. A multipole expansion usually expands  
 390 the kernel or the field about a centre and truncates it, which holds only for  
 391 well-separated cells or outside a circumscribing sphere: in the fast multipole  
 392 method [46], whose elastodynamic form leaves adjacent cells to direct inte-  
 393 gration [47], in the spherical-wave  $T$ -matrix of Waterman [45], and in the  
 394 force-dipole description of a point defect, an expansion of the Green's func-  
 395 tion far from it [48]. Here the Green's tensor is never expanded. Only the  
 396 source is resolved into Cartesian moments, and the integral over the source  
 397 cube is exact at every separation, so there is no radius of convergence, and  
 398 touching voxels and the self term, a distribution in closed form, are included.  
 399 The source side is shared with the multipole extensions of the coupled-dipole  
 400 method in optics [49, 50] and with Cartesian multipole decompositions of a  
 401 source [51]. The word hierarchy has its sense in kinetic theory and cosmology

402 [52]: a chain of coupled moment equations closed by truncation at a chosen  
403 order, whose truncation error is here in closed form.

404 *Its origin in the equivalent inclusion method.* In its static form the con-  
405 struction is the equivalent inclusion method with polynomial eigenstrains  
406 of Moschovidis and Mura [53]: the eigenstrain is expanded about the centre  
407 of the inhomogeneity, the consistency equation is expanded there in a Taylor  
408 series, and the coefficient of each monomial is equated, which amounts to  
409 imposing the Lippmann–Schwinger equation and its derivatives at the cen-  
410 tre [54, §2.2]. Johnson et al. [55, 56] applied an expansion about the origin  
411 to a cuboidal inhomogeneity, and Wu et al. [39] expanded the eigenstrain  
412 of a polyhedron to second degree about its centroid; both report that the  
413 expansion about the centre does not follow the field near the edges, corners  
414 or vertices. Fu and Mura [34] extended the method to time-harmonic waves  
415 for an ellipsoid, with a second unknown for the density mismatch, and in the  
416 cases they report evaluated it with uniform eigenstrains. Brisard et al. [54]  
417 replaced the Taylor form by a Galerkin form of the same equation, because  
418 the Taylor form need not give an invertible system and a higher degree need  
419 not improve it. What is added here, to our knowledge, is the elastodynamic  
420 hierarchy, with density and stiffness contrasts together, to any degree and for  
421 a body that is not an ellipsoid; its moments as distributions in closed form;  
422 and the structure and the error of its truncations, tested degree by degree  
423 against the exact solution for a sphere. The moments here are integrals of  
424 the Green’s tensor against monomials over the body. They are not moments  
425 in the sense of the method of moments [57], of which the hierarchy is the  
426 instance with monomials about the centre as expansion functions and the  $\delta$

427 function and its derivatives at the centre as testing functions; and the exact  
 428 static response of an inclusion of any shape to polynomial loads is described  
 429 by the polarisation and elastic moment tensors of Ammari and Kang [58].

### 430 6.1. The closed hierarchy

431 Let the voxel be  $V = [-h, h]^3$  with contrast  $\delta c_{ijkl} = \Delta\lambda \delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} +$   
 432  $\delta_{il}\delta_{jk})$  and  $\Delta\rho$ . The hierarchy is stated below for any degree; its third degree  
 433 is written out first, being the lowest at which the stiffness channel converges  
 434 at fourth order (§6.4). Expand the internal displacement about the centre to  
 435 third degree and project the Lippmann–Schwinger equation onto 1,  $x_p$ ,  $x_px_q$   
 436 and  $x_px_qx_v$ . The unknowns are  $u_i(0)$ ,  $\partial_p u_i(0)$ ,  $\partial_p \partial_q u_i(0)$  and  $\partial_p \partial_q \partial_v u_i(0)$ , and  
 437 the result is the closed system

$$(\delta_{ij} - \omega^2 \Delta\rho \mathbf{G}_{ij})u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2}\omega^2 \Delta\rho \mathbf{K}_{ij}^{rs})\partial_r \partial_s u_j = u_i^0, \quad (15)$$

$$\begin{aligned} &(\delta_{pr}\delta_{ij} + \omega^2 \Delta\rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj})\partial_r u_j \\ &+ (\frac{1}{6}\omega^2 \Delta\rho \mathbf{B}_{ij,p}^{rst} - \frac{1}{2}\mathbf{P}_{in,pk}^{st} \delta c_{nkrj})\partial_r \partial_s \partial_t u_j = \partial_p u_i^0, \end{aligned} \quad (16)$$

$$-\omega^2 \Delta\rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr}\delta_{ps}\delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2}\omega^2 \Delta\rho \mathbf{P}_{ij,pq}^{rs})\partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (17)$$

$$\begin{aligned} &(\omega^2 \Delta\rho \mathbf{Q}_{ij,pqv}^r - \mathbf{X}_{in,pqvk} \delta c_{nkrj})\partial_r u_j \\ &+ (\delta_{pr}\delta_{qs}\delta_{vt}\delta_{ij} + \frac{1}{6}\omega^2 \Delta\rho \mathbf{J}_{ij,pqv}^{rst} - \frac{1}{2}\mathbf{Z}_{in,pqvk}^{st} \delta c_{nkrj})\partial_r \partial_s \partial_t u_j = \partial_p \partial_q \partial_v u_i^0, \end{aligned} \quad (18)$$

438 all fields evaluated at the centre. Dropping the third gradients, that is  
 439 Eq. (18) and the second line of Eq. (16), leaves the system of second de-  
 440 gree; dropping the second gradients as well, that is Eq. (17) and the  $\partial\partial u$

terms of Eq. (15), leaves the uniform-strain closure. Equations (15) and (17) are the same at second and third degree, because every term the third degree would add to them vanishes by parity (§6.3). The coefficients are ten moments of the background Green's tensor  $\mathbf{G}$  over the voxel

$$\begin{aligned}
\mathbf{G}_{ij} &= \int_V G_{ij} \, dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s \, dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r \, dV, \\
\mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} \, dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s \, dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r \, dV, \\
\mathbf{B}_{ij,p}^{rst} &= \int_V \partial'_p G_{ij} x_r x_s x_t \, dV, & \mathbf{J}_{ij,pqv}^{rst} &= \int_V \partial'_v \partial'_q \partial'_p G_{ij} x_r x_s x_t \, dV, & \mathbf{X}_{in,pqvk} &= \int_V \partial'_v \partial'_q \partial'_p \partial'_k G_{in} \, dV, \\
\mathbf{Z}_{in,pqvk}^{rs} &= \int_V \partial'_v \partial'_q \partial'_p \partial'_k G_{in} x_r x_s \, dV,
\end{aligned} \tag{19}$$

the first six being those of the second degree and the last four those the third gradients add. The derivatives are taken with respect to the source point,  $\partial' = -\partial$ . That sign reaches the moments with an odd number of derivatives,  $\mathbf{N}$ ,  $\mathbf{Q}$ ,  $\mathbf{B}$  and  $\mathbf{J}$ , and no symmetry, parity or scaling check can detect it, so it is fixed by convention. The moments are graded by  $(D, W)$ , the number of derivatives and the degree of the weight:  $\mathbf{G}(0, 0)$ ,  $\mathbf{K}(0, 2)$ ,  $\mathbf{N}(1, 1)$ ,  $\mathbf{M}(2, 0)$ ,  $\mathbf{P}(2, 2)$ ,  $\mathbf{Q}(3, 1)$ , and  $\mathbf{B}(1, 3)$ ,  $\mathbf{J}(3, 3)$ ,  $\mathbf{X}(4, 0)$ ,  $\mathbf{Z}(4, 2)$ .

*The hierarchy at any degree.* Nothing in the construction is special to the third degree. Let the internal displacement be expanded to degree  $\ell$ ,

$$u_j(\mathbf{x}) = \sum_{|W| \leq \ell} \frac{x_W}{|W|!} U_{j;W}, \quad U_{j;W} = \partial_W u_j(0), \quad x_W = x_{w_1} \cdots x_{w_m}, \tag{20}$$

the sum running over ordered sets of indices  $W = (w_1, \dots, w_m)$ , and let the strain be its derivative. Write the moments with any number of derivatives and any weight as

$$\mathbf{g}_{in}[D; W] = \langle \partial_D G_{in}, x_W \mathbf{1}_V \rangle, \tag{21}$$

457 the derivatives being those of  $G_{in}$  as a function of its argument and the pairing  
 458 distributional (§6.2). The ten moments of Eq. (19) are these  $\mathbf{g}$  themselves:  
 459 they are the integrals of  $\partial'_D G_{in}(\mathbf{0} - \mathbf{x})$ , and the factor  $(-1)^{|D|}$  of the source  
 460 derivative cancels the parity  $(-1)^{|D|}$  of  $\partial_D G_{in}$ , so that, for example,  $\mathbf{N}_{in,k}^r =$   
 461  $\mathbf{g}_{in}[k; r]$  and  $\mathbf{Z}_{in,pqvk}^{rs} = \mathbf{g}_{in}[pqvk; rs]$ . Beyond the third degree the moments  
 462 are written as  $\mathbf{g}$  only. Imposing the Lippmann–Schwinger equation and its  
 463 derivatives to order  $\ell$  at the centre gives, for every set of derivative indices  
 464  $P$  with  $|P| \leq \ell$ ,

$$\begin{aligned}
 U_{i;P} - (-1)^{|P|} \omega^2 \Delta \rho \sum_{|W| \leq \ell} \frac{\mathbf{g}_{ij}[P; W]}{|W|!} U_{j;W} + (-1)^{|P|} \sum_{|W| \leq \ell-1} \frac{\mathbf{g}_{in}[Pk; W]}{|W|!} \delta c_{nkrj} U_{j;rW} = \partial_P u_i^0(0).
 \end{aligned}
 \tag{22}$$

465 Equations (15)–(18) are the case  $\ell = 3$ , with the moments of Eq. (19) put  
 466 for the  $\mathbf{g}$ ; their truncations above are  $\ell = 2$  and the uniform-strain closure  
 467  $\ell = 1$ . With the field rather than the eigenstrain as the unknown, the  
 468 moments  $\mathbf{g}[D; W]$  play the part of the polynomial Eshelby tensors of the  
 469 equivalent inclusion method [33, Ch. 4], which Brisard et al. [54] call influence  
 470 pseudotensors, here for the elastodynamic kernel and for the density channel  
 471 as well. For an ellipsoid the static self term of a polynomial eigenstrain is itself  
 472 a polynomial of the same degree, so the expansion about the centre is exact  
 473 in the self term and approximate only in the interaction with other bodies  
 474 [54]; for a cube it is approximate in both. The cube is centrosymmetric, so a  
 475 moment vanishes unless  $|D| + |W|$  is even, and the system needs the grades

$$\begin{aligned}
 (D, W) &= (m, W), \quad W \leq \ell && \text{(density),} \\
 (D, W) &= (m + 1, W), \quad W \leq \ell - 1 && \text{(stiffness),}
 \end{aligned}
 \quad m = 0, \dots, \ell, \quad D+W \text{ even.}
 \tag{23}$$

476 Table 1 lists what each degree adds. The unknowns number  $3\binom{\ell+3}{3}$ : 12, 30,  
 60 for  $\ell = 1, 2, 3$ .

Table 1: The grades  $(D, W)$  of moments,  $D$  derivatives on the Green's tensor and a weight  
 of degree  $W$ , that the hierarchy of degree  $\ell$  needs beyond those of degree  $\ell - 1$ .

| $\ell$ | new grades                                     | moments in all      |
|--------|------------------------------------------------|---------------------|
| 1      | (0, 0), (1, 1), (2, 0)                         | 3                   |
| 2      | (0, 2), (2, 2), (3, 1)                         | 6: G, K, N, M, P, Q |
| 3      | (1, 3), (3, 3), (4, 0), (4, 2)                 | 10: with B, J, X, Z |
| 4      | (0, 4), (2, 4), (4, 4), (5, 1), (5, 3)         | 15                  |
| 5      | (1, 5), (3, 5), (5, 5), (6, 0), (6, 2), (6, 4) | 21                  |

477

478 *Two independent systems.* In either sum of Eq. (22) the unknown  $U_{i;P}$  is  
 479 coupled to  $U_{j;W}$ , or to  $U_{j;rW}$ , through a moment of grade  $(|P|, |W|)$  or  $(|P| +$   
 480  $1, |W|)$ , and on a centrosymmetric body that moment vanishes unless  $|P|$  and  
 481 the degree of the unknown it multiplies have the same parity. The hierarchy  
 482 of any degree therefore separates exactly, at every frequency and contrast,  
 483 into two systems: one for the even gradients  $u, \partial\partial u, \dots$  and one for the odd  
 484 gradients  $\partial u, \partial\partial\partial u, \dots$ . Raising the degree from odd to even adds a block  
 485 to the even system alone, and from even to odd a block to the odd system  
 486 alone. At  $\ell = 3$  the even system is Eqs. (15) and (17), unchanged from  $\ell = 2$ ,  
 487 and the odd system is Eqs. (16) and (18). Of the sixty unknowns, twenty-one  
 488 belong to the even system and thirty-nine to the odd. The leading radiation  
 489 of a density contrast,  $\int_V \Delta\rho u \, dV$ , is drawn from the even system, and that of  
 490 a stiffness contrast,  $\int_V \delta c : \nabla u \, dV$ , from the odd one. The separation is thus  
 491 the structure behind the result of §6.4, that the blocks improve the density

492 and the stiffness channels alternately. At first degree it is implicit in the  
 493 dynamic equivalent inclusion method for an ellipsoid, where the force and  
 494 the strain systems uncouple at the centre [34]; here it holds at every degree,  
 495 for any centrosymmetric body.

## 496 6.2. The moments as distributions

497 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[ \delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (24)$$

498 [59, 27]. Expanding  $g_c$  in powers of  $r$  reduces every moment in Eq. (19) to a  
 499 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (25)$$

500 For  $m = -1$  and  $D \geq 2$  the classical integrand is not absolutely integrable.  
 501  $\partial^2(1/r)$  decays as  $r^{-3}$ , its angular average vanishes, and the value of the inte-  
 502 gral depends on the shape of the region excluded around the origin. This is  
 503 the source of the  $\delta$ -function term of the inclusion problem, which a numerical  
 504 quadrature misses. Without it the depolarisation is wrong; for a sphere it  
 505 vanishes altogether (§6.3). It is not added afterwards here. The derivatives  
 506 are distributional from the start, and each is moved onto the voxel's faces by  
 507 pairing with the test function  $w \mathbf{1}_V$ , using  $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$ :

$$\langle \partial_q T, w \mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (26)$$

508 The surface term is evaluated on the faces, a distance  $h$  from the origin, where  
 509 every derivative of  $r^m$  is smooth. The recursion lowers  $D$  and  $W$  together

510 and ends on convergent integrals. The face integrals are done in closed form  
 511 by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p; ] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (27)$$

512 which the face route satisfies exactly and a spherical excision returns as  
 513 0. The closed forms contain logarithms of the units  $(2 \pm \sqrt{3})^n$  of the Pell  
 514 equation  $x^2 - 3y^2 = 1$ . Left unreduced, these cancel catastrophically: for  
 515 example,  $\log(72010600134783751 - 41575339372323900\sqrt{3})$  is a cancellation  
 516 of 35 digits. They are therefore reduced to  $n \log(2 - \sqrt{3})$ , which is exact.  
 517 Because every power of  $r$  is paired this way, the dynamic moments keep the  
 518 even-power terms of the expansion  $(k^2 r, r^3, \dots)$  as well as the odd ones.  
 519 A route that integrates only the polynomial (odd-power) terms of  $\mathbf{G}$  misses  
 520 those terms, at a real  $O((kh)^2)$  cost (§6.6).

521 *Every grade, in closed form on four constants.* Closed forms for monomial-  
 522 weighted integrals of  $1/r$  over bricks are known [60], as are closed forms for  
 523 the potential and its first two derivatives over polyhedra with polynomial  
 524 density, from the gravity literature [61]. The engine used here differs in  
 525 taking derivatives of any order as distributions, in treating the kernel  $r$  as  
 526 well as  $1/r$ , and in the even-power terms of the dynamic series. The recursion  
 527 (26) does not depend on the grade, so the same engine gives every scalar  
 528 moment of Table 1. Each is evaluated once and stored. The scalar moments  
 529 of every grade through  $\ell = 5$ , with up to six derivatives and weights to  
 530 the fifth degree, have been evaluated for the kernels  $r^{-1}$  and  $r$ : 4224 closed  
 531 forms. (The second term of Eq. (24) places two further derivatives on  $r$ ;  
 532 those moments are stored through  $\ell = 3$  and are evaluated numerically, by

533 the second route below, beyond it.) Every one of them is a power of the  
 534 voxel's width times

$$q_0 + q_1 \sqrt{3} + q_2 \pi + q_3 \log(2 - \sqrt{3}), \quad q_i \text{ rational.} \quad (28)$$

535 The integration returns many of them in other forms, with  $\log(\sqrt{3} - 1)$ ,  
 536  $\log(1 + \sqrt{3})$ , inverse hyperbolic tangents and logarithms of powers of two,  
 537 which are equal in value but not canonical: two equal moments can then differ  
 538 in appearance, and sums of large logarithms that cancel lose digits. Each  
 539 stored form is therefore reduced to Eq. (28) by an integer-relation search at  
 540 400 digits, with  $\log 2$  and  $\log 3$  admitted to the search, and the reduction  
 541 is accepted only if it agrees with the stored form to 1200 digits. All 4224  
 542 reduce, and the coefficients of  $\log 2$  and  $\log 3$  are zero in every one.

543 Every grade is gated by two checks whose answers do not depend on  
 544 the engine, the Laplacian sum rules of Eq. (A.1), which test the  $\delta$  function  
 545 of the inclusion problem and its derivatives where a quadrature fails, and  
 546 confirmed by a second route that shares nothing with the faces of the cube,  
 547 to  $1.1 \times 10^{-12}$  or better (Appendix Appendix A).

### 548 *6.3. Parity, and the nine-component block*

549 The cube is centrosymmetric. A moment that couples a field quantity of  
 550 degree  $a$  to one of degree  $b$  therefore vanishes unless  $a + b$  is even.  $u$ ,  $\partial u$  and  
 551  $\partial \partial u$  have degrees 0, 1 and 2, so at second degree Eq. (16) contains the first  
 552 gradient alone:  $\partial u$  is determined by its own block and couples to neither  $u$   
 553 nor  $\partial \partial u$ . At any degree it couples to nothing of even degree; from  $\ell = 3$   
 554 it couples to  $\partial \partial \partial u$  (the second line of Eq. (16)). The leading scattering by  
 555 a stiffness contrast is the stress dipole  $\int_V \delta c : \nabla u \, dV$ , which needs only  $\partial u$ .

556 The leading far field of the stiffness contrast is therefore fixed by Eq. (16),  
 557 and *no treatment of the second-gradient block (17) can change it*. What  
 558 Eq. (17) contributes reaches the far field only through the  $O((kh)^2)$  coupling  
 559 in Eq. (15). That is the order, and the origin (the variation of the field across  
 560 the voxel), of the collocation error of §7, which the first-moment voxel of §8  
 561 removes, as does the hierarchy itself once its third-gradient block is kept  
 562 (§6.4).

563 With  $\Delta\rho = 0$ , Eq. (16) at second degree is the  $9 \times 9$  operator  $\delta_{pr}\delta_{ij} -$   
 564  $\mathbf{M}_{in,pk}\delta_{cnkrj}$ , built from  $\mathbf{M}$  alone. Under  $O_h$  the nine gradient components  
 565 decompose as  $A_{1g} \oplus E_g \oplus T_{1g} \oplus T_{2g}$ , and the operator is diagonal in that basis.  
 566 Its eigenvalues are the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : \quad 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (29)$$

$$T_{2g} \text{ (off-diagonal shear)} : \quad 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (30)$$

$$E_g \text{ (diagonal shear)} : \quad 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (31)$$

567 The density channel is the displacement block of Eq. (15) with the second  
 568 gradients dropped, an amplification  $(1 - \omega^2\Delta\rho\mathbf{G})^{-1}$  with  $\mathbf{G}$  the cube integral  
 569 of  $\mathbf{G}$ .

570 *The cube against the sphere.* For a sphere, Eshelby's tensor gives a dilata-  
 571 tional channel  $1 + \Delta K/(\lambda + 2\mu)$ , with  $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$ , and a single deviatoric  
 572 channel  $1 + 2\Delta\mu S_{\text{sph}}$ , with  $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$ . The cube's di-  
 573 latational channel, Eq. (29), is the sphere's exactly: its eigenvalue involves

574 M only through the contraction  $\int_V \partial_i \partial_n G_{in} dV$ , and the static Navier equa-  
 575 tion makes  $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$ , whose integral does not depend on  
 576 the shape. The cube's shear channels differ from each other. Their isotropic  
 577 average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (32)$$

578 The  $\sqrt{3}$  terms cancel in the average. What a cube adds to a sphere at this  
 579 order is therefore a pure cubic anisotropy. It splits the deviatoric channel  
 580 about the sphere's value, weighted 3 : 2 by the dimensions of  $T_{2g}$  and  $E_g$ ,  
 581 and the diagonal shear is the more depolarised. At the background of §7  
 582 ( $\nu = 7/32$ ),  $2\mu S_{\text{off}} = 0.4314$ ,  $2\mu S_{\text{diag}} = 0.5929$  and the sphere's value is  
 583 0.4960.

584 *Where Eshelby enters.* Equations (29)–(31) follow from the moment integrals  
 585 and the group theory alone. No ellipsoidal result is used, and no  $\delta$  term is  
 586 added by hand. What corresponds to Eshelby is the *truncation*. Keeping  
 587  $\partial u$  and discarding  $\partial \partial u$  is exactly the uniform-internal-strain ansatz, and by  
 588 the uniformity theorem that ansatz is exact if and only if the inclusion is  
 589 an ellipsoid. For a sphere the construction is therefore exact in the static  
 590 limit, and for a cube it is an approximation. The hierarchy (22) states that  
 591 approximation as the omitted blocks, rather than as an unquantified shape  
 592 correction. A polynomial expansion about the centre cannot represent the  
 593 edge and corner concentrations of an isolated cube's internal field at any finite  
 594 degree; Table 4 measures what each degree leaves. A voxel in a uniform layer  
 595 has no such concentrations: its neighbours have the same properties, so its  
 596 internal field is the continuum's, which is smooth. That is why a low-degree

597 internal field suffices in a lattice (§§7–8) although it is only an approximation  
 598 for an isolated cube.

599 *Against the exact sphere.* These statements are tested against the exact  
 600 elastic Mie solution, with no Eshelby formula in between (Appendix Ap-  
 601 pendix F). Applied to a ball, the construction’s dilatational and shear chan-  
 602 nels equal Mie’s static concentrations to  $2 \times 10^{-10}$ , and the cube’s dilata-  
 603 tional channel and isotropic shear average meet the same numbers; taken  
 604 as a principal value the moment has no depolarisation at all, so the whole  
 605 static response is the  $\delta$  term. At finite frequency the closure departs from  
 606 Mie only by a real term of order  $(ka)^2$  at first order in the contrast, given  
 607 in closed form in Eq. (F.1); the radiation damping is exact. That term is  
 608 the uniform-field ansatz’s own error, the order the hierarchy assigns to the  
 609 omitted second gradients.

#### 610 6.4. *What each block of the hierarchy buys*

611 A hierarchy invites the question of what its next block is worth. The  
 612 answer has a simple form, and it is not the one a count of unknowns suggests:  
 613 the blocks help in pairs.

614 *The experiment.* The layer of §7 at normal incidence is cut into  $n$  cells, and  
 615 in each the hierarchy of degree  $\ell$  is imposed: the displacement is its Taylor  
 616 polynomial of degree  $\ell$  about the centre, the strain the derivative of that, and  
 617 the equation and its first  $\ell$  derivatives hold at the centre, with the  $\delta$  terms  
 618 kept. In one dimension the moments are elementary and the exact layer is  
 619 known, so the order of each truncation can be measured for a density contrast,

620 a modulus contrast and both. The schemes are computed symbolically to 60  
621 digits for  $\ell = 1$  to 6 and, to  $\ell = 3$ , independently in double precision.

622 *The prediction.* A cell radiates the integral of its source against the kernel.  
623 If the source is held to a polynomial of degree  $d$  about the centre, the first  
624 power it omits is  $d + 1$ . When  $d + 1$  is odd, that term integrates to zero  
625 against the even part of the kernel and the error is two orders smaller; when  
626  $d + 1$  is even it does not. The density source is the displacement, of degree  $\ell$ ;  
627 the stiffness source is the strain, of degree  $\ell - 1$ . So a density contrast gains  
628 an order pair at every even  $\ell$ , and a stiffness contrast at every odd  $\ell$ . This is  
629 the parity of §6.3, read as an order of convergence.

Table 2: The order of convergence of the hierarchy of degree  $\ell$  on the layer, measured from 16 to 32 cells in reflection and in transmission at  $\omega = 300 \text{ rad s}^{-1}$ , and its reflection error on 16 cells with both contrasts. Every measured order equals the predicted one to the digits shown.

|                       | $\ell = 1$           | 2                    | 3                     | 4                     | 5                     | 6                     |
|-----------------------|----------------------|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| density contrast only | 2                    | 4                    | 4                     | 6                     | 6                     | 8                     |
| modulus contrast only | 2                    | 2                    | 4                     | 4                     | 6                     | 6                     |
| both                  | 2                    | 2                    | 4                     | 4                     | 6                     | 6                     |
| error, both, $n = 16$ | $5.4 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | $5.8 \times 10^{-12}$ | $9.5 \times 10^{-13}$ | $2.6 \times 10^{-18}$ | $3.1 \times 10^{-19}$ |

630 *The result.* Table 2: all thirty-six orders are as predicted. The second gradi-  
631 ents raise the density channel to fourth order and leave the stiffness channel  
632 at second, with a constant three to four times smaller in reflection; the third  
633 gradients bring the stiffness channel to fourth order; and so on in pairs.  
634 With both contrasts, fourth order therefore needs  $\ell = 3$  and sixth order

635  $\ell = 5$ . The truncation of Eqs. (15)–(18) at second gradients is one block  
 636 short of the next order for a stiffness contrast, which is why it improves the  
 637 uniform-strain closure by a constant and not by an order.

638 *The error in closed form.* At first order in the contrast a cell whose source is  
 639 held to its Taylor polynomial of degree  $d$  returns the exact term times  $1 + E_d$ ,  
 640 where, with  $h$  the half-width,

$$E_d = \begin{cases} (-1)^{(d-1)/2} \frac{(kh)^{d+1}}{(d+2)!} & d \text{ odd, } R \text{ and } T, \\ (-1)^{d/2} \frac{(kh)^{d+2}}{(d+2)!} & d \text{ even, } R, \\ (-1)^{d/2+1} \frac{(d+1)(kh)^{d+2}}{(d+3)!} & d \text{ even, } T, \end{cases} \quad (33)$$

641  $R$  and  $T$  standing for reflection and transmission, with  $d = \ell$  for the den-  
 642 sity term and  $d = \ell - 1$  for the stiffness term. It follows from the series  
 643 of  $\int_{-1}^1 P_d(ikhx) e^{\mp ikhx} dx$  with  $P_d$  the Taylor polynomial of  $e^{ikhx}$ , and it is  
 644 confirmed symbolically for  $d = 0$  to 6 and against the scheme on one cell,  
 645 at a contrast of  $10^{-20}$  and  $kh = 0.03$ , to 0.5% in all twelve cases. For  
 646  $d = 0$  it is  $(kd)^2/8$  in reflection and  $-(kd)^2/24$  in transmission: the con-  
 647 stants of the collocation scheme, Eq. (36), which are thus the first rung  
 648 of a ladder (`ContinuumLimit_GradientHierarchy.wl`, `scripts/measure_`  
 649 `layer_taylor_hierarchy.py`).

## 650 6.5. The single site against a sphere and a cube

651 The hierarchy is a statement about one scatterer, so it is tested first on  
 652 one scatterer. The assembly of Eq. (22), with the full tensor moments in  
 653 their dynamic form (the series of  $e^{ikr}/r$  taken to  $r^{11}$ ), is the same for any  
 654 centrosymmetric body; only the scalar moments change.

655 *A sphere.* For a ball the scalar moments are elementary, by the surface route  
656 of Eq. (26) on  $|\mathbf{r}| = a$ , and the exact answer is known. A homogeneous  
657 sphere of radius  $a = 10$  m with the contrast of §7 is illuminated by a P plane  
658 wave; the hierarchy’s polynomial source is radiated to nine scattering angles  
659 between  $11^\circ$  and  $169^\circ$ , and the error is the largest component of the difference  
660 from the exact displacement [62, 63] over the largest exact component. The  
661 prediction, from the pairs of §6.4 with  $kh$  replaced by  $ka$ , is an error of  
662  $O((ka)^2)$  for  $\ell = 1$  and 2 and of  $O((ka)^4)$  for  $\ell = 3$ , and  $O((ka)^4)$  already  
at  $\ell = 2$  for a density contrast alone. Table 3 bears the prediction out. At

Table 3: One homogeneous sphere by the hierarchy of degree  $\ell$ : relative far-field error against exact scattering, and the order between the two lowest frequencies. All nine orders are as predicted.

| contrast     | $\ell$ (unknowns) | $k_S a = 0.4$        | 0.2                  | 0.1                  | order | predicted |
|--------------|-------------------|----------------------|----------------------|----------------------|-------|-----------|
| density only | 1 (12)            | $6.4 \times 10^{-3}$ | $1.6 \times 10^{-3}$ | $3.9 \times 10^{-4}$ | 2.0   | 2         |
|              | 2 (30)            | $4.8 \times 10^{-5}$ | $2.6 \times 10^{-6}$ | $1.8 \times 10^{-7}$ | 3.9   | 4         |
|              | 3 (60)            | $1.5 \times 10^{-5}$ | $9.5 \times 10^{-7}$ | $6.9 \times 10^{-8}$ | 3.8   | 4         |
| moduli only  | 1                 | $1.9 \times 10^{-2}$ | $5.0 \times 10^{-3}$ | $1.1 \times 10^{-3}$ | 2.3   | 2         |
|              | 2                 | $5.4 \times 10^{-3}$ | $1.4 \times 10^{-3}$ | $3.4 \times 10^{-4}$ | 2.0   | 2         |
|              | 3                 | $6.2 \times 10^{-5}$ | $4.1 \times 10^{-6}$ | $2.0 \times 10^{-7}$ | 4.3   | 4         |
| both         | 1                 | $1.4 \times 10^{-2}$ | $3.1 \times 10^{-3}$ | $7.2 \times 10^{-4}$ | 2.1   | 2         |
|              | 2                 | $3.2 \times 10^{-3}$ | $6.9 \times 10^{-4}$ | $1.8 \times 10^{-4}$ | 1.9   | 2         |
|              | 3                 | $4.6 \times 10^{-5}$ | $2.5 \times 10^{-6}$ | $1.4 \times 10^{-7}$ | 4.2   | 4         |

663  
664  $k_S a = 0.4$  the error of the uniform-strain closure is 1.4%, that of the second-  
665 gradient system 0.32%, and that of the third-gradient system 0.0046%: three  
666 hundred times below the closure. The systems are well conditioned, the sixty

667 equations of  $\ell = 3$  having a condition number of 5.9 (`scripts/measure_`  
668 `ball_gradient_hierarchy.py`).

669 *A cube.* The same assembly with the cube’s moments gives the single site of  
670 a voxel. No exact solution exists for a cube, so the reference is the same cube,  
671 of half-width  $h = 10$  m, solved by the Legendre voxels of §8 on refined grids,  
672 a cube being the one body such a grid represents exactly. Solutions with first  
673 and second moments on  $2^3$  and  $3^3$  cells differ among themselves by at most  
 $2 \times 10^{-5}$ , which is the accuracy of the reference. Table 4 shows what distin-

Table 4: One homogeneous cube of half-width  $h$  by the hierarchy of degree  $\ell$ : relative far-field error against the converged reference.

| $\ell$ (unknowns) | $k_S h = 0.4$        | 0.2                  | 0.1                  | 0.05                 |
|-------------------|----------------------|----------------------|----------------------|----------------------|
| 1 (12)            | $2.1 \times 10^{-2}$ | $5.4 \times 10^{-3}$ | $2.8 \times 10^{-3}$ | $2.7 \times 10^{-3}$ |
| 2 (30)            | $3.2 \times 10^{-3}$ | $1.6 \times 10^{-3}$ | $1.9 \times 10^{-3}$ | $2.4 \times 10^{-3}$ |
| 3 (60)            | $3.5 \times 10^{-4}$ | $3.4 \times 10^{-4}$ | $2.9 \times 10^{-4}$ | $3.1 \times 10^{-4}$ |

674  
675 guishes a cube from a sphere. Each truncation has a floor that does not fall  
676 with the frequency, because the internal field of an isolated cube is not a poly-  
677 nomial even statically: it concentrates at the edges and corners, as the static  
678 expansions about the centre of Johnson et al. [56] and Wu et al. [39] found for  
679 the cuboid and for polyhedra. The floor is  $2.7 \times 10^{-3}$  for the uniform-strain  
680 closure and nearly the same with second gradients, as §6.3 requires, the static  
681 stress dipole being fixed by the first-gradient block and coupled next to the  
682 third. The third-gradient block lowers it ninefold, to  $3 \times 10^{-4}$ . Above the  
683 floor the error falls with the frequency as on the sphere: at  $k_S h = 0.4$  the  
684 third-gradient system is sixty times more accurate than the closure. The dip

685 of the second-gradient row at  $k_S h = 0.2$  is a cancellation between its static  
686 and dynamic terms, which have opposite signs. The condition numbers lie be-  
687 tween 11 and 13 (`scripts/measure_cube_gradient_hierarchy.py`). These  
688 results come from one implementation, checked against the exact sphere and  
689 against the converged cube; the moments it uses are those confirmed by two  
690 routes in §6.2.

### 691 6.6. The collocation closure

692 The collocation closure takes the internal field of a voxel as uniform, the  
693 truncation of §6.3 for the density and stiffness channels together. The voxel’s  
694 own field at its centre is then  $\mathbf{S}_{\text{self}} \Delta e$  for internal state  $e$ , where  $\mathbf{S}_{\text{self}}$  is the  
695 nine-component form of the moments  $\mathbf{G}$  and  $\mathbf{M}$ . By parity it is block-diagonal.  
696 So  $e = \psi + \mathbf{S}_{\text{self}} \Delta e$  and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}} \Delta)^{-1}. \quad (34)$$

697 With the kernel (14) the self term cancels. Substituting  $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}} \Delta)e$   
698 into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (35)$$

699 an identity checked on the matrices to  $4 \times 10^{-16}$ . Every block of  $\mathbf{K}_{\text{all}}$  is an  
700 *exact* cell integral of the one-dimensional continuum kernel (Eqs. (12), (13)),  
701 so Eq. (35) is precisely the collocation of the continuum Lippmann–Schwinger  
702 equation with a strain piecewise constant over voxels. It does not contain  
703  $\mathbf{S}_{\text{self}}$  at all.

704 The package’s cube  $T$ -matrix is this closure statically: both blocks agree  
705 with Eq. (34) to  $2 \times 10^{-5}$  at  $k_S h = 0.01$ , the difference falling as  $(kh)^2$

706 (`ContinuumLimit_CubeT.wl`). It departs dynamically by about  $0.25(kh)^2$   
 707 through its far-field form-factor corrections. Its self integrals are exact in the  
 708 static and radiation parts but omit the real even-power terms of the Green's  
 709 tensor's expansion ( $k^2r, r^3, \dots$ ), an error of  $0.4\text{--}0.6(kh)^2$  against the analytic  
 710 self term, whose series agrees with the closed-form elastodynamic tensor to  
 711  $10^{-14}$ . By Eq. (35) that error cannot enter the closure scheme.

### 712 *6.7. What a closed-form single-site $T$ -matrix gives*

713 The Cartesian multipole hierarchy costs more to set up than the alter-  
 714 natives: moments of the Green's tensor to fourth derivatives, evaluated as  
 715 distributions and reduced to closed form. What that buys follows from the  
 716 Lippmann–Schwinger equation of one body rather than from a model fitted  
 717 to it. *One derivation.* The static depolarisation, its dynamic corrections,  
 718 the radiation damping and every higher block come from that one equation  
 719 and one engine (§6.2), so they cannot be inconsistent with one another, and  
 720 the static, spherical and low-frequency limits are results of the construction  
 721 rather than conditions imposed on it. *Nothing is fitted.* Every coefficient  
 722 of Eq. (22) is a moment of the background Green's tensor over the body:  
 723 there is no polarisability to choose and no depolarisation factor to calibrate,  
 724 where other routes import the single site from the ellipsoid [4, 33] or close  
 725 the volume integral equation with an effective field [42, 43]. As in the equiva-  
 726 lent inclusion method [53, 34], the uniform-strain closure is the first member  
 727 (§6.3); what the hierarchy adds is the closed form at finite frequency for a  
 728 body that is not an ellipsoid, with the error of each truncation known. *The*  
 729 *singular self-interaction is exact:* the  $\delta$ -function content that a quadrature  
 730 misses is part of the moments from the start, and the sum rule (27) and the

731 traces (A.1) hold at every grade (§6.2). *The exact limits are built in:* the  
 732 cube's dilatational channel and its isotropic shear average equal the sphere's,  
 733 confirmed against the exact Mie solution to  $2 \times 10^{-10}$  and  $10^{-8}$  (§6.3), and  
 734 at finite frequency the closure departs from Mie only through a real term  
 735 of order  $(ka)^2$ , the radiation damping agreeing exactly in all three channels  
 736 (Eq. (F.1)). *The error is stated before anything is solved:* a truncation is  
 737 a statement of which blocks are omitted; the orders come in the predicted  
 738 parity pairs, all thirty-six on the layer and all nine on the sphere (Tables 2  
 739 and 3), and the leading error factor is in closed form, Eq. (33). *It improves*  
 740 *block by block, on fixed constants:* raising the degree adds moments, con-  
 741 firmed by a second route that shares nothing with the first (§6.2), and no  
 742 quadrature; against the exact sphere at  $k_S a = 0.4$  the error falls from 1.4%  
 743 for the uniform-strain closure to 0.0046% with third gradients, from a sys-  
 744 tem of sixty equations with a condition number of 5.9 (Table 3). That each  
 745 degree improves on the one below is shown here for one sphere and one cube,  
 746 not in general: in the Taylor form of the equivalent inclusion method, for  
 747 interacting inhomogeneities, a higher degree need not improve the solution  
 748 and the system need not be invertible [64, 54]. *It says what a body's shape*  
 749 *does:* at the order of the uniform-strain closure a cube is a sphere plus a pure  
 750 cubic anisotropy, which splits the deviatoric channel about the sphere's value  
 751 in the ratio 3 : 2 (Eq. (32)). *It describes a scatterer, not only a cell:* the as-  
 752 sembly is the same for every centrosymmetric body, only the scalar moments  
 753 changing (§6.5), and the same moments are the self block of a voxel scheme  
 754 (§11).

755 *What it does not give.* A polynomial about the centre cannot follow the  
756 edge and corner concentrations of an isolated cube's internal field, so each  
757 truncation there has a floor that does not fall with the frequency ( $3 \times 10^{-4}$   
758 with third gradients; Table 4). And in a lattice the Legendre voxel of §8 is the  
759 more economical in unknowns: both are fourth order on a smoothly graded  
760 sphere, the hierarchy with sixty unknowns a voxel and the first-moment voxel  
761 with thirty-six (§11). The hierarchy's case is not the number of unknowns;  
762 it is that it yields the single site itself, in closed form, with its error known.

## 763 **7. Convergence, and the closed-form error**

764 *The experiment.* Every convergence test of this paper is the same experiment,  
765 varied in one respect at a time, and it is set out here once.

- 766 1. *The medium.* The background has  $\alpha = 5000$  and  $\beta = 3000 \text{ m s}^{-1}$  and  
767  $\rho = 2500 \text{ kg m}^{-3}$ . The layer occupies  $0 < z < D$  with  $D = 2 \text{ m}$ ,  
768 and differs from the background by  $\Delta\lambda = 2 \text{ GPa}$ ,  $\Delta\mu = 1 \text{ GPa}$  and  
769  $\Delta\rho = 100 \text{ kg m}^{-3}$ : 6.4% in the P modulus, 4.4% in the shear modulus  
770 and 4% in the density.
- 771 2. *The source.* A plane force at  $z_s = -12 \text{ m}$ , above the layer, radiates a  
772 plane wave towards it. The incident field in the layer is then known in  
773 closed form, and it is smooth there because the source is outside.
- 774 3. *What is measured.* The scattered displacement, total less incident, at  
775 two observers outside the layer: at  $z = -1 \text{ m}$  above it, where it is the  
776 reflected wave, and at  $z = 4 \text{ m}$  below it, where it is the change the layer  
777 makes to the transmitted wave. Outside the layer the scattered field is

- 778 a single outgoing wave, so one observer on each side holds everything;  
 779 the two are reported separately, as reflection / transmission.
- 780 4. *The exact answer.* The reflection and transmission of the layer from  
 781 continuity of displacement and traction at its two faces (§3); for a  
 782 layer whose properties vary with depth, from the differential equation  
 783 integrated to 30 digits.
- 784 5. *The error.*  $e = |u_{\text{scheme}} - u_{\text{exact}}|/|u_{\text{exact}}|$  at each observer: relative to the  
 785 scattered field, not to the total, so that an error of  $10^{-6}$  is a millionth  
 786 of what the layer scatters.
- 787 6. *The refinement.* The layer is cut into  $n$  planes of voxels of side  $d =$   
 788  $D/n$ , with  $n$  from 1 to 32, at  $\omega = 60, 300$  and  $600 \text{ rad s}^{-1}$ , for which  
 789  $k_P D = 0.024, 0.12$  and  $0.24$  and  $k_S D = 0.04, 0.2$  and  $0.4$ . Three  
 790 frequencies show that an order is not an accident of one.
- 791 7. *The arithmetic.* The errors of the higher-degree voxels fall below the  
 792 round-off of double precision, so the symbolic implementation computes  
 793 them to 40 digits; the independent numerical implementation, in double  
 794 precision, agrees with it down to its own floor of about  $10^{-13}$ .

795 The later sections change one item each: the incident wave to SV or SH and  
 796 its angle (§8), the medium to a stratified or a graded one and the background  
 797 to a stratified one (§9), and the body to a sphere (§10).

798 *Second order.* For a layer  $D = 2 \text{ m}$  with the contrast  $\Delta\lambda = 2 \text{ GPa}$ ,  $\Delta\mu =$   
 799  $1 \text{ GPa}$ ,  $\Delta\rho = 100 \text{ kg m}^{-3}$  in a background  $\alpha = 5000$ ,  $\beta = 3000 \text{ m s}^{-1}$ ,  $\rho =$   
 800  $2500 \text{ kg m}^{-3}$ , illuminated by a P plane force, the reflected displacement of  
 801 the chain converges to the continuous layer's at order 2.000 for  $n = 4 \rightarrow$   
 802  $16$  at  $\omega = 60, 300$  and  $600 \text{ rad s}^{-1}$ , with the collocation (35) and with the

803 package *T*-matrix alike (`ContinuumLimit_Chain.wl`). The error depends on  
 804  $k_S h$  alone. Throughout, an order quoted for a pair of grids is the apparent  
 805 order  $\ln(e_1/e_2)/\ln(n_2/n_1)$  between them, with  $e$  the relative error against  
 806 the exact solution and  $n$  the refinement; the orders in the legend of Fig. 7  
 807 are least-squares slopes of  $\ln e$  against  $\ln n$  over the grids marked as fitted,  
 808 with their standard errors. The two agree where a sequence is asymptotic  
 809 and differ where it is not, which the apparent orders show.

810 *The leading error in closed form.* In Eq. (35) the kernel is integrated exactly;  
 811 the only approximation is the internal field's variation across a voxel. At first  
 812 order in the contrast the discrete scattered field is therefore a midpoint rule  
 813 applied to the continuum's Born integral, with the kernel's cell integral exact.  
 814 For reflection the integrand carries the two-way phase  $e^{2ikz}$ ; for transmission  
 815 it is flat. With  $k$  the background wavenumber of the wave concerned ( $k_P$  for  
 816 an incident P wave,  $k_S$  for an incident S wave),  $R$  the reflection coefficient  
 817 and  $T^{\text{sc}}$  the scattered part of the transmitted field, for *every*  $n$

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (36)$$

818 confirmed for  $n = 1$  to 16 to  $10^{-4}$  of the error itself (`ContinuumLimit_`  
 819 `ErrorConstant.wl`); for an incident S wave, with  $k = k_S$ , the same forms  
 820 and those of Eq. (39) hold to  $4 \times 10^{-4}$  (`ContinuumLimit_IncidentS.wl`, §8).  
 821 At the full contrast the reflection constant is unchanged (0.124 against 1/8);  
 822 the transmission constant grows to  $-0.092$ .

823 *Why a uniform-field single site stops at second order.* The two constants have  
 824 opposite signs, so no scalar correction of  $\mathbf{T}$ , no single  $O((kh)^2)$  factor, can

cancel both. The package *T*-matrix illustrates it: its form-factor corrections reduce the reflection error by a factor 4.45 and *increase* the transmission error by 1.7, at every  $n$ . The missing term in Eq. (36) is the first moment of the internal field over each voxel, which radiates through the derivative of the kernel. A fourth-order scheme must carry it: a single site with a gradient channel, and a kernel whose cell integrals include the kernel's derivative.

## 8. Fourth order and beyond: Legendre moments of the voxel field

Let each voxel carry the mean *and* the first moment of its internal field. On voxel  $j$ , with local coordinate  $s \in [-h, h]$  and Legendre functions  $\phi_b(s) = P_b(s/h)$  ( $\phi_0 = 1$ ,  $\phi_1 = s/h$ ,  $\phi_2 = (3s^2/h^2 - 1)/2$ , ...), the field is  $w = \sum_b c_{b,j} \phi_b$ ,  $w = (u, \varepsilon)$  the displacement and strain of the P channel, and the polarisation is  $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$  with  $\Delta M = \Delta \lambda + 2\Delta \mu$ . Testing the continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (37)$$

with the same functions on every voxel gives

$$\left( \int_i \phi_a^2 dz \right) c_{a,i} = \int_i \phi_a w^0 dz + \sum_{j,b} \left[ \int_i \int_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (38)$$

$\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$ . Between voxels the double integral is separable, a product of Legendre moments of  $e^{\pm i k z}$ ; within a voxel it is split at  $z' = z$  and the local term contributes  $-\int_i \phi_a \phi_b dz / M_P$ . Every entry is closed-form. Keeping only  $\phi_0$  recovers a second-order scheme; keeping  $\phi_0$  and  $\phi_1$  converges at fourth order in both reflection and transmission (`ContinuumLimit_`

FourthOrder.w1; Table 5). A single plane of first-moment voxels represents

Table 5: Relative error of the scattered displacement of the layer, reflection / transmission, at  $\omega = 300 \text{ rad s}^{-1}$ , against the number of voxel planes  $n$ : collocation, the mean-only voxel and the first-moment voxel of Eq. (38). The order is 4.00 at  $\omega = 60$  and  $600 \text{ rad s}^{-1}$  too.

| $n$   | collocation                               | mean only                                 | mean and first moment                       |
|-------|-------------------------------------------|-------------------------------------------|---------------------------------------------|
| 1     | $1.8 \times 10^{-3} / 1.3 \times 10^{-3}$ | $1.2 \times 10^{-3} / 1.8 \times 10^{-3}$ | $2.8 \times 10^{-7} / 2.8 \times 10^{-7}$   |
| 4     | $1.1 \times 10^{-4} / 8.3 \times 10^{-5}$ | $7.5 \times 10^{-5} / 1.1 \times 10^{-4}$ | $1.1 \times 10^{-9} / 1.1 \times 10^{-9}$   |
| 16    | $7.0 \times 10^{-6} / 5.2 \times 10^{-6}$ | $4.7 \times 10^{-6} / 6.9 \times 10^{-6}$ | $4.2 \times 10^{-12} / 4.2 \times 10^{-12}$ |
| order | 2 / 2                                     | 2 / 2                                     | 4 / 4                                       |

844

845 the 2 m layer 6500 times more accurately than a single plane of uniform ones.

846 The collocation column is the scheme of §6 and reproduces the chain of the

847 full  $9 \times 9$  system to all printed digits.

848 *Higher moments.* Carrying the Legendre moments to degree  $p$  changes noth-

849 ing in the construction: the moments and the same-cell double integrals of

850  $P_b(s/h)$  against exponentials are again closed-form, and for  $p \leq 1$  the general-

851 degree solver reproduces the chain above exactly. The second-moment voxel

852 ( $p = 2$ ) converges at sixth order and the third-moment voxel ( $p = 3$ ) at

853 eighth, in reflection and transmission (Table 6; `ContinuumLimit_SecondMoment.`

854 `w1`); the second-moment voxel's order is 6.00 at  $\omega = 60, 300$  and  $600 \text{ rad s}^{-1}$ .

855 The errors fall to  $10^{-22}$ , so they are computed in 40-digit arithmetic.

856 *The leading error in closed form.* Without scattering, the Galerkin solution

857 is the  $L^2$  projection  $\Pi$  of the incident field onto each voxel's basis, and the

858 kernel is integrated over each voxel exactly; at first order in the contrast the

Table 6: Relative error of the scattered displacement of the layer, reflection / transmission, at  $\omega = 300 \text{ rad s}^{-1}$ , for voxels carrying Legendre moments to degree  $p$ . The order is the mean over the halvings  $n = 2 \rightarrow 4 \rightarrow 8$ .

| $n$   | first moment ( $p = 1$ )                    | second moment ( $p = 2$ )                   | third moment ( $p = 3$ )                    |
|-------|---------------------------------------------|---------------------------------------------|---------------------------------------------|
| 1     | $2.8 \times 10^{-7} / 2.8 \times 10^{-7}$   | $2.8 \times 10^{-11} / 4.2 \times 10^{-11}$ | $1.5 \times 10^{-15} / 1.5 \times 10^{-15}$ |
| 2     | $1.7 \times 10^{-8} / 1.7 \times 10^{-8}$   | $4.4 \times 10^{-13} / 6.5 \times 10^{-13}$ | $6.0 \times 10^{-18} / 6.0 \times 10^{-18}$ |
| 4     | $1.1 \times 10^{-9} / 1.1 \times 10^{-9}$   | $6.9 \times 10^{-15} / 1.0 \times 10^{-14}$ | $2.4 \times 10^{-20} / 2.4 \times 10^{-20}$ |
| 8     | $6.7 \times 10^{-11} / 6.7 \times 10^{-11}$ | $1.1 \times 10^{-16} / 1.6 \times 10^{-16}$ | $9.2 \times 10^{-23} / 9.2 \times 10^{-23}$ |
| order | 4 / 4                                       | 6 / 6                                       | 8 / 8                                       |

error is therefore the kernel's cell integral applied to  $\Pi w^0 - w^0$ . The field varies as  $e^{iks}$  across a voxel and the kernel toward an outside observer as  $e^{iqs}$ ,  $q = +k$  for reflection and  $q = -k$  for transmission, so for every scheme and every  $n$

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] ds}{\int_{-h}^h e^{i(q+k)s} ds}, \quad (39)$$

with  $\Pi$  the centre value for collocation, the cell mean for the mean-only voxel and the projection onto  $\{P_0, \dots, P_p\}$  for the voxel of degree  $p$  (`ContinuumLimit_FourthOrderError.wl`, `ContinuumLimit_SecondMoment.wl`). The results are in Table 7. The chain of Eq. (38) reproduces the forms for collocation and degrees 0 and 1 at  $n = 1$  to 8 to  $10^{-4}$  of the error, the remainder being the second order in the contrast, and those for degrees 2 and 3 to  $4 \times 10^{-7}$  at a contrast reduced by  $10^{-6}$ . The constant for degree  $p$  follows from the Legendre expansion  $e^{iks} = \sum_j i^j (2j+1) j_j(kh) P_j(s/h)$ . The projection drops the terms  $j > p$ , led by  $j = p+1$ , whose coefficient is  $O((kh)^{p+1})$ ; its moment against  $e^{iqs}$  is  $2h i^{p+1} j_{p+1}(qh) = O((qh)^{p+1})$ . With  $j_n(x) \approx x^n / (2n+1)!!$  and

Table 7: The leading error of the discrete layer in closed form, Eq. (39): discrete over exact scattered field at first order in the contrast, for every number of voxel planes  $n$ .

|               | reflection                        | transmission                      |
|---------------|-----------------------------------|-----------------------------------|
| collocation   | $1 + (kd)^2/8 + O(d^4)$           | $1 - (kd)^2/24 + O(d^4)$          |
| mean only     | $1 + (kd)^2/12 + O(d^4)$          | $1 - (kd)^2/12 + O(d^4)$          |
| first moment  | $1 - (kd)^4/720 + O(d^6)$         | $1 - (kd)^4/720 + O(d^6)$         |
| second moment | $1 + (kd)^6/100800 + O(d^8)$      | $1 - (kd)^6/100800 + O(d^8)$      |
| third moment  | $1 - (kd)^8/25401600 + O(d^{10})$ | $1 - (kd)^8/25401600 + O(d^{10})$ |
| degree $p$    | $1 + (-1)^p c_p (kd)^{2p+2}$      | $1 - c_p (kd)^{2p+2}$             |

873  $h = d/2$ ,

$$c_p = \frac{2p+3}{4^{p+1} [(2p+3)!!]^2} = \frac{[(p+1)!]^2}{(2p+2)! (2p+3)!}, \quad c_0 = \frac{1}{12}, \quad c_1 = \frac{1}{720}, \quad c_2 = \frac{1}{100800}, \quad (40)$$

874 and the leading error carries  $(q/k)^{p+1}$ : the same sign in reflection and trans-  
875 mission for odd  $p$ , opposite signs for even  $p$ . The first-moment voxel's leading  
876 error therefore does not depend on the direction in which the field is observed.  
877 Eq. (40) is checked symbolically for  $p = 0$  to 5 (every lower power of  $kd$  van-  
878 ishes) in Mathematica and, independently, in sympy.

879 *One plane of voxels and the thin-layer series.* Refining at fixed thickness is  
880 one limit; thinning a single plane is the other. For one plane of voxels ( $n = 1$ ,  
881  $d = D$ ) the series in  $D$  of the reflection coefficient, computed exactly at the  
882 full contrast, coincides with that of the exact layer, Eq. (7), term by term  
883 through  $D^{2p+2}$  and first differs at  $D^{2p+3}$ , for  $p = 0$  to 3 and for a layer both  
884 stiffer and softer than its surroundings (`ContinuumLimit_SecondMoment`).

885 `w1`). Each further Legendre moment adds two terms of the thin-layer series.  
 886 *In three dimensions.* The three-dimensional first-moment voxel carries the  
 887 nine-component state with Legendre weights  $\{1, x/h, y/h, z/h\}$ , 36 unknowns  
 888 in all, tested with the same functions and coupled by cell-to-cell double in-  
 889 tegrals of the Green's tensor. For a laterally uniform layer at normal in-  
 890 cidence it reduces *exactly* to Eq. (38) (`ContinuumLimit_Reduction3D.w1`).  
 891 The lateral first moments are never excited: the lattice and the incident  
 892 field are symmetric under reflection about every voxel's centre planes, and  
 893  $x/h, y/h$  are odd there. The mean and the  $z$ -moment carry weights that  
 894 are constant laterally, so their lateral form factor  $\text{sinc}(g_x h) \text{sinc}(g_y h)$  van-  
 895 ishes on the reciprocal lattice and the plane sum of the double integrals  
 896 is the plate-to-plate integral in  $z$  alone. Two facts mark where genuinely  
 897 three-dimensional structure begins. The lateral first moment's form fac-  
 898 tor,  $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$ , does not vanish on the reciproc-  
 899 al lattice ( $F_1 = i(-1)^p/p\pi$ ), so the gradient channels couple through the  
 900 evanescent orders. And at oblique incidence the mean's own zeros shift,  
 901  $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h/p\pi + O(k_x^2 h^2)$ , so the tiling identity holds  
 902 only to  $O(k_{\parallel} h)$ .  
 903 *Other angles, other waves, and degree two.* The orders and their constants  
 904 hold beyond normal incidence and incident P. At oblique incidence, where  
 905 the tiling identity holds only to  $O(k_{\parallel} h)$  and the gradient channels couple  
 906 through the evanescent orders, the mean-only and first-moment voxels remain  
 907 second and fourth order in the reflected P and the converted S wave; for  
 908 incident SV and SH waves the orders, and with  $k = k_S$  the constants, are  
 909 those of incident P; and the voxel of degree two converges at sixth order

910 for P, SV and SH waves, its error on four planes at normal incidence being  
 911 the  $c_2(kd)^6$  of Eq. (40) to 4%. Appendix Appendix B gives the spectral  
 912 construction, Eq. (B.1), the tests, and the second implementations that check  
 913 them; Figure 7(a,b) shows the convergence.

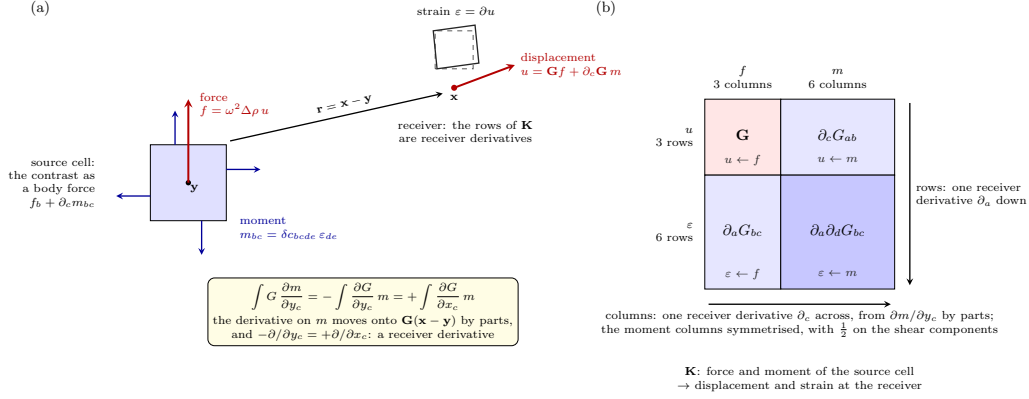


Figure 1: Where the kernel comes from. (a) A source cell at  $\mathbf{y}$  acts on the background through the body force of Eq. (3): a force  $f$  from its density contrast, and the divergence of its contrast stress  $m$ , the moment, from its stiffness contrast (arrows on the faces). At the receiver  $\mathbf{x}$  the state is the displacement  $u$  and the strain  $\varepsilon$ . The moment's derivative moves onto  $\mathbf{G}(\mathbf{x} - \mathbf{y})$  by parts, Eq. (4), and becomes a receiver derivative with a plus sign. (b) The block structure of the  $9 \times 9$  kernel  $\mathbf{K}$  that results: the force columns are  $\mathbf{G}$ , the moment columns carry one receiver derivative across, the strain rows one receiver derivative down, and the strain-from-moment block two.

*Alt text:* Two panels. On the left, a shaded square cell with a dot at its centre marked  $\mathbf{y}$ ; a thick red arrow labelled force rises from the centre, and four blue arrows labelled moment point outward from the cell's faces. A long arrow labelled  $\mathbf{r}$  equals  $\mathbf{x}$  minus  $\mathbf{y}$  runs from the cell to a red dot marked  $\mathbf{x}$  at the upper right, from which a red arrow labelled displacement leaves and above which a small dashed square deformed into a parallelogram is labelled strain. A boxed line below reads: the integral of  $G$  times the  $y$ -derivative of  $m$  equals minus the integral of the  $y$ -derivative of  $G$  times  $m$ , equals plus the integral of the  $x$ -derivative of  $G$  times  $m$ . On the right, a two-by-two grid of shaded blocks with rows labelled  $u$ , three rows, and strain, six rows, and columns labelled  $f$ , three columns, and  $m$ , six columns; the blocks read  $\mathbf{G}$ , the first derivative of  $G$ , the first derivative of  $G$ , and the second derivative of  $G$ ; an arrow down the right side reads rows: one receiver derivative down, and an arrow along the bottom reads columns: one receiver derivative across.

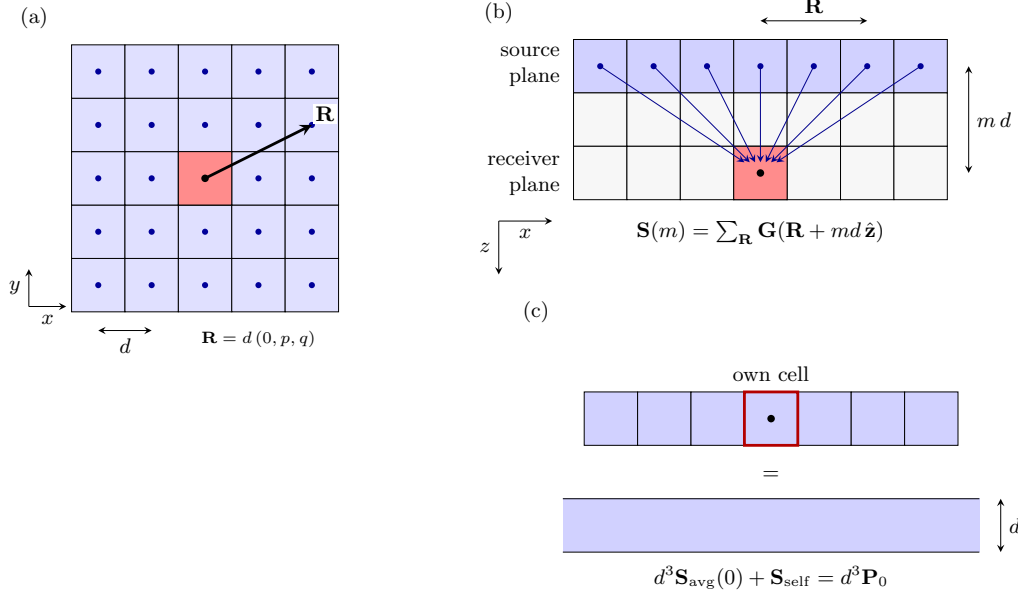


Figure 2: The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch  $d$ , and a lattice vector  $\mathbf{R} = d(0, p, q)$  joins the centre of one voxel (shaded darker) to another. (b) Cross-section ( $z$  down), between planes: the voxel of a receiver plane (shaded darker) is coupled to every voxel of a source plane  $m$  planes above it, one for each lattice vector  $\mathbf{R}$ ; the sum over  $\mathbf{R}$  is the specular lattice sum of Eq. (6). (c) Within a plane: the sum over every cell of the plane, the receiver’s own cell (outlined) included, is the field of a uniform plate of thickness  $d$ , the tiling identity of Eq. (13).

*Alt text:* Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled  $\mathbf{R}$  runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked  $d$ . At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation  $m d$  marked. Below it, a single row of voxels with one outlined as the receiver’s own cell is shown equal to a continuous shaded band of the same thickness  $d$ , a uniform plate.

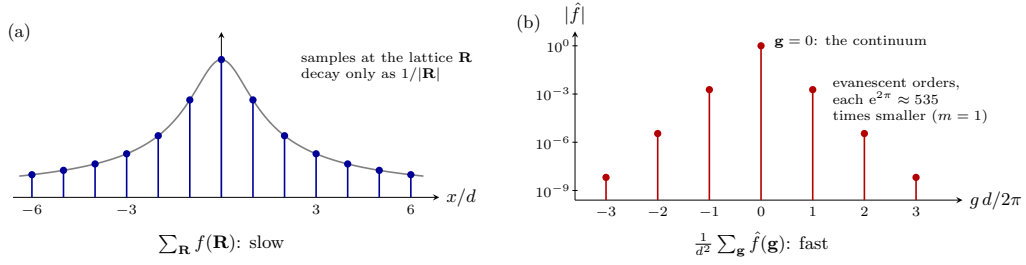


Figure 3: The Poisson summation formula (9) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as  $1/|\mathbf{R}|$ , so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders  $\mathbf{g} = 2\pi p/d$ . The order  $\mathbf{g} = 0$  is the continuum's plane wave; every other order is evanescent and, at  $m = 1$ , smaller than its neighbour by  $e^{2\pi} \approx 535$ . Magnitudes are schematic: the static decay  $e^{-|\mathbf{g}|md}$  is shown.

*Alt text:* Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

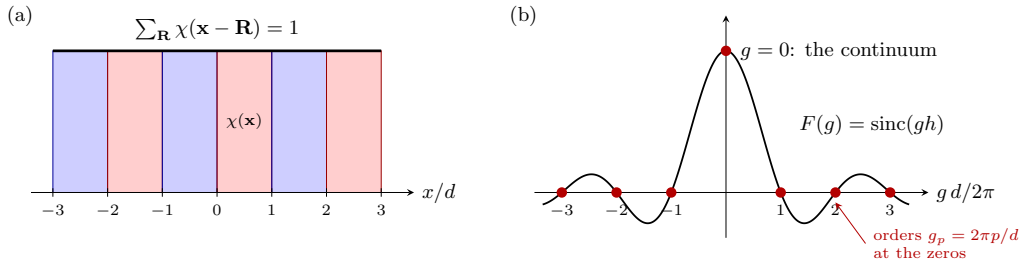


Figure 4: Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions  $\chi$  of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube's lateral form factor  $F(g) = \text{sinc}(gh)$  (shown along one axis) vanishes at every reciprocal-lattice order  $g_p = 2\pi p/d$  with  $p \neq 0$ , so of all the orders in the lattice sum (10) only  $g = 0$ , the continuum's plane wave, survives the source-cell average.

*Alt text:* Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of  $2\pi/d$ ; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

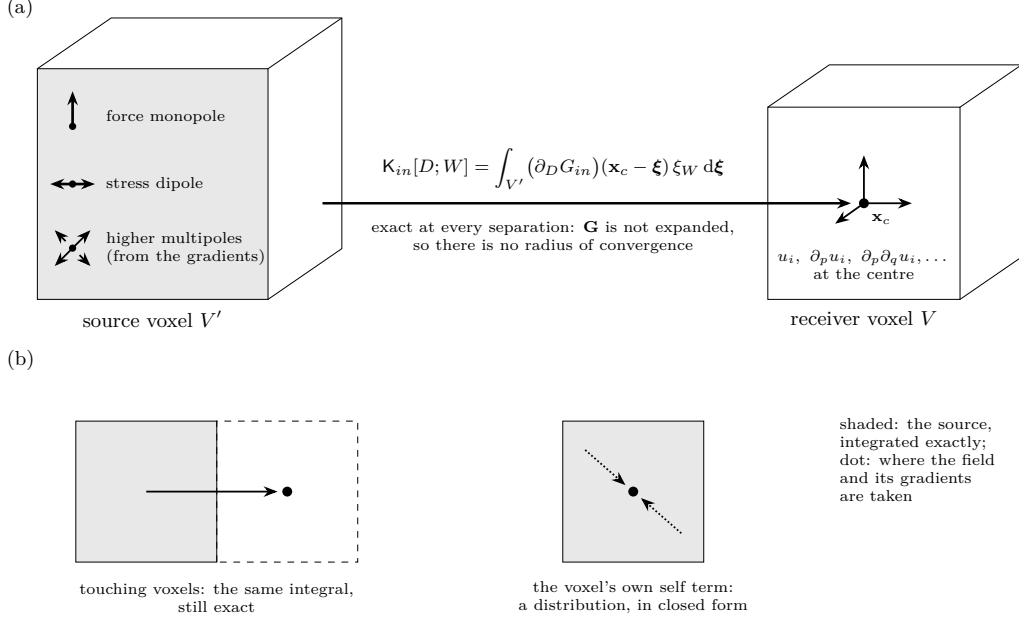


Figure 5: The Cartesian multipole hierarchy. (a) The source voxel radiates through the Cartesian multipole moments of its source: a force monopole from the density contrast, a stress dipole from the stiffness contrast, and higher multipoles from the field's gradients. The receiver voxel takes the field through its value and gradients at its centre  $\mathbf{x}_c$ . The two are coupled by the integral of the derivatives of the Green's tensor over the whole source cube, Eq. (47), which is exact at every separation because the Green's tensor is not expanded. (b) The cases a multipole expansion of the Green's tensor cannot reach are therefore included: touching voxels, by the same integral, and the voxel's own self term, a distribution in closed form.

*Alt text:* Two panels. In the upper panel a shaded cube on the left, labelled source voxel, contains three small symbols: a single arrow labelled force monopole, a pair of opposed arrows labelled stress dipole, and a cross of four arrows labelled higher multipoles, from the gradients. A long arrow runs from it to the centre of an unshaded cube on the right, labelled receiver voxel, where a dot with three short arrows marks the field and its gradients at the centre. Above the arrow is the coupling integral of the derivatives of the Green's tensor over the source cube, and below it the words exact at every separation:  $\mathbf{G}$  is not expanded, so there is no radius of convergence. In the lower panel, on the left, a shaded square touches a dashed square whose centre is marked by a dot, with an arrow from the shaded square to the dot, labelled touching voxels: the same integral, still exact; on the right a single shaded square has a dot at its centre and two dotted arrows pointing

## 914 9. A heterogeneous layer

915 A heterogeneous medium is not a layer of identical cells. The simplest  
916 heterogeneous case with an exact answer is a stratified model built from  
917 layers of constant properties, with real interfaces between them: the form of  
918 many layered crustal velocity models. The last test takes such a model. A  
919 layer  $D = 4$  m thick is divided into eight model layers of 0.5 m, and in each  
920 one  $\Delta\lambda$ ,  $\Delta\mu$  and  $\Delta\rho$  are drawn uniformly within  $\pm 15\%$  of the background  
921 (the values are listed in `ContinuumLimit_Heterogeneous.w1`). The result is  
922 a random stratification with physical contrasts. The voxels refine the model's  
923 layering, with  $m$  planes of voxels in each model layer, so no voxel straddles  
924 a model interface: the natural discretisation when the model's interfaces are  
925 known. At  $\omega = 1500 \text{ rad s}^{-1}$  the layer is  $k_P D = 1.2$  and  $k_S D = 2.0$  thick, so  
926 the field reverberates between the model interfaces. The schemes are those  
927 of §§7–8, with the contrast now changing from plane to plane; nothing else  
928 in them changes. In the terms of the experiment of §7, the test is this.

- 929 1. *The medium.* The eight sets of contrasts are drawn once, from a fixed  
930 seed, and the same eight are used in every run, so that every grid and  
931 every scheme sees the same model.
- 932 2. *Normal incidence.* The source is the plane force at  $z_s = -12$  m, and the  
933 scattered displacement is taken at  $z = -1$  m, where it is the reflected  
934 wave, and at  $z = D + 2 = 6$  m, where it is the change in the transmitted  
935 wave. The P problem has the modulus contrast  $\Delta\lambda + 2\Delta\mu$  and the wave  
936 speed  $\alpha$ ; the S problem is the same one-dimensional problem with  $\Delta\mu$   
937 and  $\beta$ .
- 938 3. *Oblique incidence.* A P plane wave is incident at  $20^\circ$  to the vertical.

939 What is measured is the pair of displacement amplitude vectors of  
 940 the reflected P and the reflected S plane waves,  $R_{PP}$  and  $R_{PS}$ , in the  
 941 specular order. The lateral sums of the coupling (§8) run over the  
 942 reciprocal-lattice orders  $|p|, |q| \leq 8$ .

943 4. *The exact answer.* The state of displacement and traction, two compo-  
 944 nents at normal incidence and four for P–SV at  $20^\circ$ , is carried across  
 945 each model layer by the matrix exponential of that layer’s system ma-  
 946 trix times its thickness, the eight matrices are multiplied in order,  
 947 and the radiation conditions of the two half-spaces are imposed on  
 948 the product. The arithmetic carries 30 to 40 digits. The reference  
 949 shares nothing with the voxel scheme but the model. It is checked  
 950 against a third algorithm, the reflection–transmission recursion of Ken-  
 951 nett [65] over the nine interfaces: the same-type coefficients,  $R_{PP}$  and  
 952  $R_{SS}$  at normal incidence and  $R_{PP}$  at  $20^\circ$ , agree to  $10^{-14}$  (`scripts/`  
 953 `gate_heterogeneous_reference_vs_kennett.py`). Only same-type  
 954 coefficients are compared, because they do not depend on how the up-  
 955 and downgoing waves are normalised; a converted coefficient does. One  
 956 sign differs, by convention: at normal incidence the one-dimensional  
 957 model’s  $R_{PP}$  is a ratio of vertical displacements, the recursion’s a ra-  
 958 tio of amplitudes along each wave’s own direction, and the reflected P  
 959 wave travels upwards.

960 5. *The error.* As in §7, the difference from the exact scattered quan-  
 961 tity relative to its size: for the amplitude vectors at  $20^\circ$ ,  $|\mathbf{R}_{\text{scheme}} -$   
 962  $\mathbf{R}_{\text{exact}}|/|\mathbf{R}_{\text{exact}}|$ .

963 6. *The refinement.* Each model layer is cut into  $m$  planes of voxels, so

964 that the layer holds  $n = 8m$ :  $m = 1, 2, 4, 8, 16$  at normal incidence  
965 and  $m = 1, 2, 4$  at  $20^\circ$ , where the lateral sums make each run dearer.  
966 The orders are apparent orders, from  $m = 4$  to 16 and from  $m = 2$  to  
967 4.

The errors are in Table 8 and Fig. 7(c).

Table 8: Relative error against the exact stratified layer, for  $m$  planes of voxels per model layer (§9). At normal incidence the order is measured from  $m = 4$  to 16, where it is also 2.00 for collocation and 4.00 in transmission; at  $20^\circ$  it is measured from  $m = 2$  to 4.

|                          |               | $m = 1$              | $m = 2$              | $m = 4$              | order |
|--------------------------|---------------|----------------------|----------------------|----------------------|-------|
| P, normal, $R$           | mean only     | $1.9 \times 10^{-3}$ | $4.7 \times 10^{-4}$ | $1.2 \times 10^{-4}$ | 2.00  |
|                          | first moment  | $8.5 \times 10^{-7}$ | $5.3 \times 10^{-8}$ | $3.3 \times 10^{-9}$ | 4.00  |
| S, normal, $R$           | mean only     | $5.3 \times 10^{-3}$ | $1.3 \times 10^{-3}$ | $3.3 \times 10^{-4}$ | 2.00  |
|                          | first moment  | $6.2 \times 10^{-6}$ | $3.9 \times 10^{-7}$ | $2.4 \times 10^{-8}$ | 4.00  |
| P, $20^\circ$ , $R_{PP}$ | mean only     | $1.4 \times 10^{-3}$ | $3.6 \times 10^{-4}$ | $9.0 \times 10^{-5}$ | 2.00  |
|                          | first moments | $3.3 \times 10^{-7}$ | $2.1 \times 10^{-8}$ | $1.3 \times 10^{-9}$ | 4.00  |
| P, $20^\circ$ , $R_{PS}$ | mean only     | $2.8 \times 10^{-3}$ | $6.9 \times 10^{-4}$ | $1.7 \times 10^{-4}$ | 2.00  |
|                          | first moments | $1.4 \times 10^{-6}$ | $8.8 \times 10^{-8}$ | $5.5 \times 10^{-9}$ | 4.00  |

968

969 *The interfaces cost nothing.* Every order is the one measured for the uniform  
970 layer. At a model interface the strain of the true field jumps. A voxel basis  
971 that is discontinuous from voxel to voxel represents that jump exactly when  
972 the interface lies on a voxel boundary, so the interface adds no error of its  
973 own. The only requirement is that voxel boundaries coincide with the model's  
974 interfaces. A voxel that straddled an interface would carry a contrast that is  
975 not uniform within it, a case not examined here.

976 *What fourth order buys.* For a reflection error of  $10^{-6}$  at normal incidence  
977 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model  
978 layer. That figure is not a run: it is the second-order law carried from the  
979 finest grid computed, 128 planes, whose error is  $7.3 \times 10^{-6}$ , to the target,  
980  $n = 128 (7.3 \times 10^{-6} / 10^{-6})^{1/2}$ . The first-moment voxel needs 8, one per model  
981 layer, and that figure is a run: the model's own layering already gives  $8.5 \times$   
982  $10^{-7}$ . Counting unknowns (9 per voxel against 36), that is 3123 against 288  
983 per vertical column of voxels, a factor of eleven. In three dimensions, with  
984 the lateral grid refined as the vertical one is, the number of voxels falls by  
985  $(347/8)^3 \approx 8 \times 10^4$  and the number of unknowns by about  $2 \times 10^4$ . The same  
986 model has been solved independently in Python, with a contrast for each  
987 plane: the raw reflected displacements agree with the Mathematica scheme  
988 to  $1.1 \times 10^{-15}$  at normal incidence and at  $20^\circ$ , with and without the first  
989 moments. Lateral heterogeneity needs the full three-dimensional solver; it is  
990 the first extension discussed below.

991 *A medium that varies within a cell.* The stratified model is exact on its own  
992 grid: every cell is homogeneous. A smoothly varying medium is not, and  
993 sampling it as cells of constant properties is an approximation of its own.  
994 Let the contrast of cell  $j$  be its Legendre projection to degree  $r$ ,  $\Delta(z) =$   
995  $\Delta_0 \sum_{c \leq r} f_c^{(j)} P_c(s/h)$ , with the voxel's field of degree  $p$  as in §8. The product  
996 of contrast and field is again a Legendre sum,  $P_c P_b = \sum_e A_{cbe} P_e$ , with  $A_{cbe} =$   
997  $(2e+1) \begin{pmatrix} c & b & e \\ 0 & 0 & 0 \end{pmatrix}^2$  the square of a Wigner  $3j$  symbol, so the coupling of Eq. (38)

998 becomes

$$\iint_j \phi_a(z) \mathbf{K}(z-z') \Delta(z') \phi_b(z') dz dz' = \sum_{c \leq r} \sum_{e \leq p+r} f_c^{(j)} A_{cbe} \left[ \iint_j \phi_a(z) \mathbf{K}(z-z') \phi_e(z') dz dz' \right] \Delta_0 : \quad (41)$$

999 the same kernel moments, now to degree  $p+r$ , and nothing else new. Against  
 1000 the exact graded layer, a high-precision solution of  $(M(z)u')' + \omega^2 \rho(z)u = 0$ ,  
 1001 the order of convergence is

$$\min(2p+2, 2r+2), \quad (42)$$

1002 or  $2p+2$  when degree  $r$  represents the profile exactly (Table 9; `ContinuumLimit_`  
 1003 `GradedContrast.wl`). A contrast that is constant in each cell therefore  
 1004 caps every voxel at second order, however rich its field basis, and for the  
 1005 smooth profile the voxel of degree 2 with a linear contrast has the error  
 1006 of the voxel of degree 1 to within 0.06%. At first order in the contrast  
 1007 this is what one expects: the contrast's projection error is orthogonal to  
 1008 the polynomials of degree  $r$ , so against the smooth product of the inci-  
 1009 dent and adjoint fields it contributes at  $O(h^{2r+2})$ . To keep order  $2p+2$   
 1010 the contrast must be represented to the degree of the field. The closed forms  
 1011 of the construction are tested one by one by an independent implementa-  
 1012 tion (`scripts/crosscheck_graded_contrast.py`): the Legendre moments  
 1013 of the kernel against  $2h i^n j_n(ch)$  and 30-digit quadrature (to  $2 \times 10^{-16}$ ), the  
 1014 linearisation coefficients against the  $3j$  symbols (exactly), and the same-cell  
 1015 double integrals against 30-digit quadrature ( $1.4 \times 10^{-16}$ ). It then assem-  
 1016 bles the scheme without linearisation, evaluating the projected contrast at  
 1017 the quadrature nodes, against its own 30-digit solution of the graded layer;  
 1018 its errors agree with those of Table 9 to  $10^{-3}$ , and at  $\omega = 1200 \text{ rad s}^{-1}$  it

Table 9: A contrast that varies within the cells: the field of each voxel to Legendre degree  $p$ , its contrast to degree  $r$ . Relative reflection error at  $n = 16$  planes and the order ( $n = 4 \rightarrow 8 \rightarrow 16$ ),  $\omega = 300 \text{ rad s}^{-1}$ , for the profiles  $1 + (2z/D - 1)/2$  (linear) and  $1 + \frac{1}{2} \sin(2\pi z/D)$  (smooth) times the contrast of §7. Predicted: Eq. (42), and  $2p + 2$  for the linear profile once  $r \geq 1$ .

| $p$ | $r$ | linear profile        |       | smooth profile        |       |
|-----|-----|-----------------------|-------|-----------------------|-------|
|     |     | error                 | order | error                 | order |
| 0   | 0   | $7.7 \times 10^{-5}$  | 2.00  |                       |       |
| 1   | 0   | $7.6 \times 10^{-5}$  | 2.00  | $5.5 \times 10^{-5}$  | 1.95  |
| 1   | 1   | $6.4 \times 10^{-11}$ | 4.00  | $1.4 \times 10^{-7}$  | 3.95  |
| 2   | 1   | $2.3 \times 10^{-17}$ | 6.00  | $1.4 \times 10^{-7}$  | 3.95  |
| 2   | 2   |                       |       | $1.6 \times 10^{-10}$ | 5.98  |

1019 measures the orders of Eq. (42) again.

1020 *Which basis sets the error.* Equation (42) gives the order. The size of the  
1021 error, and which of the two bases it belongs to, appear when the scattered  
1022 field is separated by its order in the contrast: the Born term  $T_1$ , and the  
1023 second-order term  $T_2$  in which the field scattered once is scattered again.

1024 Appendix Appendix G gives both terms for the exact layer in closed  
1025 form (Eqs. (G.2)–(G.7)) and for the scheme, where the first needs no solve  
1026 (Eq. (G.8)), and shows how both are read from solutions at four scaled  
1027 contrasts (Eq. (G.9)). The terms must be separated: at the full contrast the  
1028 terms of third and higher order are 7% of the second and carry errors of their  
1029 own.

1030 With  $f(z)$  the profile and  $\Pi_q$  the  $L^2$  projection onto the Legendre poly-  
1031 nomials of degree at most  $q$  in each cell, define the *relative projection error*

1032 of the medium at degree  $q$ ,

$$\mathcal{E}_q = \frac{\sum_{\text{cells}} \int (f - \Pi_q f)^2 dz}{\int f^2 dz} = 1 - \frac{\|\Pi_q f\|^2}{\|f\|^2}, \quad \|f\|^2 = \int f^2 dz : \quad (43)$$

1033 the fraction of the squared norm of the profile that polynomials of degree  $q$   
 1034 on the cells cannot represent. It is a property of the medium and the grid  
 1035 alone; it involves neither the wave nor the solution. For the smooth profile  
 1036 and every one of the nine pairs with  $p, r \leq 2$ , the relative error of  $T_2$  is

$$\frac{|T_2^{\text{scheme}} - T_2|}{|T_2|} = \mathcal{E}_{\min(p,r)} \quad (44)$$

1037 to within 1.3% at  $\omega = 300 \text{ rad s}^{-1}$  on 4 to 16 cells: the ratio of the two sides  
 1038 is 0.998 to 1.005 when  $\min(p, r)$  is 0 or 1, and 1.010 to 1.013 when it is 2.  
 1039 The error of the Born term is smaller by two to six orders of magnitude. The  
 1040 error of the whole is therefore that of  $T_2$ , the projection error times  $|T_2|/|T_1|$   
 1041 (0.044 here): for  $p = r = 1$  on 16 cells that predicts  $1.6 \times 10^{-7}$ , and Table 9  
 1042 has  $1.4 \times 10^{-7}$ .

1043 Equation (44) is exact in the long-wave limit (Appendix Appendix G).  
 1044 There the strain scattered once is the profile itself, a cell holds it as  $\Pi_{\min(p,r)} f$ ,  
 1045 and the second scattering takes place in the cell of the first: the law is  
 1046 a statement about the single site, whose  $T$ -matrix is exact for the cell as  
 1047 represented and whose error is that the cell is not the medium. Measured,  
 1048 the part of  $T_2$  between different cells carries at most 0.002 of the error, and  
 1049 the departure from the law falls as  $(kD)^2$ .

1050 Three consequences follow. The two bases must be matched: a contrast  
 1051 of higher degree than the field is wasted (the pairs  $(p, r) = (0, 1)$  and  $(0, 2)$   
 1052 have the errors of  $(0, 0)$  to three digits), and a field of higher degree than  
 1053 the contrast improves only the Born term. The wavefield basis is there to

1054 carry what the medium imprints on the field, not only the wave: at these  
 1055 frequencies the wave itself varies little across a cell. And the error is known  
 1056 from the medium, before anything is solved:  $\mathcal{E}_q$  is a property of the profile  
 1057 and the cells. It vanishes when the medium is piecewise constant on the cells,  
 1058 which is the stratified model above; there the contrast needs no more than a  
 1059 constant in each cell, and the degree of the field alone sets the order.

1060 *A stratified background.* Everything above takes the background to be a  
 1061 whole space. The method is meant for a background that is plane strati-  
 1062 fied, with the voxels carrying the departure from it, and the results must  
 1063 hold there. They do, in the one case tested: normal incidence, P, with the  
 1064 background two half-spaces joined at  $z = z_I$  below the layer, the lower one  
 1065 faster and denser (displacement reflection coefficient  $-0.19$ ). The voxels then  
 1066 interact through the Green's function of that background,

$$g(z, z') = \frac{i}{2M_P k} \left[ e^{ik|z-z'|} + \rho_I e^{ik(2z_I - z - z')} \right], \quad z, z' < z_I, \quad (45)$$

1067 with  $\rho_I$  the reflection coefficient of the interface: the whole-space term and  
 1068 a reverberation that is smooth and carries no local term. The incident field  
 1069 is the field of the source in the background, and the exact answer is the  
 1070 reflection of the whole stack less that of the background. With the interface  
 1071 3 m below the layer, and again with the lowest voxel against it, the scheme  
 1072 converges at order 2.00, 4.00 and 6.0 for  $p = 0, 1$  and 2 on the uniform  
 1073 layer, and at 1.98, 3.98 and 5.98 on the smooth profile with  $r = p$ . The  
 1074 first-order term is predicted to round-off ( $5 \times 10^{-16}$ ): it is the sum over  
 1075 four paths, the incident wave travelling down or up and the scattered wave  
 1076 leaving up or down, of the exact first-order term of each path times the

1077 factor  $\sum_{a \leq p} (2a + 1) j_a(k_{\text{in}} h) j_a(k_{\text{out}} h) / j_0((k_{\text{in}} - k_{\text{out}}) h)$  for that path's pair  
 1078 of wavenumbers, whose leading terms are the constants of Eq. (40). And the  
 1079 relative error of  $T_2$  over  $\mathcal{E}_p$  is 1.00 to 1.02 at  $kD = 0.12$  on 4 to 16 cells for all  
 1080 three degrees, departing from one by  $1.1 \times 10^{-4}$  and  $3 \times 10^{-5}$  at  $kD = 0.012$   
 1081 for the two positions of the interface: Eq. (44) holds as in the whole space  
 1082 (`scripts/measure_layer_stratified_background.py`; the exact stack is  
 1083 computed in 30-digit arithmetic). The reason is the one given for Eq. (G.10):  
 1084 the reverberation adds no local term, so the second scattering in the cell of  
 1085 the first is unchanged. This check is by one implementation against an exact  
 1086 solution.

1087 *Oblique incidence, and a three-layer background.* The same holds at oblique  
 1088 incidence, for P, SV and SH waves, in the two half-spaces and in a three-layer  
 1089 background in which the layer of voxels reverberates between two interfaces  
 1090 (Appendix Appendix C). At  $20^\circ$  the orders are 2, 4 and 6 for  $p = 0, 1$  and  
 1091 2 in all thirty-six cases (Table C.15), and the errors are close to those of the  
 1092 whole space: the background changes the field the voxels must represent,  
 1093 not how well they represent it. Both terms of the error remain known in  
 1094 advance. The scheme's first-order term equals a closed-form prediction, built  
 1095 from the thirty-six pairs of incident and radiated plane waves (Eq. (C.3)), to  
 1096  $2 \times 10^{-15}$ ; and the law of Eq. (44) holds to 3% for  $p \leq 1$  and 6% for  $p = 2$  at  
 1097  $k_S D = 0.2$ , and to 0.6% at  $k_S D = 0.067$ , in every background and for every  
 1098 wave type (Table C.16).

## 1099 10. A sphere: what the layer predicts in three dimensions

1100 The layer and the stratified layer are laterally uniform. A sphere is not,  
 1101 and elastic scattering by a sphere, homogeneous [62, 63] or radially varying  
 1102 [66], has an exact solution: it is the usual benchmark for volume-integral and  
 1103 discrete-dipole methods. This section solves two spheres with the full three-  
 1104 dimensional voxel scheme: the sphere is voxelised on an  $n_{\text{sub}}^3$  grid, the Foldy-  
 1105 Lax system is solved with the cell-averaged coupling by an FFT-accelerated  
 1106 iterative solver, and a P wave is incident. For the graded sphere the voxel is  
 1107 the mean-only voxel of §8 in three dimensions: each cell carries the mean of its  
 1108 nine-component state, tested with the same constant, and the  $L^2$  projection  
 1109 of its contrast, which is its cell mean; every cell that overlaps the sphere is  
 1110 kept, and the cells are coupled by cell-to-cell double integrals of the Green’s  
 1111 tensor, one block per orbit of the cube group (Appendix Appendix E). For  
 1112 the sharp sphere the voxel is the package’s cube  $T$ -matrix, which differs from  
 1113 the collocation closure only through its far-field corrections (§6.6) and has  
 1114 the same order, each cell keeping the contrast of its centre. The far field is  
 1115 computed from the solution for the incident plane wave itself, each voxel’s  
 1116 displacement and strain carrying the plane-wave phase at its centre; adding  
 1117 to that phase a linear variation about the scatterer’s centre, as a Taylor  
 1118 expansion of the incident field would, counts the wave’s variation twice and  
 1119 leaves an error of order  $(ka)^2$  that no refinement removes. In the terms of  
 1120 the experiment of §7, the test is this.

- 1121 1. *The body.* A sphere of radius  $a = 10$  m (Fig. 6) in the background of  
 1122 §7, carrying the contrast of that section, at  $\omega = 150$  and  $300 \text{ rad s}^{-1}$ ,  
 1123 for which  $k_S a = 0.5$  and  $1$  and  $k_P a = 0.3$  and  $0.6$ .

- 1124 2. *The incident wave.* A P plane wave travelling along  $+z$ .
- 1125 3. *The grid.* The cube of side  $2a$  that circumscribes the sphere is cut into  
1126  $n_{\text{sub}}^3$  cells of side  $2a/n_{\text{sub}}$ ; which of them are kept is stated with each  
1127 voxel above.
- 1128 4. *The solve.* The iteration is carried to a relative residual of  $10^{-11}$ , far  
1129 below every discretisation error reported, so that the errors are the  
1130 scheme's and not the solver's.
- 1131 5. *The exact answer.* The partial-wave series of the sphere, summed to  
1132 order 10 at  $k_S a = 0.5$  and 11 at  $k_S a = 1$ , and evaluated as the exact  
1133 scattered displacement at the observers, not as its asymptotic form.
- 1134 6. *What is measured.* The scattered displacement vector at nine scattering  
1135 angles, equally spaced from  $11^\circ$  to  $169^\circ$  in the  $z$ - $x$  plane: forward and  
1136 backward scattering and everything between, P and S together.
- 1137 7. *The error.* The largest component of the difference between the scheme's  
1138 far field and the exact displacement over the nine angles, divided by the  
1139 largest component of the exact one. It is a maximum, not an average,  
1140 so that no angle is hidden.
- 1141 8. *The distance.* The solution of the scheme is the field inside the voxels,  
1142 and the scattered field outside follows from it at any distance. For the  
1143 far-field test it is read with the asymptotic formula, which keeps the  
1144 term in  $1/r$  alone: a choice of this test, not a property of the scheme.  
1145 The reference is the exact field, so the two differ by the near-field terms,  
1146 of relative size  $1/(kr)$ , however good the scheme. The graded sphere  
1147 is therefore observed at  $5 \times 10^8$  radii, where those terms are  $10^{-8}$  of  
1148 the peak: at  $5 \times 10^4$  radii they are  $1.4 \times 10^{-4}$ , which is 4% of the

1149 discretisation error at 12 voxels across and most of it at 24, and would  
 1150 be read as a failure to converge. The sharp sphere, whose errors are  
 1151 a hundred times larger, is observed at  $5 \times 10^4$  radii. The near field is  
 1152 tested separately, with a readout that drops nothing (Table D.17).

1153 *A sphere without a staircase.* A voxelised sphere with a sharp surface changes  
 1154 its shape with every grid. A sphere whose contrast falls smoothly to zero at  
 1155 its surface does not: the voxels that cross the surface carry a vanishing con-  
 1156 trast. The body here has a homogeneous core  $r < a/2$  carrying the contrast  
 1157 of §7, and across the shell  $a/2 < r < a$  the contrast falls to zero as the  
 1158 smoothstep  $s(x) = 10x^3 - 15x^4 + 6x^5$ ,  $x = (a - r)/(a - a/2)$ , which vanishes  
 1159 like  $(a - r)^3$  at the surface. Its exact scattering follows, order by order, from  
 1160 the radial equations of Takeuchi and Saito [66], integrated across the shell  
 1161 from the core’s regular solutions and matched to outgoing waves at  $r = a$ .  
 1162 The equations are derived symbolically from the equations of motion, and  
 1163 the solution is checked against a sphere of homogeneous concentric shells  
 1164 in closed form, by unitarity and reciprocity, and by an independent imple-  
 1165 mentation that agrees to  $2 \times 10^{-11}$  (`TakeuchiSaito.wl`, `ContinuumLimit_`  
 1166 `GradedSphere.wl`, `scripts/crosscheck_graded_sphere.py`). Each voxel  
 1167 carries the mean of the contrast over its cell. Table D.18 shows the error  
 1168 falling at second order from  $n_{\text{sub}} = 6$  to 24 at both frequencies (`scripts/`  
 1169 `pilot_graded_voxel_sphere.py`): the apparent order between successive  
 1170 grids is 2.02 to 2.06 at  $k_S a = 0.5$  and 2.00 at  $k_S a = 1$ , and the least-squares  
 1171 orders of Fig. 7(d) are  $2.045 \pm 0.004$  and  $1.994 \pm 0.004$ . With 24 voxels across  
 1172 the error is  $1.8 \times 10^{-4}$  and  $8.3 \times 10^{-4}$ . At  $n_{\text{sub}} = 4$  a voxel is as wide as the  
 1173 graded shell and the order is not yet asymptotic. This is the order the layer

1174 predicts for the mean-only voxel, and the first test of that prediction on a  
 1175 body with genuine three-dimensional structure.

1176 What makes the sequence this clean is that the voxel carries a single error.  
 1177 The collocation voxel of the sharp sphere applied to this body, each cell at  
 1178 the profile's value at its centre and only the cells whose centres lie inside the  
 1179 sphere kept, carries two: its own internal-field error and the error of sampling  
 1180 the profile, both  $O(h^2)$ . Its error falls from  $1.0 \times 10^{-1}$  at  $n_{\text{sub}} = 4$  to  $4.7 \times 10^{-3}$ ,  
 1181  $2.6 \times 10^{-3}$ ,  $6.4 \times 10^{-4}$  and  $2.3 \times 10^{-4}$  at 6, 8, 12 and 16, apparent orders 7.6,  
 1182 2.0, 3.5 and 3.6 at  $k_{\text{sa}} = 0.5$ , and at  $k_{\text{sa}} = 1$  the apparent orders are 1.7, 2.5  
 1183 and 2.4 from 6 to 16: on these grids its error is below the mean-only voxel's,  
 1184 but its apparent order is not that of any single term and cannot be read from  
 1185 them. The same collocation voxel with the cell mean of the contrast, and  
 1186 every overlapping cell kept, gives  $3.4 \times 10^{-3}$ ,  $1.9 \times 10^{-3}$  and  $8.2 \times 10^{-4}$  at 6,  
 1187 8 and 12, apparent orders 2.03 and 2.05: the sampling of the profile was the  
 1188 second term, and with it removed the collocation and mean-only voxels differ  
 1189 only by their constants, as in Table 7. The cell mean, with every overlapping  
 1190 cell kept, represents the same body on every grid, and the order the layer  
 1191 predicts is then the one observed.

1192 The first-moment voxel of §8, extended to three dimensions with a con-  
 1193 trast linear in each cell, converges at fourth order on this same sphere (Ap-  
 1194 pendix Appendix E): its apparent orders are 4.01 and 4.13 where the dis-  
 1195 cretisation error is isolated at weak contrast, and at the full contrast its error  
 1196 with 16 voxels across is  $4.9 \times 10^{-6}$  and  $5.9 \times 10^{-6}$ , 82 and 317 times below  
 1197 the mean-only voxel's on the same grid.

1198 *The near field, an impedance march, and a sharp sphere.* Three further tests  
 1199 are in Appendix Appendix D. Read at finite distance, down to a quarter  
 1200 of a radius from the surface, the graded sphere converges at the orders of  
 1201 the far field, second for the mean-only voxel and 3.8 to 3.95 for the first-  
 1202 moment voxel, while the asymptotic readout would be in error there by 48%  
 1203 to 85% (Table D.17). Against the impedance march of Haines et al. [67],  
 1204 which needs far fewer unknowns, the first-moment voxel reaches the march's  
 1205 accuracy with six voxels across and continues at close to fourth order below  
 1206 it, where the march is limited by the scattering between the spheres of the  
 1207 periodic array it must solve (Table D.19). And a sphere with a sharp surface  
 1208 does not converge: its error tracks the volume of its staircase, which changes  
 1209 with the grid (Table D.20); the smooth sphere, like the layer, measures the  
 1210 scheme.

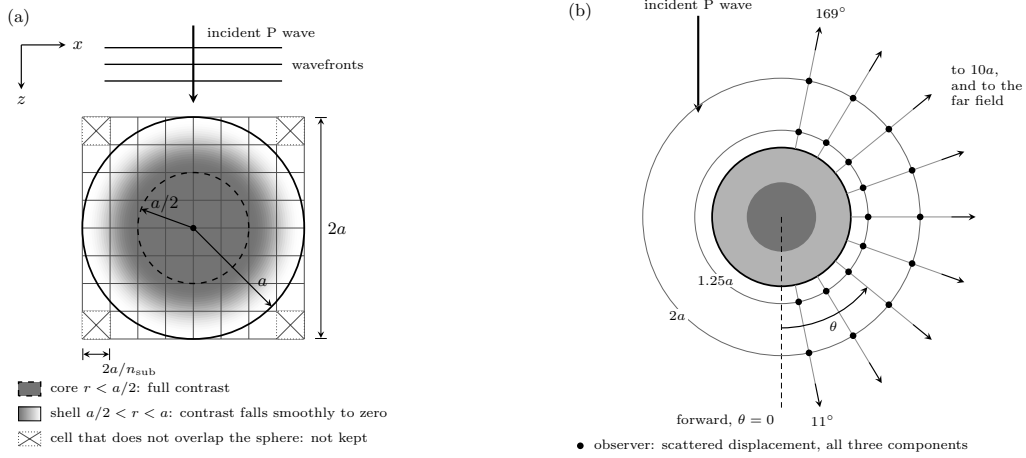


Figure 6: The sphere experiment, in the section through the centre that contains the direction of incidence. (a) The body and the grid. The contrast is full in the core  $r < a/2$  and falls smoothly to zero across the shell  $a/2 < r < a$ . The circumscribing cube of side  $2a$  is cut into  $n_{\text{sub}}^3$  cells, here  $n_{\text{sub}} = 8$ ; every cell that overlaps the sphere is kept and carries the projection of the contrast it contains. A P plane wave is incident along  $+z$ . (b) The observers. The scattered displacement is taken at nine scattering angles  $\theta$ , measured from the forward direction and equally spaced from  $11^\circ$  to  $169^\circ$ , on spheres of several radii about the centre, and compared with the exact displacement at the same points.

*Alt text:* Two panels. On the left, a disc on a square grid of eight by eight cells: the inner half of the disc is uniformly dark, and the outer half fades from dark to white at the rim, which is drawn as a solid circle; the four corner cells of the grid, which the disc does not reach, are crossed out. Horizontal lines above the grid with a downward arrow mark the incident wave. On the right, the same disc drawn smaller, with two concentric circles around it at one and a quarter and at two radii; nine dots on each circle lie on nine rays from the disc, fanning from just off the downward, forward direction to just off the upward, backward one, and each ray ends in an outward arrow.

## 1211 11. The Cartesian multipole hierarchy as a voxel scheme

1212 The Legendre voxels of §8 represent the field of a voxel by its moments  
 1213 over the voxel. The hierarchy of §6 represents it by its gradients at the  
 1214 centre. Both are polynomial, and §6.4 shows that on the layer the hierarchy  
 1215 of degree  $\ell = 3$  is fourth order, as the first-moment voxel is. This section  
 1216 builds the hierarchy into a voxel scheme in three dimensions and compares  
 1217 the two.

1218 *The scheme.* Each voxel  $c$ , centred at  $\mathbf{x}_c$ , carries the coefficients  $U_{j;W}^c$  of  
 1219 Eq. (20). The equation and its derivatives to order  $\ell$  are imposed at ev-  
 1220 ery centre (with  $\ell = 1$  and no density contrast this is the static numerical  
 1221 equivalent inclusion method on cuboidal elements, with uniform eigenstrains,  
 1222 equivalence at the element centres and a fast Fourier transform, of Zhou et al.  
 1223 [68]):

$$1224 \quad U_{i;P}^c - \sum_{c'} \left\{ \omega^2 \Delta \rho \sum_W \frac{K_{ij}[P; W](\mathbf{x}_c - \mathbf{x}_{c'})}{|W|!} U_{j;W}^{c'} + \sum_W \frac{K_{in}[Pk; W](\mathbf{x}_c - \mathbf{x}_{c'})}{|W|!} \delta c_{nkrj} U_{j;rW}^{c'} \right\} = \partial_P u_i^0(\mathbf{x}_c), \quad (46)$$

$$1225 \quad K_{in}[D; W](\Delta) = \int_V (\partial_D G_{in})(\Delta - \xi) \xi_W d\xi. \quad (47)$$

1226 The two kinds of block are of different character. For  $c' = c$  the integrand is  
 1227 singular and the integral is a distribution: it is the moment of the hierarchy,  
 1228  $K_{in}[D; W](0) = (-1)^{|D|} \mathbf{g}_{in}[D; W]$ , in closed form on the cube. For  $c' \neq c$   
 1229 the field point is the centre of another voxel, at least half a voxel from the  
 1230 source voxel, so the integrand is smooth and a tensor Gauss rule integrates  
 1231 it; the derivatives of the Green's tensor, to any order, come from the family  
 1232  $x^a y^b z^c r^p e^{ikr}$ , which is closed under differentiation. All the singular work  
 of the scheme is thus in the self block, where it is done once and in closed

1233 form; the Legendre voxel's self block is a double integral of the same singular  
 1234 kernel. The blocks depend on the offset alone, so the product of the system  
 1235 with a vector is a convolution over the lattice and is done exactly by the fast  
 1236 Fourier transform. Where the contrast varies within a voxel it is projected  
 1237 there on polynomials of degree  $r$ , as in §9, and the source is the product of  
 1238 that polynomial and the field's: the moments then carry weights to degree  
 1239  $\ell + r$ , and a block is a sum, over the monomials of the contrast, of blocks  
 1240 with the weight shifted.

1241 *Checks of the construction.* The derivatives of the Green's tensor to fourth  
 1242 order agree with the closed forms of its value, gradient and second derivative,  
 1243 and with central differences beyond them, to  $8 \times 10^{-7}$ . A lattice of one voxel  
 1244 reproduces the single site of §6.5 to  $10^{-14}$ . Only the offsets of one octant  
 1245 are computed, the others following from the parity of each index; a block  
 1246 obtained that way equals one computed directly to  $10^{-15}$ . The iterative  
 1247 solution equals a dense solution to  $7 \times 10^{-11}$  where both can be run.

1248 *A body with corners.* The cube of §6.5, at  $k_{sh} = 0.2$ , is cut into  $n^3$  voxels.  
 1249 Table 10 gives the far-field error against the converged reference, with the  
 1250 Legendre voxels on the same body. On this body every degree of the hierarchy  
 1251 converges at about the same rate, an order near two between the two finest  
 1252 grids, and the degree buys a constant: thirty times from  $\ell = 1$  to  $\ell = 3$   
 1253 on 64 voxels. That rate is the body's and not the scheme's, for the mean-  
 1254 only Legendre voxel shows it too: a homogeneous cube has edges and corners,  
 1255 whose fields no polynomial follows. The uniform-strain closure and the mean-  
 1256 only voxel agree on one cell, as they must, being the same approximation.  
 1257 And the first-moment Legendre voxel is far ahead: one cell of 36 unknowns is

Table 10: A homogeneous cube cut into  $n^3$  voxels,  $k_S h = 0.2$ : relative far-field error of the hierarchy of degree  $\ell$  and of the Legendre voxels, with the unknowns of each voxel.

|                        | unknowns a voxel | $n = 1$              | 2                    | 3                    | 4                    |
|------------------------|------------------|----------------------|----------------------|----------------------|----------------------|
| hierarchy, $\ell = 1$  | 12               | $5.4 \times 10^{-3}$ | $1.7 \times 10^{-3}$ | $8.3 \times 10^{-4}$ | $5.0 \times 10^{-4}$ |
| hierarchy, $\ell = 2$  | 30               | $1.6 \times 10^{-3}$ | $6.7 \times 10^{-4}$ | $3.7 \times 10^{-4}$ | $2.4 \times 10^{-4}$ |
| hierarchy, $\ell = 3$  | 60               | $3.4 \times 10^{-4}$ | $7.0 \times 10^{-5}$ | $3.0 \times 10^{-5}$ | $1.7 \times 10^{-5}$ |
| Legendre, mean only    | 9                | $5.4 \times 10^{-3}$ | $1.3 \times 10^{-3}$ | $5.8 \times 10^{-4}$ | $3.3 \times 10^{-4}$ |
| Legendre, first moment | 36               | $4.8 \times 10^{-5}$ | $6.0 \times 10^{-6}$ |                      |                      |

more accurate than eight voxels of the hierarchy of degree three with 480. A projection, which the Legendre voxel is, follows a field that is not polynomial better than an expansion about a point: the advantage of a Galerkin over a Taylor discretisation of the Lippmann–Schwinger equation that Brisard et al. [54] established for the static equivalent inclusion method.

*A smooth medium.* The comparison that bears on voxel schemes is in a medium without corners. First in one dimension: the layer of §6.4 with a contrast that varies in depth, projected in each cell to the degree of the field, against the exact graded layer. With the profile  $\sin^2(\pi z/D)$  the orders from 16 to 32 cells are 1.99, 1.98 and 3.99 for  $\ell = 1, 2$  and 3: a varying contrast leaves the pairs of §6.4 as they are. With the smoothstep of §10 as the profile they are 1.97, 1.99 and 3.84, and on coarser grids they are erratic, the third-gradient scheme being worse than the second on four cells. The orders appear only when about four cells cross the graded part of the profile (`scripts/measure_layer_taylor_hierarchy_graded.py`). That fixes the conditions of the three-dimensional test. The graded sphere of §10 has a shell

1274 5 m wide, and a solver that reaches fourteen voxels across puts fewer than  
1275 four voxels across it; so the core is reduced here from  $a/2$  to  $a/10$ , which  
1276 widens the shell to 9 m, and a second profile,  $\sin^2(\pi x/2)$  in the fraction  $x$  of  
1277 the shell's width measured from the surface, is run beside the smoothstep.  
1278 The exact reference is the radial solution of §10 for each body. The contrast  
1279 is linear in each voxel for  $\ell = 3$ , which suffices for fourth order (§9), and of  
degree  $\ell$  for  $\ell = 1$  and 2.

Table 11: The graded sphere with a shell 9 m wide,  $k_{Sa} = 0.5$ : relative far-field error against the exact solution for  $n_{\text{sub}}$  voxels across, with the apparent order between the two finest grids of each row. The lower block is the profile  $\sin^2$ .

|                        | $n_{\text{sub}} = 4$ | 6                    | 8                    | 10                   | 12                   | 14                   | order |
|------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|-------|
| hierarchy, $\ell = 1$  | $1.5 \times 10^{-3}$ | $8.4 \times 10^{-4}$ | $4.9 \times 10^{-4}$ | $3.2 \times 10^{-4}$ |                      |                      | 1.94  |
| hierarchy, $\ell = 2$  | $3.3 \times 10^{-4}$ | $2.1 \times 10^{-4}$ | $1.5 \times 10^{-4}$ | $1.0 \times 10^{-4}$ |                      |                      | 1.56  |
| hierarchy, $\ell = 3$  | $2.6 \times 10^{-4}$ | $5.0 \times 10^{-5}$ | $1.7 \times 10^{-5}$ | $7.2 \times 10^{-6}$ | $3.6 \times 10^{-6}$ | $2.0 \times 10^{-6}$ | 3.80  |
| Legendre, mean only    | $7.3 \times 10^{-3}$ | $3.2 \times 10^{-3}$ | $1.8 \times 10^{-3}$ | $1.1 \times 10^{-3}$ |                      |                      | 2.02  |
| Legendre, first moment | $2.9 \times 10^{-4}$ | $6.4 \times 10^{-5}$ | $2.2 \times 10^{-5}$ | $9.4 \times 10^{-6}$ | $4.6 \times 10^{-6}$ | $2.5 \times 10^{-6}$ | 3.92  |
| hierarchy, $\ell = 3$  | $1.9 \times 10^{-4}$ | $3.8 \times 10^{-5}$ | $1.3 \times 10^{-5}$ | $5.2 \times 10^{-6}$ | $2.7 \times 10^{-6}$ | $1.5 \times 10^{-6}$ | 3.84  |
| Legendre, first moment | $2.2 \times 10^{-4}$ | $4.8 \times 10^{-5}$ | $1.6 \times 10^{-5}$ | $6.8 \times 10^{-6}$ | $3.4 \times 10^{-6}$ | $1.8 \times 10^{-6}$ | 4.02  |

1280

1281 Table 11 gives the result. The first-gradient scheme converges at second  
1282 order, and is three to four times more accurate than the mean-only Legendre  
1283 voxel on every grid, for twelve unknowns a voxel against nine. The second-  
1284 gradient scheme is more accurate again but gains no order, as §6.4 predicts;  
1285 its apparent order is still rising towards two. The third-gradient scheme  
1286 converges at an apparent order between 3.7 and 4.1 from grid to grid, 3.8

1287 over the range from six to fourteen voxels across, for both profiles; the first-  
 1288 moment Legendre voxel does the same, between 3.7 and 4.0 and 3.85 over  
 1289 the range. Neither has settled on four at fourteen voxels across, and the  
 1290 second profile, gentler than the first, changes neither: the two fourth-order  
 1291 schemes behave alike. On every grid from six voxels across the hierarchy's  
 1292 error is the smaller, by a fifth to a quarter. It carries sixty unknowns a voxel  
 1293 against thirty-six, so that for a given number of unknowns the Legendre voxel  
 1294 remains somewhat the more accurate; the wide margin of Table 10 was the  
 1295 corners.

1296 *What the hierarchy is for.* The two schemes reach the same order by different  
 1297 closures of the same equation, and they differ in what they cost and in what  
 1298 they give besides. The Legendre voxel is the more economical in unknowns.  
 1299 The hierarchy's blocks are simpler: its self block is known in closed form on  
 1300 four constants (§6.2), and its coupling is a single integral of a smooth func-  
 1301 tion where the Legendre voxel's is a double integral. The iteration converged  
 1302 in five steps at every size; the time goes into the tables of blocks, which are  
 1303 computed once for a grid and a frequency. Formed on the same footing, one  
 1304 block for each orbit of offsets under the 48 symmetries of the cube, the others  
 1305 following exactly, and every point value of the Green's tensor or its derivatives  
 1306 from the same compiled radial ladder, the hierarchy's tables are the cheaper:  
 1307 0.6, 1.9 and 4.5 s for 6, 10 and 14 voxels across, against 4.0, 9.3 and 23.3 s for  
 1308 the first-moment Legendre voxel, with eight times fewer kernel evaluations.  
 1309 Both have a one-off cost besides, the hierarchy's the smaller (0.3 s against  
 1310 2.2 s): the moments of its self block depend on neither the frequency nor the  
 1311 cell size, and are computed once and stored (`scripts/build_cube_unit_`

1312 `moments.py`; `scripts/measure_hierarchy_table_cost.py`; one machine,  
 1313 five threads). And the hierarchy is, in addition, a single-site  $T$ -matrix in  
 1314 closed form, whose accuracy improves block by block (§6.5), which no voxel  
 1315 defined by moments over the cell provides. The lattice results of this section  
 1316 come from one implementation, checked against exact solutions and, where  
 1317 it can be, against a dense solve; they are at one frequency and on one fam-  
 1318 ily of bodies (`scripts/gradient_voxel_lattice.py`, `scripts/measure_`  
 1319 `lattice_gradient_hierarchy.py`, `scripts/measure_graded_sphere_gradient_`  
 1320 `hierarchy.py`, `scripts/measure_graded_sphere_gradient_hierarchy_fft.`  
 1321 `py`).

## 1322 12. Discussion

1323 Figure 7 collects the convergence of every test, with the order fitted to  
 1324 each by least squares (`scripts/plot_convergence_orders.py`).

1325 *What each basis is for.* A voxel holds two functions, and the results give  
 1326 each basis its part. The basis of the wavefield sets how well the voxel carries  
 1327 the field across itself: in a uniform medium the scheme converges at order  
 1328  $2p + 2$  with the constant  $c_p$  of Eq. (40), for P and S waves and at normal and  
 1329 oblique incidence alike. The basis of the medium sets how well the voxel *is*  
 1330 the medium: with degree  $r$  the order is capped at  $2r + 2$  (Eq. (42)). Neither  
 1331 can make up for the other, and the second-order term shows why. The field  
 1332 scattered once is a copy of the medium’s profile, so the wavefield basis must  
 1333 carry the profile as well as the wave, and the error of that term is the relative  
 1334 projection error of the profile at the lower of the two degrees (Eq. (44)). At  
 1335 the frequencies of interest the wave varies little across a cell and the medium

1336 may vary much, so it is the medium, not the wavelength, that decides the  
 1337 degree and the size of the cells. Two cases bracket the choice. Where the  
 1338 medium is piecewise constant on the cells, as in a layered model with voxel  
 1339 faces on its interfaces, a constant is exact for the medium, the projection  
 1340 error vanishes, and the degree of the field is spent entirely on the wave.  
 1341 Where the medium varies smoothly within the cells, the two degrees should  
 1342 be equal: a higher degree for the contrast than for the field is wasted, and a  
 1343 higher degree for the field improves only the Born term. That the two degrees  
 1344 can be chosen independently is the practice of higher-order electromagnetic  
 1345 volume methods [1]; what the present results add is the rule for choosing  
 1346 them, and the error each choice leaves.

1347 *Where the error of a voxel scheme lives.* The results separate the two ap-  
 1348 proximations a voxel scheme makes. The *coupling* between voxels makes  
 1349 none: averaged over the source cell it is the continuum's cell integral exactly,  
 1350 between planes by the form factor's zeros and within a plane by the tiling  
 1351 identity, and the self term it omits is exactly the one the single site carries.  
 1352 Every error therefore comes from the *single site*: from the internal field it  
 1353 assumes and, where the medium varies within the cell, from the medium it  
 1354 is given. Its  $T$ -matrix sums the rescattering within the voxel exactly, but  
 1355 for the voxel as represented, and the split of the second-order term in §9  
 1356 measures the consequence: the error sits in the part whose two scatterings  
 1357 lie in one cell, and the part between cells carries at most 0.2% of it. For a  
 1358 uniform field that error is the midpoint rule's, known in closed form; for a  
 1359 linear field it is the next term of the same expansion. The converse is the  
 1360 practical warning. A kernel that is not the cell integral (the point Green's

1361 tensor, or a cell average that does not match the single site’s closure) leaves  
1362 a local term that does not vanish as the voxels shrink (§4): the evanescent  
1363 lattice orders carry a static contribution of fixed size per voxel volume. A  
1364 scheme built that way converges to the wrong limit, and refinement cannot  
1365 reveal it, because the error is scale-invariant.

1366 *What the single site is.* The Cartesian multipole hierarchy of §6 turns the  
1367 uniform-field single site from a modelling choice into a derived object with  
1368 a stated error (§6.7). Its one departure from exact scattering by a sphere,  
1369 the real  $(ka)^2$  term of Eq. (F.1), and the collocation error of Eq. (36) have  
1370 the same origin, the variation of the field across the scatterer, and the same  
1371 remedy, a richer internal field. In a lattice the remedy is cheap: the field  
1372 a voxel must represent is the smooth continuum field, not the field of an  
1373 isolated cube, with its concentrations at the edges and corners.

1374 *Relation to other methods.* The partition of the error is the coupled-dipole  
1375 method’s choice of polarisability seen from the other side: for space-filling  
1376 cubes the tiling identity fixes the local field exactly, and the polarisability  
1377 is neither needed nor a free parameter. The voxel of degree  $p$  is close kin  
1378 to the spectral element of seismology; the two share a Galerkin condition  
1379 on polynomial bases, and differ as a differential formulation from an integral  
1380 one, and so suit different problems. Appendix Appendix H sets out both  
1381 relations, and where each method is the better choice.

1382 *The tension between the two error sources.* The coupled-dipole method has  
1383 two sources of error, the single site and the coupling between sites, and each  
1384 has been corrected on its own: the polarisability by the Clausius–Mossotti

1385 relation, the lattice dispersion relation or the exact static limit, and the  
 1386 coupling by integrating or filtering the Green tensor. The two are not in-  
 1387 dependent. The point kernel's evanescent orders carry a static local term  
 1388 whose size per voxel volume does not shrink with the voxels (§4), and the  
 1389 single site must cancel exactly that term. A polarisability tuned for one  
 1390 kernel is therefore wrong for another: the lattice dispersion relation corrects  
 1391 the point-dipole lattice, and changing the kernel to an integrated or a fil-  
 1392 tered one changes the term it must cancel. Tuning either alone moves the  
 1393 error between the two, and refinement cannot reveal a mismatch, because  
 1394 the surviving term is scale-invariant. For space-filling cubes the tension is  
 1395 resolved rather than balanced: the cell-averaged kernel omits exactly the self  
 1396 term that the closure carries (Eq. (35)), so the coupling error vanishes iden-  
 1397 tically and every remaining error is the single site's, the variation of the field  
 1398 across the cell. That error has a second tension of its own. It is a midpoint-  
 1399 rule error whose constants have opposite signs in reflection and transmission  
 1400 (Eq. (36)), so any scalar adjustment of the single site that improves one wors-  
 1401 ens the other; the package's far-field corrections reduce the reflection error  
 1402 by a factor 4.45 and increase the transmission error by 1.7. No polarisability,  
 1403 however chosen, can remove it. What removes it is a richer internal field:  
 1404 the first moment lowers the error by two orders in  $kd$  in both observables at  
 1405 once.

1406 *The cost of higher order, and a fast solver.* A first-moment voxel carries four  
 1407 times as many unknowns as a uniform one, but by the closed-form constants it  
 1408 reaches a reflection error of  $10^{-6}$  with voxels 58 times larger in each direction;  
 1409 the stratified model of §9 measures it directly, 347 planes of uniform-field

1410 voxels against 8 first-moment voxels. And because the discretisation error  
 1411 is known in advance and the coupling between cells is exact, a fast product  
 1412 need be no more accurate than that error, and can be budgeted separately  
 1413 from it. Appendix Appendix I sets out the cost, and what the error analysis  
 1414 offers a fast solver.

1415 *Limitations and outlook.* The results concern laterally uniform media, a ho-  
 1416 mogeneous layer and a randomly stratified one, for which the discrete system  
 1417 is exactly a chain of planes; a laterally heterogeneous medium needs the first-  
 1418 moment voxel in a full three-dimensional solver, where the gradient channels  
 1419 couple through the evanescent lattice orders (§8) and the tiling identity holds  
 1420 only to  $O(k_{\parallel}h)$ . The smoothly graded sphere of §10 shows the mean-only  
 1421 voxel converging at second order on a laterally heterogeneous body against  
 1422 an exact solution, and the first-moment voxel converges at fourth order there  
 1423 (§11). A contrast that varies within a cell, which §9 shows is needed beyond  
 1424 second order in a smoothly varying medium, has a further consequence in  
 1425 three dimensions: its linear part is odd about the cell’s centre, so it couples  
 1426 the single site’s uniform-field and first-moment blocks, which parity keeps  
 1427 apart for a homogeneous cube (§6.3). A graded voxel therefore needs at least  
 1428 the first-moment block; that voxel is built, and tested on the graded sphere,  
 1429 in the companion paper. The numerical evidence is at moderate, physically  
 1430 typical contrasts (up to 15% in density and moduli) and at  $k_S h \leq 0.3$ , the  
 1431 range in which the single-site  $T$ -matrix is validated; the closed forms are first  
 1432 order in the contrast, and the departures from them at the full contrast are of  
 1433 second order. What is established is the *order* of convergence and its leading  
 1434 constant, not the behaviour at strong contrast, where the single site’s own

1435 closure may limit the accuracy before the lattice coupling does. The single  
 1436 site's dynamic error is in closed form for a sphere only; for a cube, for which  
 1437 no exact solution exists, it is known through the layer (§§7–8) and, for the  
 1438 isolated cube, by comparison with a converged solution (§6.5). The hier-  
 1439 archy beyond the first gradient is established less widely than the rest: its  
 1440 orders and its error factor on the layer are derived symbolically and checked  
 1441 independently to the third degree, and its moments are confirmed by two  
 1442 routes, but its three-dimensional results, the single site against the sphere  
 1443 and the cube and the voxel scheme of §11, come from one implementation, at  
 1444 one frequency for the voxel scheme, checked against exact solutions. There  
 1445 the fourth order is approached and not attained on the grids reached (3.8  
 1446 at fourteen voxels across, as for the first-moment voxel), the fifth-gradient  
 1447 system, whose moments are in hand, has not been assembled in three di-  
 1448 mensions, and the two fourth-order schemes have been compared in the cost  
 1449 of forming their tables only (§11). Two extensions follow directly. The first  
 1450 is the three-dimensional solver itself. The second is a stratified *background*,  
 1451 a layered reference medium, as distinct from the stratified contrast of §9.  
 1452 For a laterally uniform layer of voxels the orders, the first-order factor and  
 1453 the projection-error law all survive it, at normal and at oblique incidence,  
 1454 for P, SV and SH waves, in a background of two half-spaces and in one of  
 1455 three layers (§9). What remains is a body of finite lateral extent in such a  
 1456 background, where the image of a voxel in an interface it touches is singular.

### 13. Conclusions

A voxel of a volume-integral scheme must represent the medium within it and the wavefield within it. This paper has determined what each basis must be and what error a given choice leaves, by an experiment in which nothing else is approximated: a layer of cubes is the layer itself, so its exact solution isolates the discretisation error.

With the wavefield expanded to Legendre degree  $p$  and the contrast to degree  $r$  in each voxel, the scheme converges at order  $\min(2p+2, 2r+2)$ , with the leading error  $c_p(kd)^{2p+2}$  in closed form (Eq. (40)) for P, SV and SH waves at normal and oblique incidence, in a whole space and in a plane-stratified background. A contrast held constant in each cell caps every voxel at second order. The error of the term of second order in the contrast is the relative projection error of the medium at degree  $\min(p, r)$ : it is the single site's, whose  $T$ -matrix sums the rescattering within a voxel exactly for the voxel as represented, so the two bases must be matched, and where the medium is piecewise constant on the cells it vanishes. A sphere whose contrast falls smoothly to zero at its surface converges at the orders the layer predicts.

The single site is derived rather than assumed. Three properties of the cube make the experiment exact: the source-cell-averaged coupling is the continuum's cell integral exactly, the moments of the Green's tensor over the cube give the single site with no polarisability to choose, and its self term cancels the kernel's, so the simplest scheme is a collocation of the continuum equation. The uniform-strain closure is the first member of a Cartesian multipole hierarchy, the elastodynamic extension of the Taylor form of the equivalent inclusion method [53] to bodies that are not ellipsoids, whose

1482 moments are in closed form on four constants. Its blocks gain accuracy  
1483 in pairs, with the error of each truncation in closed form: as a single site  
1484 the third-gradient system is three hundred times more accurate than the  
1485 uniform-strain closure against the exact sphere, and as a voxel scheme it  
1486 is fourth order, as the first-moment voxel is, on a smoothly graded sphere,  
1487 where in a lattice the Legendre voxel is the more economical. The results  
1488 and the evidence for each are collected in Appendix Appendix J.

1489 *The error is known in advance.* For every discretisation studied here the  
1490 error is not something to be discovered by refining until the answer stops  
1491 changing. Its order and its leading constant are known before anything is  
1492 computed:  $+(kd)^2/8$  in reflection and  $-(kd)^2/24$  in transmission for the  
1493 collocation voxel (Eq. (36)),  $(kd)^2/12$ ,  $(kd)^4/720$  and  $(kd)^6/100800$  for the  
1494 voxels of degree 0, 1 and 2 (Eq. (40)), and the order  $\min(2p + 2, 2r + 2)$   
1495 where the medium varies within a cell (Eq. (42)), with the size of the error  
1496 there given by the relative projection error of the medium (Eq. (44)), which  
1497 is computed from the model and the grid alone. That is worth more than  
1498 the order alone, for three reasons. First, the cell size and the degree of the  
1499 voxel that a required accuracy needs follow from the wavenumber and the  
1500 medium, before the computation and without a refinement study. Secondly,  
1501 it turns a refinement study from a hope into a test. A scheme of this kind can  
1502 converge smoothly to the wrong limit (§4), and smooth convergence alone  
1503 cannot reveal it; but the difference between two grids must have the known  
1504 order and the known constant, and a computation whose differences do not  
1505 is in error from some cause other than the discretisation. Thirdly, it shows  
1506 where effort pays. The error is that of representing the field and the medium

1507 within a cell, so raising the degree of the cell does what multiplying the  
 1508 cells cannot: in the stratified model one first-moment voxel per layer reaches  
 1509  $10^{-6}$  where the uniform-field voxel needs about forty. Whether the error of  
 1510 a body with three-dimensional structure can be known in advance in the  
 1511 same way is the question taken up in a following paper [69], where it decides  
 1512 where the cells of an adaptive grid are placed. That paper continues the  
 1513 paper's first strand, the two bases and their error, with the Legendre voxels.  
 1514 The second strand, the Cartesian multipole hierarchy, whose coefficients are  
 1515 integrals of the Green's tensor over the cube, is taken further in a paper on  
 1516 the coupling integrals of polynomial cubes in closed form, of which it is the  
 1517 worked application.

## 1518 **Appendix A. The moment engine: gates and a second route**

1519 Two checks, whose answers do not depend on the engine, gate every grade.  
 1520 Because  $\nabla^2 r^m = m(m+1)r^{m-2}$  and  $\nabla^2(1/r) = -4\pi\delta$ ,

$$\sum_p E[m; ppR; W] = m(m+1) E[m-2; R; W], \quad \sum_p E[-1; ppR; W] = -4\pi (-1)^{|R|} (\partial_R x_W)(0),$$

(A.1)

1521 for any further derivatives  $R$  and any weight  $W$ . The second is the sum rule  
 1522 (27) at every grade: it tests the  $\delta$  function of the inclusion problem and its  
 1523 derivatives, which is where a quadrature fails. Both hold for every grade of  
 1524 Table 1 with two or more derivatives, 910 traces of each kind; the grades with  
 1525 fewer, for which the integrand is integrable, agree with numerical integration  
 1526 over the cube.

1527 The closed forms are also confirmed by a route that shares nothing with  
 1528 the faces of the cube. The cube is its inscribed ball plus the remainder.

1529 On the ball the derivatives are moved onto the sphere, where the surface  
1530 integrals of monomials are Gamma functions, and all the distributional con-  
1531 tent is there; on the remainder the integrand is an ordinary function, in-  
1532 tegrated radially in closed form and over directions through the six faces  
1533 by Gauss–Legendre quadrature. This evaluation reproduces all 4224 stored  
1534 values, the largest difference being  $1.1 \times 10^{-12}$ , and satisfies Eq. (A.1) by  
1535 itself to  $5 \times 10^{-10}$  or better up to eight derivatives (`CubeMomentCore.wl`,  
1536 `CubeMomentHigherGrades.wl`, `CubeMomentStore.wl`, `CubeMomentCanonical.`  
1537 `wl`; `scripts/crosscheck_cube_moments_ball_shell.py`).

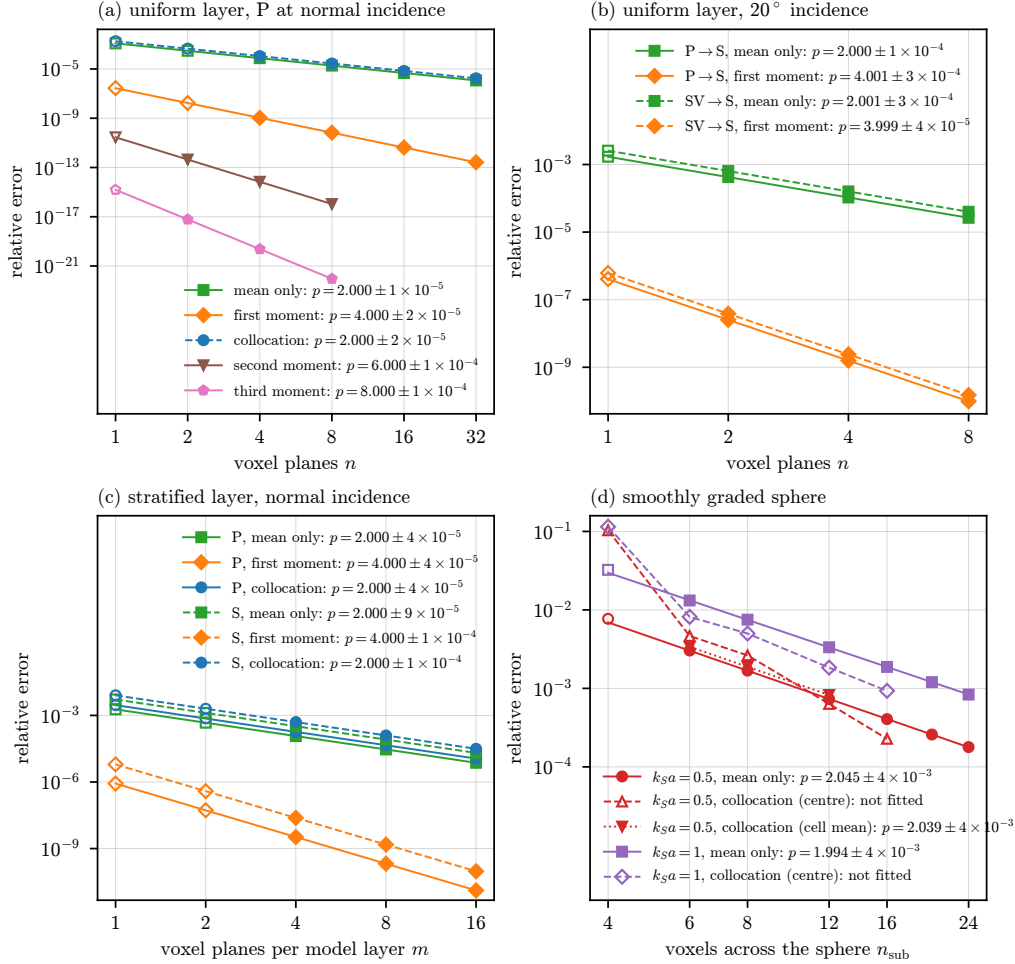


Figure 7: Convergence of every test, with least-squares regression lines. Each line fits  $\log e = c - p \log n$  to the filled points; open points are pre-asymptotic and are not fitted. The legend gives the order  $p$  and its standard error. The marker, and in colour the hue, marks the scheme, so that the figure reads in grey scale: collocation (circles; §6), the mean-only voxel (squares) and the voxels carrying Legendre moments to degree  $p = 1$  (diamonds), 2 (triangles) and 3 (pentagons) (Eqs. (38) and (B.1)). (a) The uniform layer, P reflection at normal incidence ( $D = 2$  m,  $\omega = 300$  rad s $^{-1}$ , the contrast of §7), fitted from  $n = 4$ ; the collocation line is dashed and lies on the mean-only line, from which its constant differs only by the factor 3/2 of Table 7; the second- and third-moment voxels, computed in 40-digit arithmetic, fitted from  $n = 2$ . (b) The uniform layer at  $20^\circ$ , fitted from  $n = 2$ : the S wave converted from incident P (solid) and reflected from incident SV (dashed). (c) The stratified layer of §9, reflection at normal incidence for P (solid) and S (dashed), against the number  $m$  of voxel planes per model layer, fitted from  $m = 4$ . (d) The smoothly graded sphere of §10 against its exact solution, for  $n_{\text{sub}}$  voxels across: the mean-only voxel, fitted from  $n_{\text{sub}} = 6$  (at 4 a voxel is as wide as the graded shell); the collocation voxel with the contrast sampled at the cell centre, joined by dashed lines and

## 1538 Appendix B. Further tests of the Legendre voxel

1539 *Oblique incidence.* The scheme is built spectrally. By Poisson summation  
 1540 the Bloch coupling between planes  $i$  and  $j$  at lateral wavenumber  $\mathbf{k}_{\parallel}$  is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z-z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g}, \quad (\text{B.1})$$

1541 with  $L_a$  the lateral Legendre factors. The whole-space tensor decays as  
 1542  $1/k_z^2$  and the nine-component state adds at most two powers of  $k_z$ , so each  
 1543 entry of  $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$  is a  $\delta(z)$  of constant weight plus, per mode,  $[\mathrm{i}\alpha/2\gamma +$   
 1544  $\mathrm{i}\beta \operatorname{sgn}(z)/2] \mathrm{e}^{\mathrm{i}\gamma|z|}$ ,  $\gamma^2 = k_W^2 - \kappa^2$ : every  $z$  integral is closed-form. Double  
 1545 averaging makes the lattice sum converge without Ewald splitting, to three  
 1546 digits at  $|p|, |q| \leq 8$ . Against the exact layer, for a P wave incident at  $20^\circ$   
 and  $40^\circ$  (`ContinuumLimit_Oblique.wl`; Table B.12). The mean-only voxel

Table B.12: Relative error of the reflected P ( $R_{PP}$ ) and converted S ( $R_{PS}$ ) waves for a P wave incident at  $20^\circ$ , with the mean-only and first-moment voxels of Eq. (B.1). At  $40^\circ$  the orders are again 2 and 4.

| $n$   | mean only            |                      | mean and first moments |                       |
|-------|----------------------|----------------------|------------------------|-----------------------|
|       | $R_{PP}$             | $R_{PS}$             | $R_{PP}$               | $R_{PS}$              |
| 1     | $9.3 \times 10^{-4}$ | $1.7 \times 10^{-3}$ | $7.2 \times 10^{-8}$   | $4.1 \times 10^{-7}$  |
| 2     | $2.3 \times 10^{-4}$ | $4.2 \times 10^{-4}$ | $4.5 \times 10^{-9}$   | $2.5 \times 10^{-8}$  |
| 8     | $1.5 \times 10^{-5}$ | $2.7 \times 10^{-5}$ | $1.7 \times 10^{-11}$  | $9.9 \times 10^{-11}$ |
| order | 2                    | 2                    | 4                      | 4                     |

1547  
 1548 is second order and the first-moment voxel fourth order in both the reflected  
 1549 P and the converted S wave: the breakdown of the tiling identity at  $O(k_{\parallel}h)$

1550 and the evanescent coupling of the gradient channels do not cost an order.

1551 At  $\mathbf{k}_{\parallel} = 0$  the construction reproduces Eq. (38) to all printed digits.

1552 *Incident S.* Nothing in the construction is specific to P. For a normally in-  
 1553 cident S wave the layer is the one-dimensional problem of Eq. (38) with  
 1554  $M_P \rightarrow \mu$ ,  $k_P \rightarrow k_S$  and  $\Delta M \rightarrow \Delta\mu$ ; the collocation and mean-only schemes  
 1555 converge at order 2.00 and the first-moment voxel at 4.00, in reflection and  
 1556 transmission, and the full  $9 \times 9$  construction with an incident SH wave re-  
 1557 produces that one-dimensional result to all printed digits. At oblique inci-  
 1558 dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders  
 are those of Table B.13. The orders, and with  $k = k_S$  the closed-form con-

Table B.13: Observed order of convergence ( $n = 2 \rightarrow 8$ ,  $\omega = 300 \text{ rad s}^{-1}$ ) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle  $\arcsin(\beta/\alpha) = 36.9^\circ$ .

|                                            | mean only   | mean and first moments |
|--------------------------------------------|-------------|------------------------|
| SV at $20^\circ$ : $R_{SS} / R_{SP}$       | 2.00 / 2.00 | 4.00 / 3.99            |
| SV at $30^\circ$ : $R_{SS} / R_{SP}$       | 1.98 / 2.00 | 3.97 / 4.00            |
| SH at $20^\circ$ and $40^\circ$ : $R_{SH}$ | 2.00        | 4.00                   |

1559

1560 stants of Eqs. (36) and (39), are those of the incident P wave.

1561 *Degree two at oblique incidence.* In three dimensions the voxel of degree  
 1562 2 carries, on each of the nine components of its state, the ten Legendre  
 1563 products of total degree at most two in  $(z, x, y)$ . In the construction of  
 1564 Eq. (B.1) it converges at sixth order against the exact layer for incident  
 1565 P, SV and SH waves (Table B.14; `scripts/measure_oblique_degree.py`).

1566 The test is run at  $\omega = 3000 \text{ rad s}^{-1}$  ( $k_P D = 1.2$ ,  $k_S D = 2$ ), ten times the  
1567 frequency of the other tables, because at  $300 \text{ rad s}^{-1}$  a sixth-order error on  
1568 these grids lies below the round-off of double precision. The exact reflection  
1569 coefficient is that of Kennett’s recursion, an independent algorithm, and the  
1570 same computation with the bases of degree 0 and 1 returns orders 2 and 4  
1571 in all five cases. At normal incidence the error on four planes,  $6.95 \times 10^{-9}$ , is  
1572 the  $c_2(kd)^6 = 7.2 \times 10^{-9}$  of Eq. (40) to 4%. Doubling the number of lattice  
1573 orders retained in Eq. (B.1) changes the errors by less than 1%. Unlike the  
1574 degrees 0 and 1, this degree is implemented once at oblique incidence; its  
check is the exact solution.

Table B.14: The voxel of degree 2 (ten Legendre products per state component) at  $\omega = 3000 \text{ rad s}^{-1}$ : relative error of the same-type reflection coefficient against the exact layer on  $n$  planes, and the order over the last halving, with the orders of the degrees 0 and 1 from the same computation.

| incident wave               | error, $p = 2$       |                      |                      | order   | orders          |
|-----------------------------|----------------------|----------------------|----------------------|---------|-----------------|
|                             | $n = 1$              | $n = 2$              | $n = 4$              | $p = 2$ | $p = 0 / p = 1$ |
| P, normal: $R_{PP}$         | $3.4 \times 10^{-5}$ | $4.6 \times 10^{-7}$ | $7.0 \times 10^{-9}$ | 6.05    | 2.04 / 4.05     |
| P at $20^\circ$ : $R_{PP}$  | $7.3 \times 10^{-6}$ | $1.0 \times 10^{-7}$ | $1.5 \times 10^{-9}$ | 6.04    | 2.04 / 4.06     |
| SV at $20^\circ$ : $R_{SS}$ | $2.2 \times 10^{-4}$ | $2.3 \times 10^{-6}$ | $3.4 \times 10^{-8}$ | 6.11    | 2.10 / 4.14     |
| SH at $20^\circ$ : $R_{SH}$ | $2.3 \times 10^{-4}$ | $2.5 \times 10^{-6}$ | $3.6 \times 10^{-8}$ | 6.11    | 2.10 / 4.14     |
| SH at $40^\circ$ : $R_{SH}$ | $3.0 \times 10^{-4}$ | $3.6 \times 10^{-6}$ | $5.2 \times 10^{-8}$ | 6.09    | 2.12 / 4.05     |

1575

1576 The spectral construction is implemented twice. The second implementa-  
1577 tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the defi-  
1578 nition of the scheme: it takes the depth dependence of the transformed kernel  
1579 from an independent residue construction, obtains the local weight as a nu-

1580 merical large- $k_z$  limit of the inverted Christoffel matrix, evaluates every dou-  
 1581 ble integral by quadrature and separates P from S by sampling the upgoing  
 1582 field at two depths. The raw reflected displacements of the two agree to  $9 \times$   
 1583  $10^{-16}$  in all 36 cases compared: incident P, SV and SH, normal incidence and  
 1584  $20^\circ$ ,  $n = 1, 2, 4$ , with and without the first moments. The one-dimensional  
 1585 scheme of any degree is also implemented twice. The second implementation  
 1586 (`scripts/crosscheck_second_moment_voxel.py`) evaluates every integral  
 1587 by Gauss quadrature, split at the receiver within a voxel, and the exact layer  
 1588 from its own interface system, in double precision. Its errors agree with the  
 1589 40-digit ones within its round-off floor ( $3 \times 10^{-14}$  in reflection,  $1.6 \times 10^{-13}$  in  
 1590 transmission); at  $\omega = 1200 \text{ rad s}^{-1}$  it measures orders 4.02 / 3.99 and 6.02 /  
 1591 6.00; for one plane as the layer thins its differences from the exact layer fall  
 1592 as  $D^{3.00}$ ,  $D^{5.00}$  and  $D^{7.00}$  for  $p = 0, 1$  and  $2$ ; and it rederives  $c_p$  for  $p \leq 5$ .  
 1593 Figure 7(a,b) shows the convergence, with the orders fitted to it.

## 1594 **Appendix C. The stratified-background construction**

1595 The experiment is built in four steps.

1596 *The kernel.* At each lateral wavenumber  $\boldsymbol{\kappa}$  the whole-space kernel between  
 1597 two depths factorises over the plane-wave modes, three going down and three  
 1598 going up (P, SV, SH):  $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z - z') = \mathbf{D}_\downarrow \mathbf{E}(z - z') \mathbf{S}_\downarrow$  for  $z > z'$ , with  $\mathbf{S}_\downarrow$  the  
 1599  $3 \times 9$  matrix of mode amplitudes radiated downwards by a nine-component  
 1600 source,  $\mathbf{E}(L) = \text{diag}(e^{i\gamma_m L})$  the propagation over a distance  $L$  with  $\gamma_m$  the  
 1601 vertical wavenumber of mode  $m$ , and  $\mathbf{D}_\downarrow$  the  $9 \times 3$  matrix of the modes'  
 1602 states; and likewise upwards. With the interfaces at  $z_A$  above and  $z_B$  below,  
 1603  $H = z_B - z_A$ , and  $\mathbf{R}_A, \mathbf{R}_B$  the  $3 \times 3$  reflection matrices of the two interfaces

1604 seen from inside the layer, a source at  $z'$  sends  $\mathbf{s}_\downarrow = \mathbf{S}_\downarrow q$  down and  $\mathbf{s}_\uparrow = \mathbf{S}_\uparrow q$   
 1605 up, and the reverberation it sets up is

$$\begin{aligned} \mathbf{d}_A &= (\mathbf{I} - \mathbf{R}_A \mathbf{E}(H) \mathbf{R}_B \mathbf{E}(H))^{-1} \mathbf{R}_A [\mathbf{E}(z' - z_A) \mathbf{s}_\uparrow + \mathbf{E}(H) \mathbf{R}_B \mathbf{E}(z_B - z') \mathbf{s}_\downarrow], \\ \mathbf{u}_B &= \mathbf{R}_B [\mathbf{E}(z_B - z') \mathbf{s}_\downarrow + \mathbf{E}(H) \mathbf{d}_A], \\ \hat{\mathbf{G}}_{\text{rev}}(\boldsymbol{\kappa}; z, z') q &= \mathbf{D}_\downarrow \mathbf{E}(z - z_A) \mathbf{d}_A + \mathbf{D}_\uparrow \mathbf{E}(z_B - z) \mathbf{u}_B, \end{aligned} \tag{C.1}$$

1606 with  $\mathbf{d}_A$  the downgoing amplitudes leaving the upper interface and  $\mathbf{u}_B$  the  
 1607 upgoing ones leaving the lower. It is added to the whole-space kernel in  
 1608 every block of Eq. (B.1), at every order of the reciprocal lattice. It is smooth  
 1609 and separable in  $z$  and  $z'$ , so its integrals over the cells are one-dimensional  
 1610 moments of exponentials. For two half-spaces  $\mathbf{R}_A = 0$  and only the second  
 1611 line survives.

1612 *The interfaces.*  $\mathbf{R}_A$ ,  $\mathbf{R}_B$  and the transmission matrices into and out of the  
 1613 layer follow from continuity of displacement and traction at each interface:  
 1614 six equations in the states of the six modes on either side, solved for each  
 1615 lateral wavenumber, evanescent orders included.

1616 *The incident field and the output.* A plane wave of unit amplitude arrives  
 1617 from the upper half-space. Its field in the layer, a downgoing part and an  
 1618 upgoing part with the reverberation summed as in Eq. (C.1), is the right-  
 1619 hand side of the scheme. The output is the upgoing wave of the same type  
 1620 as the incident one, arriving at  $z_A$  from the voxels and transmitted into the  
 1621 upper half-space: the change that the layer of voxels makes to the same-type  
 1622 reflection coefficient ( $R_{PP}$ ,  $R_{SS}$  or  $R_{SH}$ ), a quantity that does not depend on  
 1623 how the waves are normalised.

1624 *The reference, and a check of the construction.* The exact change is the

1625 same-type reflection coefficient of the stack with the layer, less that of the  
 1626 background, both from Kennett’s recursion [65], an independent algorithm.  
 1627 Before any voxel is introduced, the background’s own reflection coefficient  
 1628 computed from the modes and the interface matrices above agrees with Ken-  
 1629 nett’s to  $5 \times 10^{-15}$  in all twelve cases.

1630 It is computed by two independent algorithms. The first is invariant  
 1631 imbedding, the impedance march of Haines et al. [67]: the six-component  
 1632 state of displacement and traction obeys  $dy/dz = A(z)y$ , and with the four  
 1633  $3 \times 3$  blocks  $A_{ij}$  of  $A$  the impedance  $Y$ , which gives the traction from the  
 1634 displacement, obeys the Riccati equation

$$\frac{dY}{dz} = A_{21} + A_{22}Y - YA_{11} - YA_{12}Y. \quad (C.2)$$

1635 It is started below the stack with the impedance of the downgoing waves  
 1636 of the lower half-space, which is the radiation condition, marched upwards,  
 1637 exactly through the homogeneous gaps and by fourth-order Runge–Kutta  
 1638 with the medium sampled at the stages through the graded layer, and at  
 1639 the top the reflection matrix follows from continuity of displacement and  
 1640 traction with the modes of the upper half-space.  $Y$  stays bounded, so no  
 1641 growing exponential is formed. The second is the propagator: the same  
 1642 state is integrated across the graded layer as a  $6 \times 6$  matrix, with  $A(z) =$   
 1643  $F(z)\text{diag}(i\gamma_m)F(z)^{-1}$  built at each depth from the plane-wave modes of the  
 1644 medium there,  $F$  the matrix of the six modes’ displacements and tractions  
 1645 and  $\gamma_m$  their vertical wavenumbers, and the radiation conditions of the two  
 1646 half-spaces are imposed on the result. For the graded layer the two agree  
 1647 to  $10^{-12}$  in all twelve cases with 400 steps of the march, whose error falls  
 1648 sixteenfold for each doubling of the steps; for a uniform layer the propagator

1649 equals Kennett's recursion to  $3 \times 10^{-14}$ .

1650 *Oblique incidence, and a three-layer background.* At oblique incidence the  
1651 test is the three-dimensional scheme of Eq. (B.1), with P, SV and SH waves,  
1652 in two backgrounds: the two half-spaces above, and a three-layer background  
1653 in which the voxels lie in a layer between a slower half-space above ( $\alpha = 4200$ ,  
1654  $\beta = 2400 \text{ m s}^{-1}$ ,  $\rho = 2300 \text{ kg m}^{-3}$ ) and the faster one below (6500, 3700,  
1655 2800), so that the layer reverberates between two interfaces. The construc-  
1656 tion is the one set out above: the reverberation added to the whole-space  
1657 kernel at every order of the reciprocal lattice, Eq. (C.1), the interface ma-  
1658 trices, the incident field, and the reference, Kennett's recursion [65], against  
1659 which the background's own reflection coefficient is reproduced to  $5 \times 10^{-15}$   
1660 in all twelve cases before any voxel is introduced.

1661 The results are in Table C.15, at  $\omega = 3000 \text{ rad s}^{-1}$  ( $k_S D = 2$ ; at  $300 \text{ rad s}^{-1}$   
1662 the sixth-order errors lie below round-off, as in Table B.14) and  $20^\circ$  in the  
1663 upper half-space. In all thirty-six cases the order is that of the whole space,  
1664 with the interfaces 3m from the layer and with the voxels against them  
1665 (`scripts/measure_oblique_stratified_background.py`). The errors are  
1666 close to those of the whole space at the same frequency (Table B.14): the  
1667 background changes the field the voxels must represent, not how well they  
1668 represent it.

1669 *The two terms of the error at oblique incidence.* The orders say that the  
1670 scheme converges; the two-term analysis says what its error is. Both terms  
1671 are tested at  $20^\circ$ , in the whole space and in the stratified backgrounds, for  
1672 P, SV and SH waves.

Table C.15: The layer of voxels in a stratified background at  $20^\circ$ ,  $\omega = 3000 \text{ rad s}^{-1}$ : relative error of the change of the same-type reflection coefficient on  $n = 4$  planes, and the order over the last halving ( $n = 2 \rightarrow 4$ ), for field degrees  $p = 0, 1$  and  $2$ .

| background                                    | wave | $p = 0$                     | $p = 1$                     | $p = 2$                     |
|-----------------------------------------------|------|-----------------------------|-----------------------------|-----------------------------|
| two half-spaces, interface 3 m below          | P    | $6.1 \times 10^{-3}$ (2.03) | $2.7 \times 10^{-6}$ (4.02) | $2.3 \times 10^{-9}$ (6.04) |
|                                               | SV   | $1.5 \times 10^{-2}$ (2.11) | $2.6 \times 10^{-5}$ (4.13) | $3.7 \times 10^{-8}$ (6.09) |
|                                               | SH   | $1.6 \times 10^{-2}$ (2.11) | $2.7 \times 10^{-5}$ (4.13) | $4.1 \times 10^{-8}$ (6.09) |
| two half-spaces, interface against the voxels | P    | $5.1 \times 10^{-3}$ (2.04) | $3.8 \times 10^{-6}$ (4.04) | $2.1 \times 10^{-9}$ (6.00) |
|                                               | SV   | $1.6 \times 10^{-2}$ (2.10) | $2.5 \times 10^{-5}$ (4.14) | $3.3 \times 10^{-8}$ (6.11) |
|                                               | SH   | $1.7 \times 10^{-2}$ (2.10) | $2.4 \times 10^{-5}$ (4.15) | $3.4 \times 10^{-8}$ (6.11) |
| three layers, interfaces 3 m away             | P    | $6.3 \times 10^{-3}$ (2.03) | $8.7 \times 10^{-7}$ (4.04) | $2.4 \times 10^{-9}$ (6.06) |
|                                               | SV   | $1.0 \times 10^{-2}$ (2.10) | $5.8 \times 10^{-6}$ (4.08) | $5.2 \times 10^{-8}$ (6.11) |
|                                               | SH   | $1.3 \times 10^{-2}$ (2.10) | $3.0 \times 10^{-6}$ (3.95) | $9.3 \times 10^{-8}$ (6.10) |
| three layers, interfaces against the voxels   | P    | $4.8 \times 10^{-3}$ (2.04) | $2.4 \times 10^{-6}$ (4.04) | $4.1 \times 10^{-9}$ (6.03) |
|                                               | SV   | $8.5 \times 10^{-3}$ (2.09) | $7.1 \times 10^{-6}$ (4.01) | $5.0 \times 10^{-8}$ (6.09) |
|                                               | SH   | $1.4 \times 10^{-2}$ (2.10) | $5.8 \times 10^{-6}$ (3.92) | $8.5 \times 10^{-8}$ (6.12) |

1673 *The first-order term.* The scheme is written as  $(\mathbf{M} - \mathbf{K})\mathbf{c} = \mathbf{b}$  with the  
 1674 output  $u = \mathbf{r} \mathbf{c}$ , as in Eq. (G.8), so that  $T_1^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{b}$  with no differencing.  
 1675 The prediction is built from plane waves. The background field in the layer  
 1676 is a sum of six, P, SV and SH going down and going up, and the voxels  
 1677 radiate into six. For a pair made of an incident wave of wavevector  $\mathbf{k}_{\text{in}}$  and  
 1678 a radiated wave of wavevector  $\mathbf{k}_{\text{out}}$ , a cubic voxel of half-side  $h$  whose field is  
 1679 held to total degree  $p$  gives the exact first-order term of that pair times

$$1 + E_p(\mathbf{k}_{\text{in}}, \mathbf{k}_{\text{out}}) = \frac{\sum_{|a| \leq p} \prod_{i=1}^3 (2a_i + 1) j_{a_i}(k_{\text{in},i}h) j_{a_i}(k_{\text{out},i}h)}{\prod_{i=1}^3 j_0((k_{\text{in},i} - k_{\text{out},i})h)}, \quad (\text{C.3})$$

1680 the sum being over the Legendre products  $P_{a_1} P_{a_2} P_{a_3}$  of the voxel's basis and  
 1681  $j_n$  the spherical Bessel functions: the numerator is the projection of the inci-  
 1682 dent wave onto the basis paired with that of the outgoing wave, the denomi-  
 1683 nator the exact pairing. Its leading departure from one is  $-d^2 \mathbf{k}_{\text{in}} \cdot \mathbf{k}_{\text{out}}/12$  for  
 1684  $p = 0$ , with  $d = 2h$ , and along one axis it is the factor of Eq. (39). The pre-  
 1685 dicted first-order term is the sum over the thirty-six pairs of the exact term  
 1686 of each pair, an integral of two exponentials over the voxel, times its factor,  
 1687 the radiated waves then being carried to the observer by the reverberation  
 1688 of Eq. (C.1) exactly. On one and two planes of voxels of degree 0, 1 and  
 1689 2, at  $\omega = 3000 \text{ rad s}^{-1}$ , the scheme's first-order term equals this prediction  
 1690 to  $2 \times 10^{-15}$  in all twelve combinations of background and wave type. The  
 1691 first-order error is therefore known in closed form at oblique incidence, in a  
 1692 stratified background, for every wave type.

1693 *The second-order term.* For the law of Eq. (44) the contrast must vary

1694 within the voxels. Each plane of voxels is given the smooth profile of §9  
 1695 in depth, projected to Legendre degree  $r = p$ , and the coupling is that of  
 1696 Eq. (41) with the reverberation added. The second-order term of the scheme  
 1697 is  $T_2^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{K} \mathbf{M}^{-1} \mathbf{b}$ . The exact  $T_2$  needs the exact stack with a layer  
 1698 whose properties vary continuously in depth, at oblique incidence, for which  
 1699 a recursion over homogeneous layers is not exact. It is computed by two  
 1700 independent algorithms, invariant imbedding [67] and the propagator of the  
 1701 six-component state, which agree to  $10^{-12}$  in all twelve cases (above).  $T_2$  then  
 1702 follows from Eq. (G.9) at  $\epsilon = 0.1$ , where halving  $\epsilon$  changes it by  $2 \times 10^{-8}$ .  
 1703 The ratio of the relative error of  $T_2$  to the relative projection error  $\mathcal{E}_p$  is in  
 1704 Table C.16. On four planes the law holds to 3% for  $p = 0$  and 1 and to 6%  
 1705 for  $p = 2$  at  $k_S D = 0.2$ , and to 0.6% for all three degrees at  $k_S D = 0.067$ , in  
 1706 every background and for every wave type: the departure falls about ninefold  
 1707 as the frequency falls threefold, as  $(kD)^2$ . On two planes the ratios for  $p = 0$   
 1708 and 1 lie within 0.03 of those in the table; for  $p = 2$  they are lower, 0.89 to 0.99  
 1709 at  $k_S D = 0.2$  and 0.988 to 0.999 at  $k_S D = 0.067$ . The law is loosest for the  
 1710 highest degree at the highest frequency, because the projection error of degree  
 1711 2 is so small ( $1.6 \times 10^{-5}$  on four planes) that the departure from the long-wave  
 1712 limit is a larger share of it. Both terms of the error are thus known in advance  
 1713 at oblique incidence and in a stratified background, as they are at normal  
 1714 incidence in the whole space (`scripts/measure_oblique_two_term.py`).

1715 Two limits stand. The oblique construction is implemented once, and  
 1716 its checks are the exact solutions. And the voxels here fill their plane, so  
 1717 that the image of a voxel in an interface it touches is the image of a whole  
 1718 plane; a body of finite lateral extent against an interface, where that image

Table C.16: The relative error of the second-order term over the relative projection error of the medium, at  $20^\circ$ , on  $n = 4$  planes with  $r = p$ ; the law of Eq. (44) predicts one in the long-wave limit.

|                                             |      | $k_S D = 0.2$ |         |         | $k_S D = 0.067$ |         |         |
|---------------------------------------------|------|---------------|---------|---------|-----------------|---------|---------|
| background                                  | wave | $p = 0$       | $p = 1$ | $p = 2$ | $p = 0$         | $p = 1$ | $p = 2$ |
| whole space                                 | P    | 1.000         | 0.998   | 1.011   | 1.000           | 1.000   | 1.000   |
|                                             | SV   | 0.998         | 0.992   | 1.032   | 1.000           | 0.999   | 1.004   |
|                                             | SH   | 0.999         | 0.992   | 1.031   | 1.000           | 0.999   | 1.003   |
| two half-spaces, interface 3 m below        | P    | 1.010         | 1.009   | 1.019   | 1.001           | 1.001   | 1.001   |
|                                             | SV   | 1.012         | 1.008   | 1.046   | 1.001           | 1.001   | 1.005   |
|                                             | SH   | 1.027         | 1.025   | 1.059   | 1.003           | 1.003   | 1.006   |
| three layers, interfaces 3 m away           | P    | 1.004         | 1.003   | 1.013   | 1.000           | 1.000   | 1.000   |
|                                             | SV   | 0.979         | 0.973   | 1.002   | 0.997           | 0.996   | 1.000   |
|                                             | SH   | 1.007         | 1.004   | 1.035   | 1.001           | 1.001   | 1.004   |
| three layers, interfaces against the voxels | P    | 1.002         | 1.001   | 1.011   | 1.000           | 1.000   | 1.000   |
|                                             | SV   | 0.995         | 0.988   | 1.018   | 0.999           | 0.999   | 1.002   |
|                                             | SH   | 1.004         | 1.002   | 1.032   | 1.000           | 1.000   | 1.003   |

1719 is singular, has not been run.

## 1720 **Appendix D. The sphere: near field, an impedance march and a** 1721 **sharp surface**

1722 *The near field.* A receiver is seldom at infinity, and the scheme does not  
1723 require it to be. The scattered field at a point  $\mathbf{x}$  outside the voxels is the  
1724 propagator applied to the source that each voxel carries, the contrast times  
1725 the field found in it:

$$\psi^{\text{sc}}(\mathbf{x}) = \sum_j \int_{V_j} \mathbf{K}(\mathbf{x} - \mathbf{x}') \boldsymbol{\Delta}(\mathbf{x}') e_j(\mathbf{x}') d\mathbf{x}', \quad (\text{D.1})$$

1726 with  $\mathbf{K}$  the nine-component Green's tensor of §2, the same one that couples  
1727 the voxels in the solve,  $\boldsymbol{\Delta}$  the contrast operator of Eq. (5) and  $e_j$  the poly-  
1728 nomial internal state of voxel  $j$ . Its first three rows are the displacement and  
1729 its last six the strain. Nothing is dropped: the terms in  $1/r^2$  and  $1/r^3$  are in  
1730  $\mathbf{K}$ . The integrand's source is a polynomial in each voxel and is integrated by  
1731 a tensor Gauss rule with six nodes a side; the rule's own error is measured  
1732 by repeating the readout with four nodes, and from 8 voxels across the two  
1733 differ by less than  $4 \times 10^{-6}$  at  $1.25a$  and by less than  $10^{-10}$  at  $2a$ . The ob-  
1734 servers are the nine angles of Fig. 6(b) on spheres of radius  $1.25a$ ,  $2a$  and  
1735  $10a$ , where  $k_S r$  runs from 0.6 to 10; the reference is the exact partial-wave  
1736 displacement at the same points; and the error is the measure of the far-field  
1737 test. Table D.17 gives it for the mean-only voxel and for the first-moment  
1738 voxel (Appendix Appendix E). The orders of the far field hold in the near  
1739 field: second for the mean-only voxel at every distance and both frequencies,  
1740 and 3.8 to 3.95 for the first-moment voxel, still rising towards four as in the

Table D.17: The graded sphere read in the near field: relative error of the scattered displacement at 1.25, 2 and 10 radii from the centre, against the exact displacement at the same points, and the apparent order between the two finest grids.

| $n_{\text{sub}}$           | mean only            |                      |                      | first moment         |                      |                      |
|----------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                            | 1.25a                | 2a                   | 10a                  | 1.25a                | 2a                   | 10a                  |
| $k_S a = 0.5$              |                      |                      |                      |                      |                      |                      |
| 4                          | $3.3 \times 10^{-2}$ | $1.2 \times 10^{-2}$ | $5.3 \times 10^{-3}$ | $1.4 \times 10^{-2}$ | $1.6 \times 10^{-3}$ | $6.3 \times 10^{-4}$ |
| 6                          | $1.4 \times 10^{-2}$ | $5.2 \times 10^{-3}$ | $1.9 \times 10^{-3}$ | $1.2 \times 10^{-3}$ | $3.8 \times 10^{-4}$ | $1.9 \times 10^{-4}$ |
| 8                          | $8.0 \times 10^{-3}$ | $2.9 \times 10^{-3}$ | $1.1 \times 10^{-3}$ | $4.7 \times 10^{-4}$ | $1.4 \times 10^{-4}$ | $7.3 \times 10^{-5}$ |
| 12                         | $3.5 \times 10^{-3}$ | $1.2 \times 10^{-3}$ | $4.7 \times 10^{-4}$ | $7.8 \times 10^{-5}$ | $3.1 \times 10^{-5}$ | $1.7 \times 10^{-5}$ |
| 16                         | $2.0 \times 10^{-3}$ | $6.7 \times 10^{-4}$ | $2.6 \times 10^{-4}$ | $2.5 \times 10^{-5}$ | $1.0 \times 10^{-5}$ | $5.7 \times 10^{-6}$ |
| order, 12 $\rightarrow$ 16 | 2.03                 | 2.06                 | 2.02                 | 3.95                 | 3.85                 | 3.77                 |
| $k_S a = 1.0$              |                      |                      |                      |                      |                      |                      |
| 4                          | $4.8 \times 10^{-2}$ | $3.5 \times 10^{-2}$ | $3.3 \times 10^{-2}$ | $9.9 \times 10^{-3}$ | $1.3 \times 10^{-3}$ | $6.0 \times 10^{-4}$ |
| 6                          | $2.0 \times 10^{-2}$ | $1.4 \times 10^{-2}$ | $1.4 \times 10^{-2}$ | $9.3 \times 10^{-4}$ | $3.0 \times 10^{-4}$ | $2.0 \times 10^{-4}$ |
| 8                          | $1.2 \times 10^{-2}$ | $8.1 \times 10^{-3}$ | $7.8 \times 10^{-3}$ | $3.5 \times 10^{-4}$ | $1.1 \times 10^{-4}$ | $7.4 \times 10^{-5}$ |
| 12                         | $5.2 \times 10^{-3}$ | $3.6 \times 10^{-3}$ | $3.5 \times 10^{-3}$ | $6.0 \times 10^{-5}$ | $2.5 \times 10^{-5}$ | $1.7 \times 10^{-5}$ |
| 16                         | $2.9 \times 10^{-3}$ | $2.0 \times 10^{-3}$ | $1.9 \times 10^{-3}$ | $1.9 \times 10^{-5}$ | $8.3 \times 10^{-6}$ | $5.7 \times 10^{-6}$ |
| order, 12 $\rightarrow$ 16 | 2.01                 | 2.01                 | 2.00                 | 3.95                 | 3.84                 | 3.78                 |

1741 far field at this contrast. A quarter of a radius from the surface, with 16 vox-  
 1742 els across, the first-moment voxel's error is  $2.5 \times 10^{-5}$  and  $1.9 \times 10^{-5}$ , 78 and  
 1743 153 times below the mean-only voxel's. The error grows towards the body,  
 1744 by a factor of 1.5 to 7 between  $10a$  and  $1.25a$ , because the near field carries  
 1745 the higher multipoles of the source, which a given grid resolves less well; the  
 1746 order does not change. On the coarsest grid a voxel is as wide as the graded  
 1747 shell and twice as wide as the gap to the nearest observer, and the sequence  
 1748 is not yet asymptotic. Read at these distances with the asymptotic formula,  
 1749 the same solutions would be in error by 48% to 85%: the far-field readout  
 1750 is for the far field (`scripts/measure_graded_sphere_near_field.py`; the  
 1751 readout is `graded_field` of the package, tested against the propagator of a  
 1752 point source, against the gradient of its own displacement, and against the  
 1753 far-field readout at great distance).

1754 *The voxels against an impedance march.* A second route to the same body  
 1755 is invariant imbedding. The impedance of §9, Eq. (C.2), is marched up-  
 1756 wards through the depth range of the sphere with the lateral variation of the  
 1757 medium carried inside the march, on a laterally periodic grid, and a second  
 1758 sweep carries the field down [67]; the march returns the reflected plane waves  
 1759 on a plane above the body and the transmitted ones on a plane below. It is  
 1760 run here on the graded sphere and set beside the voxels, to ask how many  
 1761 voxels a given accuracy costs against an established alternative. Four things  
 1762 are fixed so that the comparison is of like with like.

- 1763 1. *The planes.* Both methods are read on the planes  $z = -1.25a$  and  
 1764  $z = +1.25a$ . On the tangent planes  $z = \pm a$  themselves the observers lie  
 1765 on the faces of the voxels at the poles, where the Gauss rule of the voxel

Table D.18: The smoothly graded sphere, voxelised with the mean-only voxel on grids of  $n_{\text{sub}}$  voxels across, against its exact solution: relative far-field error, and the apparent order between successive grids. Nine unknowns per cell, every cell overlapping the sphere kept.

| $n_{\text{sub}}$ | unknowns | $k_{\text{sa}} = 0.5$ |       | $k_{\text{sa}} = 1.0$ |       |
|------------------|----------|-----------------------|-------|-----------------------|-------|
|                  |          | error                 | order | error                 | order |
| 4                | 576      | $7.70 \times 10^{-3}$ |       | $3.24 \times 10^{-2}$ |       |
| 6                | 1 872    | $3.04 \times 10^{-3}$ | 2.30  | $1.32 \times 10^{-2}$ | 2.22  |
| 8                | 3 672    | $1.68 \times 10^{-3}$ | 2.05  | $7.52 \times 10^{-3}$ | 1.95  |
| 12               | 11 520   | $7.30 \times 10^{-4}$ | 2.06  | $3.34 \times 10^{-3}$ | 2.00  |
| 16               | 24 984   | $4.06 \times 10^{-4}$ | 2.04  | $1.88 \times 10^{-3}$ | 2.00  |
| 20               | 47 304   | $2.58 \times 10^{-4}$ | 2.03  | $1.20 \times 10^{-3}$ | 2.00  |
| 24               | 79 344   | $1.79 \times 10^{-4}$ | 2.02  | $8.34 \times 10^{-4}$ | 2.00  |

1766 readout (Eq. (D.1)) cannot integrate the singular kernel: its change  
1767 from four to six nodes is 2% to 10% there, more than the scheme's error,  
1768 and below  $10^{-4}$  a quarter of a radius away. The march's amplitudes are  
1769 carried across the same homogeneous gap by each order's own phase,  
1770 which is exact.

1771 2. *The periodic array.* The march needs a laterally periodic grid, so it  
1772 solves a square array of spheres. Its answer is scored against the exact  
1773 solution of one sphere summed over the array, by Poisson summation,  
1774 which is exact but for scattering between the spheres. That part, and  
1775 the lateral resolution, are errors of the method as used on a single body,  
1776 and are counted.

1777 3. *The error.* For the march, the norm of the difference over the plane-

1778 wave orders of the grid, relative to the norm of the exact scattered  
1779 orders; for the voxels, the norm of the difference of the scattered dis-  
1780 placement over a patch of side  $5a$  on each plane, relative to the norm  
1781 of the exact one. The two are the same quantity in wavenumber and  
1782 in space, for the array and for the single body.

1783 4. *The refinement.* For the march, the lateral grid, with the depth step  
1784 refined until the error no longer changes; for the voxels, the grid and  
1785 the degree.

Table D.19: The graded sphere at  $k_S a = 0.5$  read on the planes  $z = \mp 1.25a$ : relative error in reflection and in transmission. The march's unknowns are those of one plane, on which it carries a dense matrix; the period is  $5a$  unless stated.

| method              | setting                        | unknowns | reflection           | transmission         |
|---------------------|--------------------------------|----------|----------------------|----------------------|
| impedance march     | $12 \times 12$ , 16 steps      | 432      | $1.2 \times 10^{-2}$ | $9.9 \times 10^{-3}$ |
|                     | $16 \times 16$ , 32 steps      | 768      | $4.3 \times 10^{-3}$ | $3.6 \times 10^{-3}$ |
|                     | $20 \times 20$ , 32 steps      | 1200     | $7.6 \times 10^{-4}$ | $7.3 \times 10^{-4}$ |
|                     | $28 \times 28$ , 48 steps      | 2352     | $2.5 \times 10^{-4}$ | $5.6 \times 10^{-4}$ |
|                     | $30 \times 30$ , period $7.5a$ | 2700     | $7.2 \times 10^{-4}$ | $4.7 \times 10^{-4}$ |
| mean-only voxel     | 8 across                       | 3672     | $5.0 \times 10^{-3}$ | $7.9 \times 10^{-3}$ |
|                     | 16 across                      | 24 984   | $1.2 \times 10^{-3}$ | $2.0 \times 10^{-3}$ |
| first-moment voxel  | 6 across                       | 7488     | $3.3 \times 10^{-4}$ | $5.2 \times 10^{-4}$ |
|                     | 8 across                       | 14 688   | $1.4 \times 10^{-4}$ | $2.2 \times 10^{-4}$ |
|                     | 16 across                      | 99 936   | $1.1 \times 10^{-5}$ | $1.8 \times 10^{-5}$ |
| second-moment voxel | 8 across                       | 36 720   | $1.4 \times 10^{-5}$ | $2.3 \times 10^{-5}$ |
|                     | 10 across                      | 69 120   | $2.6 \times 10^{-6}$ | $4.2 \times 10^{-6}$ |

1786 Table D.19 gives the result. The march converges quickly in the lateral grid,

1787 because the body is smooth, and with far fewer unknowns than any voxel  
 1788 scheme: at  $20 \times 20$  it is at  $7 \times 10^{-4}$  with 1200 unknowns on a plane. The  
 1789 mean-only voxel does not reach that with sixteen voxels across and twenty  
 1790 times the unknowns; on a sharp sphere, where it also carries a staircase, it  
 1791 fares worse. The first-moment voxel reaches the march's accuracy with six  
 1792 voxels across, and below about  $2 \times 10^{-4}$  it continues at close to fourth order  
 1793 where the march does not: in transmission the march's error at a period  
 1794 of  $5a$  stops near  $5 \times 10^{-4}$ , which is the scattering between the spheres of  
 1795 its array, since lengthening the period lowers it and refining the grid does  
 1796 not. The costs are of different kinds. The march's unknowns are few but its  
 1797 operator is a dense matrix on the plane, marched through the depth, and a  
 1798 finer lateral grid requires a smaller depth step (with eight steps the  $20 \times 20$   
 1799 march diverged); the voxel schemes have many unknowns and a product done  
 1800 by fast Fourier transform. In the timings of these runs, taken on one machine  
 1801 with other work in progress and therefore only indicative, the first-moment  
 1802 voxel at six across was solved in about a sixth of the time of the  $20 \times 20$  march.  
 1803 The march has what the comparison does not use: it returns the reflection  
 1804 and transmission for every incident plane wave of the grid at once, and a  
 1805 laterally periodic or laterally extended medium is its natural problem and  
 1806 not a cost. The comparison is at one frequency (`scripts/measure_graded_`  
 1807 `sphere_march.py`, `scripts/measure_graded_sphere_planes.py`).

1808 *A sphere with a sharp surface.* The homogeneous sphere is the familiar test,  
 1809 and its voxelisation changes with every grid. Table D.20 shows its error for  
 1810 grids of 4, 6 and 8 voxels across: 4.6%, then 19%, then 4.2% at  $k_S a = 0.5$ .  
 1811 The error tracks the staircase's volume, which is 5%, 17% and 4% too large

1812 on these grids. Rescaled by the ratio of the sphere’s volume to the staircase’s,  
1813 it falls below 1% at  $k_S a = 0.5$  and to at most 3% at  $k_S a = 1.0$ , but it still  
1814 does not converge: what remains is the staircase’s shape. A refinement study  
1815 of a sharp sphere measures its staircase; the smooth sphere, like the layer,  
1816 measures the scheme.

Table D.20: A homogeneous sphere voxelised on grids of  $n_{\text{sub}}$  voxels across, against exact elastic scattering: relative far-field error, raw and rescaled by the ratio of the sphere’s volume to the staircase’s. It does not converge with refinement, because the staircase changes with the grid.

| $n_{\text{sub}}$ | voxels | $k_S a = 0.5$         |                      | $k_S a = 1.0$         |                      |
|------------------|--------|-----------------------|----------------------|-----------------------|----------------------|
|                  |        | raw                   | volume-rescaled      | raw                   | volume-rescaled      |
| 4                | 32     | $4.6 \times 10^{-2}$  | $1.4 \times 10^{-3}$ | $5.0 \times 10^{-2}$  | $5.4 \times 10^{-3}$ |
| 6                | 136    | $1.94 \times 10^{-1}$ | $7.9 \times 10^{-3}$ | $1.68 \times 10^{-1}$ | $3.0 \times 10^{-2}$ |
| 8                | 280    | $4.2 \times 10^{-2}$  | $2.3 \times 10^{-3}$ | $3.5 \times 10^{-2}$  | $8.9 \times 10^{-3}$ |

## 1817 Appendix E. The three-dimensional voxel solver

1818 The three-dimensional results of §§10 and 11 use the Legendre voxels  
1819 of §8 on a cubic grid, with the following construction (`cubic_scattering/`  
1820 `graded_voxel`).

1821 *The voxel.* Each cell carries the nine-component state of §2, displacement  
1822 and engineering strain, with the strain an *independent* unknown and not the  
1823 gradient of the displacement: a strain derived from a cell’s projected dis-  
1824 placement jumps across the faces between cells and leaves a spurious source  
1825 there (§8). The state is expanded in the Legendre products of degree at

1826 most  $p$  in the cell's local coordinates  $\boldsymbol{\xi} \in [-1, 1]^3$ , and the contrast opera-  
 1827 tor  $\boldsymbol{\Delta}$  in those of degree at most  $r$ , its coefficients the  $L^2$  projection of the  
 1828 medium;  $p = r = 1$  gives the first-moment voxel of 36 weights. The source  
 1829  $\boldsymbol{\Delta}\psi$  is then a polynomial of degree  $p + r$ , expanded in monomials  $m_c$ . The  
 1830 Lippmann–Schwinger equation is tested with each basis function over each  
 1831 cell (Galerkin), and cells  $m$  and  $n$  couple through the blocks

$$\mathbf{K}_{ac}(\mathbf{R}) = \int_{V_m} \int_{V_n} L_a(\boldsymbol{\xi}) \mathbf{P}(\mathbf{x} - \mathbf{x}') m_c(\boldsymbol{\xi}') d^3x d^3x', \quad \mathbf{R} = \mathbf{x}_m - \mathbf{x}_n, \quad (\text{E.1})$$

1832 with  $\mathbf{P}$  the  $9 \times 9$  point propagator of §2.

1833 *One integral in place of two.* With  $\mathbf{s} = \mathbf{u} - \mathbf{u}'$  the difference of the two  
 1834 local positions, Eq. (E.1) becomes  $\mathbf{K}_{ac}(\mathbf{R}) = \int W_{ac}(\mathbf{s}) \mathbf{P}(\mathbf{R} + \mathbf{s}) d^3s$  over  
 1835  $\mathbf{s} \in [-2h, 2h]^3$ , where  $W_{ac}$  is a product over the three axes of one-dimensional  
 1836 cross-correlations of the test and source polynomials: continuous, polynomial  
 1837 on  $[-2h, 0]$  and  $[0, 2h]$ , with kinks at 0 and  $\pm 2h$ . For cells that do not touch,  
 1838 the kernel is smooth on each of the eight pieces, and a three-dimensional  
 1839 Gauss rule integrates it, with 14, 10 or 8 points per axis at a Chebyshev  
 1840 distance of 2, 3 or 4, and 5 or more cells; this agrees with six-dimensional  
 1841 quadrature of Eq. (E.1) to  $10^{-9}$  one cell beyond contact.

1842 *The self cell and its 26 neighbours.* The static part of  $\mathbf{P}$  is a sum of derivatives  
 1843 of  $r^{-1}$  and  $r$ , distributions at the origin. Derivatives are moved onto  $W$  until  
 1844 the kernel is at most as singular as  $1/r$ , and the kinks of  $W$  then carry  
 1845 plane  $\delta$  functions; every piece is a box or a rectangle with its corners on the  
 1846 breakpoints, so the singular point is either a vertex of a piece, integrated  
 1847 with Duffy's pyramids, or at least  $2h$  away. The  $\delta$  function of the inclusion  
 1848 problem is carried by those plane  $\delta$  functions and is not added by hand.

1849 The construction reproduces the sum rule  $\sum_p \int_V \int_V \partial_p \partial_p |\mathbf{x} - \mathbf{x}'|^{-1} d^3x d^3x' =$   
1850  $-4\pi V$ , the biharmonic rule for  $r$  and the cube's Coulomb self-energy to  $10^{-11}$ ,  
1851 and every block equals the sum of the 64 blocks between the two cells' half-  
1852 size sub-cells, with the polynomials re-expressed on them, to  $2 \times 10^{-15}$ . The  
1853 dynamic remainder of  $\mathbf{P}$  is at most weakly singular and is integrated by the  
1854 Gauss rule of the pieces.

1855 *Symmetry, and the solve.* A block depends only on the offset  $\mathbf{o}$  between the  
1856 two cells. The background is isotropic and the cube is mapped to itself  
1857 by its 48 signed permutations  $\mathbf{Q}$ , under which  $\mathbf{P}(\mathbf{Q}\mathbf{r}) = \mathbf{S}\mathbf{P}(\mathbf{r})\mathbf{S}^\top$ , with  
1858  $\mathbf{S} = \text{diag}(\mathbf{Q}, \mathbf{R}_6(\mathbf{Q}))$  and  $\mathbf{R}_6$  the representation of  $\mathbf{e} \mapsto \mathbf{Q}\mathbf{e}\mathbf{Q}^\top$  on Voigt  
1859 vectors. With the test and source polynomials transforming by matrices  $\mathbf{T}$   
1860 and  $\mathbf{U}$ ,  $\mathbf{K}_{ac}(\mathbf{Q}\mathbf{o}) = \sum_{b,d} T_{ab} U_{cd} \mathbf{S} \mathbf{K}_{bd}(\mathbf{o}) \mathbf{S}^\top$ , so one block is computed for each  
1861 orbit of offsets and the rest follow exactly. The coupling is then a convolution  
1862 over the grid, applied by the fast Fourier transform on a  $(2n-1)^3$  circular  
1863 embedding, and the system is solved by GMRES preconditioned with each  
1864 cell's own block. Where both can be run, the solution equals that of a dense  
1865 solve to  $10^{-10}$  relative (`tests/test_graded_voxel_fft.py`). Every cell that  
1866 overlaps the body is kept, each with the  $L^2$  projection of its contrast by a  
1867 six-point Gauss rule per axis; keeping only the cells whose centre lies inside  
1868 the body drops the contrast of the partly filled cells, and the mean-only  
1869 voxel then converges at order one instead of two. The far field is read at  
1870  $5 \times 10^8$  radii, where the near-field terms that the asymptotic form omits are  
1871 negligible.

1872 *The first-moment voxel on the graded sphere.* The voxel with  $p = r = 1$  is  
1873 applied to the graded sphere of §10. To isolate the discretisation error from

1874 the rescattering the grid cannot yet resolve, the contrast is first scaled by  
 1875  $10^{-6}$ , and the exact sphere is solved at the same scaled contrast, so that no  
 1876 Born approximation enters the reference: the relative error is then that of  
 1877 the scheme's first-order term, in which the field is the incident one carried  
 1878 across each cell. With 4, 6 and 8 voxels across, 2304 to 14688 unknowns, the  
 1879 error is  $7.5 \times 10^{-5}$ ,  $5.0 \times 10^{-6}$  and  $1.6 \times 10^{-6}$  at  $k_S a = 0.5$ , and  $2.3 \times 10^{-4}$ ,  
 1880  $9.1 \times 10^{-5}$  and  $2.8 \times 10^{-5}$  at  $k_S a = 1$ . The apparent orders between 6 and  
 1881 8 voxels are 4.01 and 4.13; between 4 and 6 they are not yet asymptotic, a  
 1882 voxel at  $n = 4$  being as wide as the graded shell (`scripts/pilot_graded_`  
 1883 `voxel_sphere.py--weak`). At the full contrast the fast solve is carried to 16  
 1884 voxels across, where the error is  $4.9 \times 10^{-6}$  at  $k_S a = 0.5$  and  $5.9 \times 10^{-6}$  at  
 1885  $k_S a = 1$ , and the apparent order approaches four at a rate set by the number  
 1886 of voxels across the graded shell.

## 1887 **Appendix F. The uniform-strain closure against the exact sphere**

1888 *Against the exact sphere.* These statements are tested against the exact elas-  
 1889 tic Mie solution for a sphere, with no Eshelby formula in between. The  
 1890 concentration of Mie order  $n$  is  $E_n = a_n/a_n^{\text{Born}}$  (orders 0, 1, 2 are the dilata-  
 1891 tional, density and shear channels), extrapolated to  $ka \rightarrow 0$ . The formulation  
 1892 is applied to a ball of radius  $a$ , with its moment  $\mathbf{M}$  evaluated by the surface  
 1893 route of Eq. (26) on  $|\mathbf{r}| = a$ . Its dilatational and shear channels equal Mie's  
 1894 to  $2 \times 10^{-10}$ , and the density channel is 1. The cube's closed forms meet  
 1895 the same numbers. Its dilatational channel is Mie's  $E_0$  to  $2 \times 10^{-10}$ , and its  
 1896 isotropic shear average, Eq. (32), is the depolarisation read from Mie's  $E_2$   
 1897 to  $10^{-8}$ . The same moment taken as a principal value, over the ball with

1898 a small sphere excluded around the origin, has zero depolarisation in every  
 1899 channel. The whole static response of the sphere is the  $\delta$  term.

1900 At finite frequency the closure uses the dynamic ball moments, which are  
 1901 in closed form. It departs from Mie at second order in  $ka$  and first order in  
 1902 the contrast, and the departure is closed-form too: the ball's moments are  
 1903 elementary, and Mie's low-frequency series with the contrast kept exact gives  
 1904  $E_n$  to relative order  $(ka)^3$ . To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta\lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta\rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta\mu}{\mu}, & n = 2, \end{cases} \quad (\text{F.1})$$

1905 with no  $O(ka)$  term and no  $O((ka)^3)$  term. The imaginary  $(ka)^3$  terms,  
 1906 the radiation damping, agree exactly in all three channels. So the closure's  
 1907 only error at this order is the real  $(ka)^2$  term, the uniform-field ansatz's  
 1908 own. That is the order the hierarchy assigns to the omitted second gradi-  
 1909 ents (§6.3), and it has the same origin, the variation of the field across the  
 1910 scatterer, as the collocation error of §7. Eq. (F.1) is derived symbolically  
 1911 (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from  
 1912 the numerical Mie solve and an independent implementation of the closure  
 1913 to  $5 \times 10^{-4}$ , which is the size of the next order in the contrast.

## 1914 **Appendix G. The scattering series of the layer, and the projection-** 1915 **error law**

1916 *The scattering series.* Multiply the whole contrast,  $\Delta\lambda$ ,  $\Delta\mu$  and  $\Delta\rho$   
 1917 together, by a number  $s$ , and let  $u(s)$  be the scattered displacement at an

1918 observer at depth  $z_o$  outside the layer. It vanishes at  $s = 0$  and is analytic  
 1919 in  $s$  there, and its expansion is the Born series,

$$u(s) = s T_1 + s^2 T_2 + s^3 T_3 + \dots, \quad (\text{G.1})$$

1920 so that  $T_n$  is the part of the field at the full contrast,  $s = 1$ , that has been  
 1921 scattered  $n$  times.

1922 *The first two terms of the exact layer.* They are integrals over the layer.  
 1923 With the incident state  $w^0 = (u^0, \varepsilon^0)$ , the contrast  $\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$   
 1924 with profile  $f(z)$ , and the kernel  $\mathbf{K}$  of Eq. (38), whose first row  $(g, g')(z_o - z)$   
 1925 gives the displacement at the observer,

$$T_1 = \int_0^D (g, g')(z_o - z) \Delta_1 f(z) w^0(z) dz, \quad (\text{G.2})$$

1926

$$w^1(z) = \int_0^D \mathbf{K}(z - z') \Delta_1 f(z') w^0(z') dz', \quad T_2 = \int_0^D (g, g')(z_o - z) \Delta_1 f(z) w^1(z) dz. \quad (\text{G.3})$$

1927 Here  $w^1$  is the state scattered once, inside the layer; its strain includes the  
 1928 local term of the kernel,  $-f(z) \Delta M \varepsilon^0(z)/M_P$ . Each further term repeats  
 1929 the step,  $w^n(z) = \int_0^D \mathbf{K}(z - z') \Delta_1 f(z') w^{n-1}(z') dz'$ . For a plane force at  $z_s$   
 1930 above the layer,  $w^0(z) = c e^{ik(z-z_s)}(1, ik)$  with  $c = i/(2M_P k)$ , every integrand  
 1931 is an exponential times the profile. Writing  $a = \omega^2 \Delta \rho$  and  $b = \Delta M$ , and  
 1932 carrying out the products of Eqs. (G.2) and (G.3), the terms seen in reflection  
 1933 ( $z_o < 0$ ) and in transmission ( $z_o > D$ ) are, for any profile,

$$T_1^R = c^2 e^{-ik(z_o + z_s)} (a + bk^2) \int_0^D f(z) e^{2ikz} dz, \quad T_1^T = c^2 e^{ik(z_o - z_s)} (a - bk^2) \int_0^D f(z) dz, \quad (\text{G.4})$$

1934

$$\begin{aligned}
T_2^R &= c^2 e^{-ik(z_o+z_s)} \left[ c(a^2 - b^2 k^4) \int_0^D \int_0^D f(z) f(z') e^{ik(z+z'+|z-z'|)} dz' dz - \frac{b^2 k^2}{M_P} \int_0^D f(z)^2 e^{2ikz} dz \right], \\
T_2^T &= c^2 e^{ik(z_o-z_s)} \left[ c(a - bk^2)^2 \int_0^D \int_0^z f(z) f(z') dz' dz + c(a + bk^2)^2 \int_0^D \int_z^D f(z) f(z') e^{2ik(z'-z)} dz' dz \right. \\
&\quad \left. + \frac{b^2 k^2}{M_P} \int_0^D f(z)^2 dz \right].
\end{aligned} \tag{G.5}$$

1935 The first-order reflection is the Fourier transform of the profile at the wavenum-  
1936 ber  $2k$ , and the first-order transmission its integral. In each second-order  
1937 term the last integral is the local term of the kernel, in which both scat-  
1938 terings are at one point and the profile appears squared; the others are the  
1939 two scatterings at different depths. In transmission the wave scattered twice  
1940 forwards carries the contrast of velocity twice,  $(a - bk^2)^2$ , and the wave scat-  
1941 tered backwards and then forwards again the contrast of impedance twice,  
1942  $(a + bk^2)^2$ . For the uniform layer ( $f = 1$ ) the integrals give

$$T_1^R = c^2 e^{-ik(z_o+z_s)} (a + bk^2) \frac{e^{2ikD} - 1}{2ik}, \quad T_1^T = c^2 e^{ik(z_o-z_s)} (a - bk^2) D, \tag{G.6}$$

1943

$$\begin{aligned}
T_2^R &= \frac{c^2 e^{-ik(z_o+z_s)}}{4k^3 M_P} \left[ -i(a^2 + b^2 k^4) + e^{2ikD} (a^2(i + 2kD) + b^2 k^4(i - 2kD)) \right], \\
T_2^T &= \frac{i c^2 e^{ik(z_o-z_s)}}{8k^3 M_P} \left[ a^2(1 - e^{2ikD} + 2kD(i + kD)) - 2abk^2(e^{2ikD} - 1 + 2kD(kD - i)) \right. \\
&\quad \left. + b^2 k^4(1 - e^{2ikD} + 2kD(kD - 3i)) \right].
\end{aligned} \tag{G.7}$$

1944 The reflection sees the contrast of impedance,  $a + bk^2 = k^2 M_P(\Delta\rho/\rho +$   
1945  $\Delta M/M_P)$ , and the transmission that of velocity,  $a - bk^2$ . As  $kD \rightarrow 0$ ,  $T_2^R \rightarrow$   
1946  $-c^2 e^{-ik(z_o+z_s)} b^2 k^2 D/M_P$ : the local term alone, which is the starting point of  
1947 the argument at the end of this appendix. For the smooth profile the integrals

1948 of Eqs. (G.4) and (G.5) are evaluated in closed form symbolically, and by  
 1949 quadrature they return the same values to  $10^{-16}$ . Equations (G.6) and (G.7)  
 1950 equal, identically, the first two Taylor coefficients in  $s$  of the exact reflection  
 1951 and transmission of the layer of §3 (`ContinuumLimit_BornTerms.wl`). At  
 1952  $\omega = 300 \text{ rad s}^{-1}$ ,  $|T_2|/|T_1|$  is 0.039 in reflection and 0.17 in transmission for  
 1953 the uniform layer.

1954 *The first two terms of the scheme.* Write Eq. (38) as  $(\mathbf{M} - \mathbf{K}) \mathbf{c} = \mathbf{b}$ , with  
 1955  $\mathbf{M}$  the diagonal matrix of the  $\int_i \phi_a^2 dz$ ,  $\mathbf{K}$  the matrix of the coupling blocks  
 1956 times  $\Delta_1$ , and  $\mathbf{b}$  the moments of the incident field, and let  $\mathbf{r}$  be the row that  
 1957 gives the scattered displacement at the observer from the coefficients,  $u = \mathbf{r} \mathbf{c}$ .  
 1958 Scaling the contrast by  $s$  scales  $\mathbf{K}$  and  $\mathbf{r}$  by  $s$ , so  $u(s) = s \mathbf{r} (\mathbf{M} - s \mathbf{K})^{-1} \mathbf{b}$  and

$$T_1^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{b}, \quad T_2^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{K} \mathbf{M}^{-1} \mathbf{b}. \quad (\text{G.8})$$

1959  $\mathbf{M}^{-1} \mathbf{b}$  is the projection of the incident field onto each cell's functions: the first  
 1960 term needs no solve. Splitting  $\mathbf{K}$  into its blocks within a cell and its blocks  
 1961 between cells splits  $T_2^{\text{scheme}}$  into the part scattered twice in one cell and the  
 1962 part scattered in two; the same split of the inner integral of Eq. (G.3) splits  
 1963 the exact  $T_2$ .

1964 *A check by differences.* The terms can also be read from solutions at four  
 1965 scaled contrasts,  $s = \pm\epsilon$  and  $\pm 2\epsilon$ , with no knowledge of the formulas above.  
 1966 The odd and even parts of  $u$  in  $s$  separate the odd orders from the even,  
 1967 and the two step sizes remove the next term of each (the five-point central  
 1968 differences for  $u'(0)$  and  $\frac{1}{2}u''(0)$ ):

$$\begin{aligned} T_1 &= \frac{8[u(\epsilon) - u(-\epsilon)] - [u(2\epsilon) - u(-2\epsilon)]}{12\epsilon} + 4\epsilon^4 T_5 + \dots, \\ T_2 &= \frac{16[u(\epsilon) + u(-\epsilon)] - [u(2\epsilon) + u(-2\epsilon)]}{24\epsilon^2} + 4\epsilon^4 T_6 + \dots \end{aligned} \quad (\text{G.9})$$

1969 A negative  $s$  is a medium slower and lighter than the background by the  
 1970 same fractions, as legitimate a medium as the other. With  $\epsilon = 0.01$  the  
 1971 remainders  $4\epsilon^4 T_5/T_1$  and  $4\epsilon^4 T_6/T_2$  are below  $2 \times 10^{-12}$  for this layer, and the  
 1972 formulas applied to the exact layer return Eqs. (G.6) and (G.7) less exactly  
 1973 those remainders, to  $10^{-17}$ . The measurements of §9 use the differences for  
 1974 the scheme and the exact layer alike, each in its own arithmetic, and they  
 1975 agree with Eqs. (G.2), (G.3) and (G.8).

1976 *Why the terms must be separated.* The law of Eq. (44) is a statement  
 1977 about  $T_2$ . At the full contrast the terms of third and higher order are 7% of  
 1978 the second and carry errors of their own, and if every order above the first  
 1979 is taken together the ratio of Eq. (44) reads 0.90 to 0.94 where the second-  
 1980 order term alone gives 1.00 to 1.01. The shortfall is itself the single site's. In  
 1981 the long-wave limit the kernel is local, the exact strain is  $\varepsilon_0/(1 - cf)$  with  
 1982  $c = -\Delta M/M_P$ , and the scheme's is the Galerkin solution of  $\varepsilon = \varepsilon_0 + cf\varepsilon$  in  
 1983 each cell on that cell's polynomials; the relative error of the term of order  
 1984  $n$  is  $(\int f^n dz - \int (\Pi f)^n dz)/\int f^n dz$ , 2.455 times  $\mathcal{E}_q$  for the third term of  
 1985 this profile, which has the opposite sign to the second. At  $\omega = 30 \text{ rad s}^{-1}$   
 1986 the whole nonlinear part has 0.896 times  $\mathcal{E}_q$ , and the cell-by-cell Galerkin  
 1987 solution reproduces it to four digits with  $c$  from the contrast (`scripts/`  
 1988 `measure_layer_t3_low.py`).

1989 Equation (44) is exact in the long-wave limit, for a reason that shows what  
 1990 each basis is for. The strain-strain entry of the kernel,  $g'' = -k^2 g - \delta(z)/M_P$ ,  
 1991 is local as  $kD \rightarrow 0$ , and the incident strain  $\varepsilon_0$  is then uniform across the layer,  
 1992 so the strain scattered once is  $-(\Delta M/M_P) f(z) \varepsilon_0$ : it is the profile itself. A  
 1993 cell holds the contrast as  $\Pi_r f$  and the field to degree  $p$ , so it holds the once-

1994 scattered strain as  $\Pi_p(\Pi_r f) = \Pi_q f$  with  $q = \min(p, r)$ . The second-order  
 1995 term pairs the contrast with that strain:

$$T_2 \propto \int f \cdot f \, dz = \|f\|^2, \quad T_2^{\text{scheme}} \propto \int (\Pi_r f)(\Pi_q f) \, dz = \|\Pi_q f\|^2, \quad (\text{G.10})$$

1996 the last because  $\Pi_q f$  lies in the range of  $\Pi_r$ . The relative error is  $1 -$   
 1997  $\|\Pi_q f\|^2 / \|f\|^2 = \mathcal{E}_q$ . Because the kernel is local in this limit, the second  
 1998 scattering takes place in the cell of the first:  $T_2$  is the second-order term of  
 1999 the single-site  $T$ -matrix, and Eq. (44) is a statement about the single site.  
 2000 The  $T$ -matrix sums the rescattering within a cell exactly for the cell as it  
 2001 is represented, a contrast  $\Pi_r f$  and a field of degree  $p$ ; its error is that the  
 2002 cell is not the medium. The split is measured directly (`scripts/measure_`  
 2003 `layer_bases.py`): at  $\omega = 300 \text{ rad s}^{-1}$  on 4, 8 and 16 cells,  $T_2$  of the scheme  
 2004 and of the exact layer are each divided into the part whose two scatterings  
 2005 lie in one cell and the part between different cells. For all nine pairs the  
 2006 error of the same-cell part is 0.998 to 1.013 times  $|T_2| \mathcal{E}_{\min(p, r)}$ , and that of  
 2007 the part between cells at most 0.002 times it. The coupling between cells  
 2008 contributes only the departure from the limit. Measured, the ratio of the  
 2009 two sides of Eq. (44) for  $p = r = 1$  on 8 cells departs from one by  $1.7 \times 10^{-2}$ ,  
 2010  $4.3 \times 10^{-3}$ ,  $4.8 \times 10^{-4}$  and  $4 \times 10^{-5}$  at  $kD = 0.24, 0.12, 0.04$  and  $0.012$ :  
 2011 as  $(kD)^2$  (`scripts/measure_layer_bases.py`). The pairing in Eq. (G.10)  
 2012 is verified symbolically for all nine pairs, and the projection errors are com-  
 2013 puted exactly (`ContinuumLimit_LayerDefect.wl`); they fall by  $4^{q+1}$  for each  
 2014 halving of the cells.

## Appendix H. Relation to other methods

*Relation to the coupled-dipole method.* The same partition appears in the coupled-dipole method as the choice of polarisability. The Clausius–Mossotti relation, the lattice dispersion relation [11], the polarisability taken from the exact static solution [12] and the integrated Green tensor [13] all adjust the single site so that the lattice reproduces the continuum. For space-filling cubes the adjustment is neither needed nor a free parameter: the tiling identity fixes the local field exactly, the self term of the collocation closure is the one that makes it cancel, and what is left is a closed-form error rather than an empirical one. That the method is a collocation of the volume integral equation was shown for the electromagnetic case by Lakhtakia [9]; Eq. (35) is the same statement for the elastic lattice with the self term evaluated over the finite cube, and it is what makes the error a midpoint-rule error. The filtered coupled dipoles of Piller and Martin [14] and Yurkin et al. [15] band-limit the source to the grid; the source-cell average does the complementary thing exactly, its form factor vanishing on the reciprocal lattice, and the spectral construction of Eq. (B.1) is the periodic coupled-dipole method of Chaumet et al. [16] with the Galerkin double average in place of the point dipole. The second-order convergence reported for the DDA by Yurkin et al. [3] corresponds to the uniform-field voxel’s, and the opposite-signed constants of Eq. (36) explain why a polarisability correction can improve one observable only at the expense of another, as the package’s own form-factor corrections do in reflection and transmission. Going beyond second order requires a richer internal field, which is the point of higher-order bases for volume integral equations [70, 2]; the first-moment voxel is the lowest such

2040 basis, and in its spectral form the Galerkin double average makes the lattice  
2041 sum converge without Ewald splitting.

2042 *Relation to the spectral element method.* The voxel of degree  $p$  is close kin to  
2043 the spectral element [71], which was taken up for seismic wave propagation in  
2044 the 1990s, is set out for three-dimensional models by Komatitsch and Tromp  
2045 [72], and is now a standard tool for it. Both divide the medium into hexahe-  
2046 dra, represent the field in each by polynomials of high degree, and determine  
2047 the coefficients by a Galerkin condition on a weak form. The spectral element  
2048 uses Lagrange interpolants through the Gauss–Lobatto–Legendre points of  
2049 the element; the voxel uses the Legendre polynomials themselves. Four of  
2050 the present results have their counterparts there.

2051 1. *A diagonal mass matrix.* The spectral element obtains one by integrat-  
2052 ing at the Gauss–Lobatto–Legendre points, at which each interpolant  
2053 vanishes but one [72]. That rule is not exact for the product of two  
2054 interpolants, so that the diagonal form is reached, in their words, by a  
2055 process of subintegration, as in the finite-element methods that lump  
2056 the mass. Here the Gram matrix  $\mathbf{M}$  of Eq. (G.8) is diagonal with no  
2057 approximation, because the Legendre functions are orthogonal on the  
2058 cell; this is why the first-order term needs no solve. The two routes  
2059 serve different ends: theirs an explicit march in time, ours the terms of  
2060 the scattering series.

2061 2. *The doubled order.* A scattered amplitude is a functional of the field,  
2062 and a Galerkin method gives a functional to twice the order of the field  
2063 itself. The order  $2p + 2$  is this doubling: the leading error of Eq. (39) is  
2064 the product of the part of the incident wave that the cell’s polynomials

miss and the part of the outgoing wave that they miss, each of order  $h^{p+1}$ .

3. *The medium to the degree of the field.* In a spectral element the density and the elastic moduli are evaluated at the element's own grid points and may vary from one to the next, so that a fully heterogeneous medium is held to the degree of the field [72]. The rule  $\min(2p+2, 2r+2)$  says why that is the right choice, and what a coarser representation of the medium would cost.

4. *Elements that honour the interfaces.* A spectral-element mesh is designed to honour the free surface and the main geological boundaries of the model [72]. The stratified layer of §9 is the same requirement seen from the integral side: with voxel faces on the interfaces the projection error vanishes, and a jump inside a voxel costs the order.

The two methods differ as a differential formulation differs from an integral one, and they suit different problems. The spectral element discretises the weak form of the equations of motion in the time domain. Its coupling is local and its matrices are sparse; it treats heterogeneity, anisotropy and topography of any form throughout a three-dimensional model; and the price is that the whole model is meshed, that the outgoing waves are absorbed at an artificial boundary, and that over a long simulation the accuracy is governed by the time scheme [72]. The voxel scheme discretises the volume integral equation in the frequency domain. Only the departure from a stratified background is meshed, the field of each voxel is independent of its neighbours', and propagation through the background, the radiation condition included, is carried exactly by the background's Green's tensor; the price is a cou-

2090 pling between every pair of voxels, and a background restricted to one whose  
 2091 Green's tensor is known. It differs also from the boundary integral methods,  
 2092 which are restricted to a finite number of homogeneous regions [72]: a vol-  
 2093 ume integral equation admits a medium that varies within the scatterer. The  
 2094 voxel scheme is therefore a candidate where the heterogeneity is localised in  
 2095 a layered medium, and not a replacement for the spectral element in a model  
 2096 that is heterogeneous throughout. A spectral-element mesh is designed by a  
 2097 rule of resolution, about five grid points to the shortest wavelength at poly-  
 2098 nomial degrees between five and ten [72]. The error analysis of this paper is  
 2099 the counterpart of that rule for the integral formulation: it gives the size and  
 2100 the degree of the voxels for a required accuracy in closed form, and it adds a  
 2101 term, the projection error of the medium.

## 2102 **Appendix I. The cost of higher order, and what the error analysis** 2103 **offers a fast solver**

2104 *The cost of fourth order.* A first-moment voxel carries four times as many  
 2105 unknowns as a uniform one (36 against 9 in three dimensions). The closed  
 2106 forms show what that buys. To reach an error  $\varepsilon$  in reflection, the uniform  
 2107 voxel needs  $kd = \sqrt{8\varepsilon}$  and the first-moment voxel  $kd = (720\varepsilon)^{1/4}$ ; at  $\varepsilon = 10^{-6}$   
 2108 that is  $kd = 2.8 \times 10^{-3}$  against 0.16, voxels 58 times larger in each direction.  
 2109 In a three-dimensional model the voxel count falls by  $58^3 \approx 2 \times 10^5$  and the  
 2110 number of unknowns by about  $5 \times 10^4$  (these are the Born-order constants;  
 2111 at the contrast studied the first-moment voxel's measured errors lie within  
 2112 5% of them, while the uniform voxel's transmission constant grows with the  
 2113 contrast, so the comparison is if anything conservative). The stratified model

2114 of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-  
 2115 moment voxels for the same accuracy. Higher moments continue the trend  
 2116 in one dimension: by Eq. (40) the second-moment voxel reaches  $\varepsilon = 10^{-6}$  at  
 2117  $kd = 0.68$  and the third-moment voxel at  $kd = 1.5$ , where a voxel is already  
 2118 a quarter of a wavelength. In three dimensions, with the Legendre products  
 2119 of total degree at most  $p$ , a voxel carries 10 ( $p = 2$ ) or 20 ( $p = 3$ ) weights per  
 2120 state component, against 4 for the first-moment voxel. The voxel of degree  
 2121 2 converges at sixth order in the layer at oblique incidence (Table B.14);  
 2122 that of degree 3 has not been built in three dimensions. The gain is largest  
 2123 precisely where it matters: when the model varies on scales that are coarse  
 2124 compared with the accuracy the wavefield must have.

2125 *What the error analysis offers a fast solver.* A voxel scheme is dense, and  
 2126 for a large medium it is solved iteratively with a fast product. The adaptive  
 2127 integral method [73] and the precorrected fast Fourier transform [74] are  
 2128 the standard forms: the sources are projected onto a regular grid, in the  
 2129 first by matching their multipole moments and in the second by matching  
 2130 the potentials they produce at a set of test points, the interactions between  
 2131 distant cells are evaluated on that grid by fast Fourier transforms, and those  
 2132 between near cells are computed directly and corrected. Such a method has  
 2133 three accuracies to set, that of the discretisation, that of the projection, and  
 2134 the radius of the near zone, and they are usually set by trial. The results of  
 2135 this paper set them beforehand, in five ways.

- 2136 1. *The target is known.* The discretisation error is known before the solve,  
 2137 in order and in size: the first-order factor of a cell, Eq. (39) and its  
 2138 form for any pair of wavenumbers (§9), and the projection error of the

2139 medium, Eq. (44). A fast product need be no more accurate than that,  
2140 and it is known when it is accurate enough.

2141 2. *The two errors do not mix.* The error of the scheme is the single site's:  
2142 the coupling between cells, averaged over the cells, is exact (§5). An  
2143 approximation of the far coupling to a set tolerance therefore adds its  
2144 own error to the discretisation's and does not alter it, so the two can  
2145 be budgeted separately.

2146 3. *For cubes on a lattice there is nothing to project.* The cells are the reg-  
2147 ular grid. The coupling depends on the separation of the cells alone, so  
2148 the product is a convolution, block by block in the polynomial degrees,  
2149 and the fast Fourier transform applies to it exactly, as in Georgakis  
2150 et al. [2] for the linear voxel. What the projection onto the grid ap-  
2151 proximates in general is here an identity.

2152 4. *For cells of several sizes the moments are already in hand.* Where the  
2153 cells differ in size or degree and are mapped to a uniform auxiliary  
2154 grid, the quantities to be matched are the polynomial moments of a  
2155 cell, which are the unknowns of the scheme; and the first-order factor  
2156 says how many of them matter at a given  $kd$ .

2157 5. *A stratified background splits as the method requires.* The Green's ten-  
2158 sor of a plane-stratified background is the whole-space tensor, which is  
2159 invariant under translation in three dimensions and carries all the sin-  
2160 gularity, plus a reverberation, Eq. (C.1), which is smooth, is invariant  
2161 under lateral translation, and separates in depth into one-dimensional  
2162 moments. The first is the convolution above; the second is a fast Fourier  
2163 transform laterally and a product in depth, with no near-zone correc-

tion except where a voxel touches an interface. A splitting of the kernel into a term that carries the singularity and a nonsingular remainder is used to a related end by Sharma and Triverio [75], for surface integral equations in homogeneous and in layered backgrounds: there the singular term is independent of frequency, so that the direct near-zone integrals are computed once for a whole sweep in frequency. And because the orders and the error law are unchanged by the background (§9), the error budget of the whole space carries over to it.

No such solver is built here. These are consequences of the analysis, and they are to be tested.

## **Appendix J. The evidence for each result, and where to reproduce it**

Table J.21 lists each result with the evidence for it. Every numerical value is reproducible from the Mathematica notebooks `Mathematica/ContinuumLimit_X.wl`, for X in Reference, ThinLayer, SpecularSums, StaticLocal, AveragedSums, CubeT, Chain, ErrorConstant, FourthOrder, FourthOrderError, Reduction3D, Oblique, IncidentS, Heterogeneous, SecondMoment, GradedContrast, LayerDefect and Figure (data for the figure), and their reference dumps `scripts/continuum_limit_*.py`; the moment integrals are `Mathematica/CubeMomentCore.wl` (gate `CubeMomentCoreTest.wl`), `CubeA22Block.wl`, `CubeMomentArchiveCheck.wl` and `CubeShearSplit.wl`, with the Python half of the last in `scripts/gate_cube_shear_split.py`; the stratified reference against Kennett’s recursion is `scripts/gate_heterogeneous_reference_vs_kennett.py`; the voxelised spheres are `scripts/pilot_sphere_voxel_`

2188 `vs_mie.py` (sharp) and `scripts/pilot_graded_voxel_sphere.py` (graded,  
 2189 with the mean-only voxel of `cubic_scattering/graded_voxel`), the graded  
 2190 sphere's exact solution `Mathematica/ContinuumLimit_GradedSphere.wl` with  
 2191 the library `Mathematica/TakeuchiSaito.wl` and `scripts/crosscheck_graded_  
 2192 sphere.py`; the test against exact Mie is `scripts/gate_sphere_closure_  
 2193 vs_mie.py`, with the closed forms in `Mathematica/SphereClosureDynamic.  
 2194 wl`; the exact kernel is `cubic_scattering/cell_averaged_lattice.py`; the  
 2195 independent implementation of the first-moment voxel is `scripts/crosscheck_  
 2196 first_moment_voxel.py`, and of the voxel of any degree `scripts/crosscheck_  
 2197 second_moment_voxel.py`, and of the graded cell `scripts/crosscheck_  
 2198 graded_contrast.py`; the projection-error law and its split are measured by  
 2199 `scripts/measure_layer_bases.py`, in a stratified background by `scripts/  
 2200 measure_layer_stratified_background.py`, `scripts/measure_oblique_  
 2201 stratified_background.py` and `scripts/measure_oblique_two_term.py`,  
 2202 and the voxel of degree 2 at oblique incidence by `scripts/measure_oblique_  
 2203 degree.py`; the convergence figure is drawn by `scripts/plot_convergence_  
 2204 orders.py`; the near field of the sphere is measured by `scripts/measure_  
 2205 graded_sphere_near_field.py`; the moments of every grade are in `Mathematica/  
 2206 CubeMomentHigherGrades.wl`, `Mathematica/CubeMomentStore.wl` and `Mathematica/  
 2207 CubeMomentCanonical.wl`, with their independent evaluation in `scripts/  
 2208 crosscheck_cube_moments_ball_shell.py`; the hierarchy on the layer is  
 2209 `Mathematica/ContinuumLimit_GradientHierarchy.wl` and `scripts/measure_  
 2210 layer_taylor_hierarchy.py`, and in three dimensions `scripts/measure_  
 2211 ball_gradient_hierarchy.py`, `scripts/measure_cube_gradient_hierarchy.  
 2212 py` and `scripts/measure_graded_sphere_gradient_hierarchy_fft.py`. All

<sup>2213</sup> are available at <https://github.com/tmnestor/MultipleScatteringCalculations>.

Table J.21: The results of the paper, and the evidence for each.

| Result                                                     | Evidence                                                          |
|------------------------------------------------------------|-------------------------------------------------------------------|
| Continuous layer, exact and to $O(D^k)$                    | reference to $2.5 \times 10^{-8}$ ; series order $k$              |
| Thin-layer series, Eq. (7)                                 | each coefficient symbolically, to $D^4$                           |
| Point-kernel local term survives $d \rightarrow 0$         | $11 \times$ the continuum term at $m = 1$                         |
| Averaged coupling between planes = continuum $\times$ sinc | $2.8 \times 10^{-16}$ , 81 entries                                |
| Tiling identity, Eq. (13)                                  | 21/21 components to $4 \times 10^{-9}$                            |
| Exact kernel, Eq. (14)                                     | replaces a $9.4 \times 10^{-4}$ static error                      |
| Moments as distributions, Eq. (26)                         | sum rule (27) exact; M, Q equal an independent archive, 81/81     |
| Channels (29)–(31)                                         | a production code by another route, $2 \times 10^{-15}$           |
| Cube vs sphere, Eq. (32)                                   | symbolic; and numerically at five $\nu$ , $4 \times 10^{-16}$     |
| Ball closure = exact Mie, static                           | $2 \times 10^{-10}$ ; cube's bulk and shear average $10^{-8}$     |
| Ball closure vs Mie, dynamic, Eq. (F.1)                    | symbolic; numerical Mie $5 \times 10^{-4}$ ; $(ka)^3$ terms exact |
| Self term cancels, Eq. (35)                                | $4 \times 10^{-16}$                                               |
| Package $T$ = closure statically                           | $2 \times 10^{-5}$ at $k_S h = 0.01$                              |
| Second-order convergence                                   | order 2.000, three frequencies                                    |
| Closed-form error, Eq. (36)                                | $n = 1$ –16, to $10^{-4}$ of the error                            |
| First-moment voxel, Eq. (38)                               | order 4.00, three frequencies                                     |
| First-moment voxel, oblique, Eq. (B.1)                     | order 4, P and S, $20^\circ$ and $40^\circ$                       |

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2216 the Research School of Earth Sciences of the Australian National University,  
2217 Canberra [21].

## 2218 **Declaration of competing interest**

2219     The author declares no known competing financial interests or personal  
2220 relationships that could have appeared to influence the work reported in this  
2221 paper.

## 2222 **Declaration of generative AI and AI-assisted technologies in the** 2223 **manuscript preparation process**

2224 *Software development..* Software development followed a human-directed,  
2225 AI-assisted workflow. The author designed the software originally, but only  
2226 directed and validated recent improvements, which build on the author’s  
2227 original Fortran implementation (the `PhD_fortran_code` folder of the repos-  
2228 itory given under Data availability) and an extensive library of Mathematica  
2229 notebooks, developed and extended since 1992, that carry out and verify the  
2230 symbolic derivations; an agentic coding assistant (Anthropic Claude, Opus  
2231 5.5) operated under human-in-the-loop supervision to implement specified  
2232 changes and diagnostic tests. The design, the choice of numerical methods,  
2233 and the resolution of the principal technical difficulties were the author’s. All  
2234 AI-generated changes were subject to code review by the author and validated  
2235 by automated tests and independent physical checks such as reciprocity, op-  
2236 tical theorem, Hamiltonian symmetry, analytic limits and convergence.

2237 *Manuscript preparation..* During the preparation of this work, the author  
2238 used Claude (Anthropic) in order to search for relevant open-access research,  
2239 to write the TikZ code of the diagrams and to help redraft the paper into more  
2240 coherent sections for readability. After using this tool, the author reviewed  
2241 and edited the content as needed and takes full responsibility for the content  
2242 of the published article.

## 2243 **Data availability**

2244 The Mathematica notebooks and Python scripts that derive and check  
2245 every result, and the package they test, are available at [https://github.](https://github.com/tmnestor/MultipleScatteringCalculations)  
2246 [com/tmnestor/MultipleScatteringCalculations](https://github.com/tmnestor/MultipleScatteringCalculations).

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